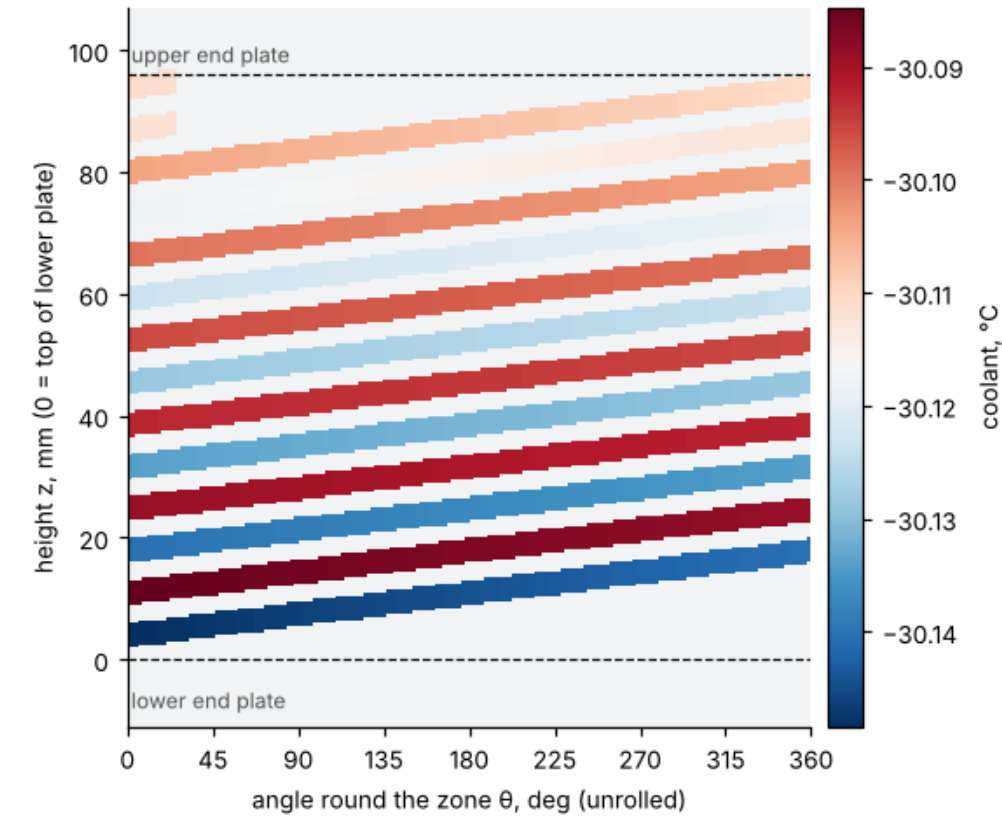


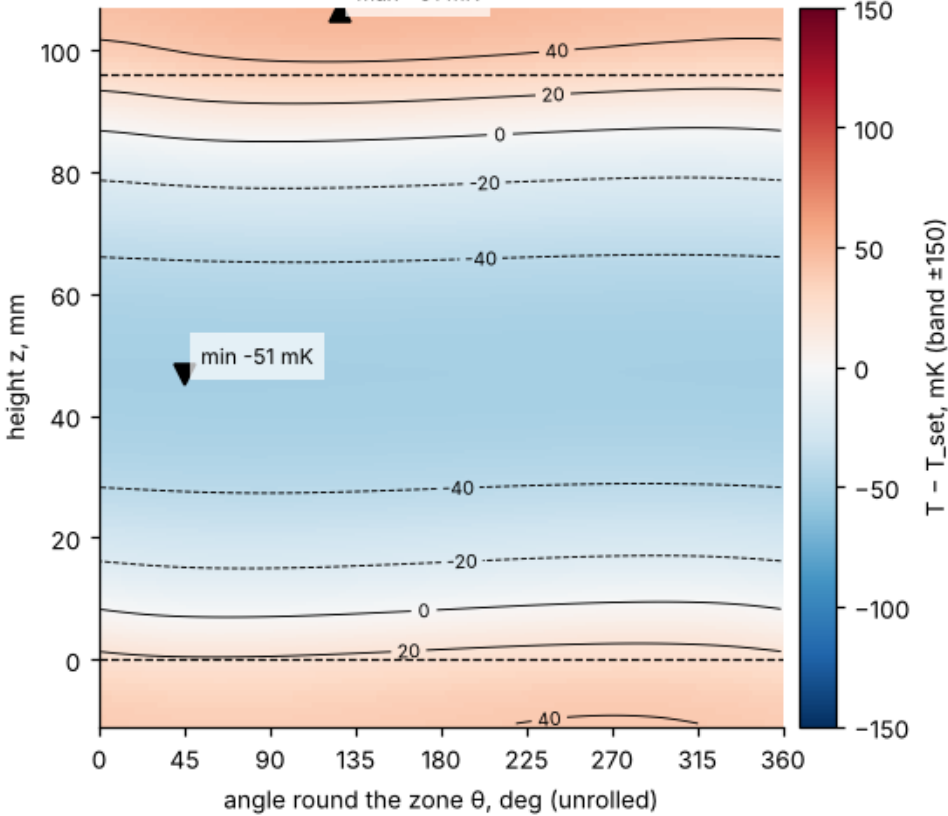
Layout A* — as built: bifilar tube on buffer

Zone 1 · total coolant flow 17.5 g/s · heat 2.55 W · inlet -30.149 °C · 1,271,160 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



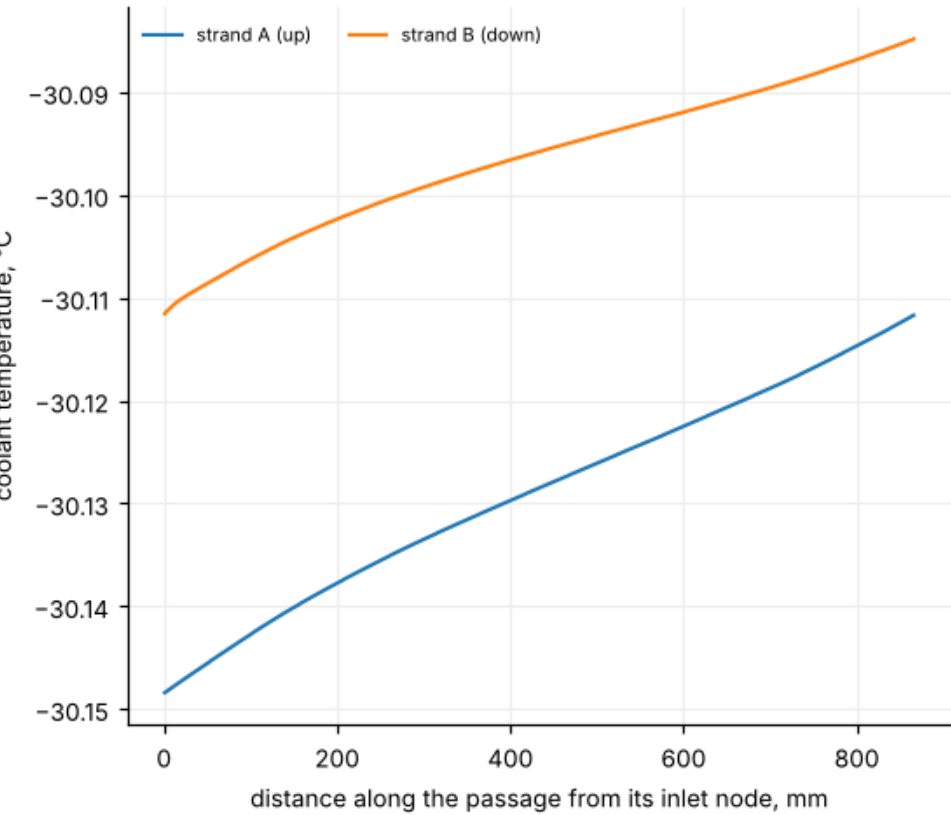
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.136 °C	-30.149 °C
Allowed inlet window, °C	-30.234 to -30.03	-30.248 to -30.049
Inner-wall spread	104.4 mK	101.3 mK
Margin inside ±0.15 K	97.8 mK	99.4 mK
Spread at 5 W (same pattern)	—	199 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	32.59 kPa	32.12 kPa*
Reynolds number, passages	3907 – 3907	3895 – 3895
Coolant-copper UA	40.0 W/K	46.1 W/K
Energy balance error		-1.9e-12 W

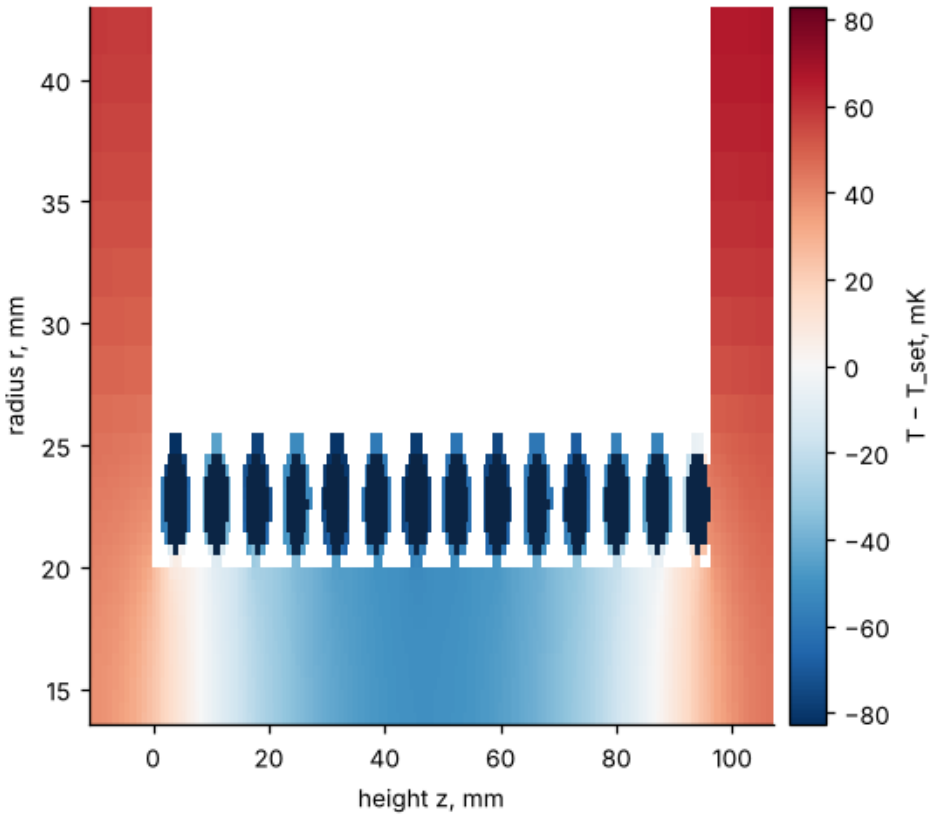
* friction of developed flow only (no bends / fittings)

Δ spread -3.1 mK · Δ inlet -13 mK → SAME

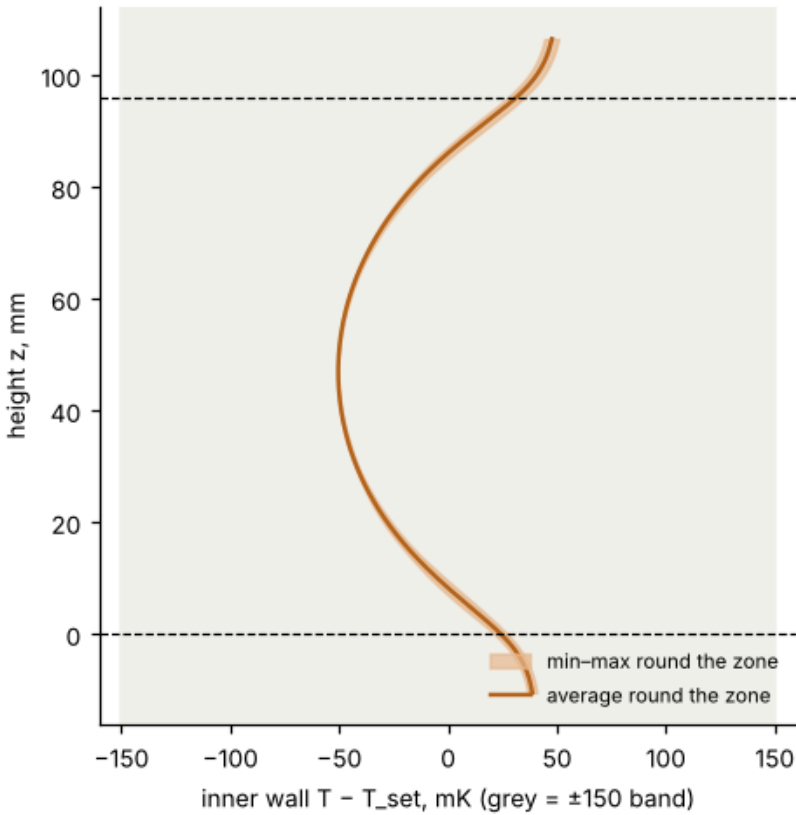
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

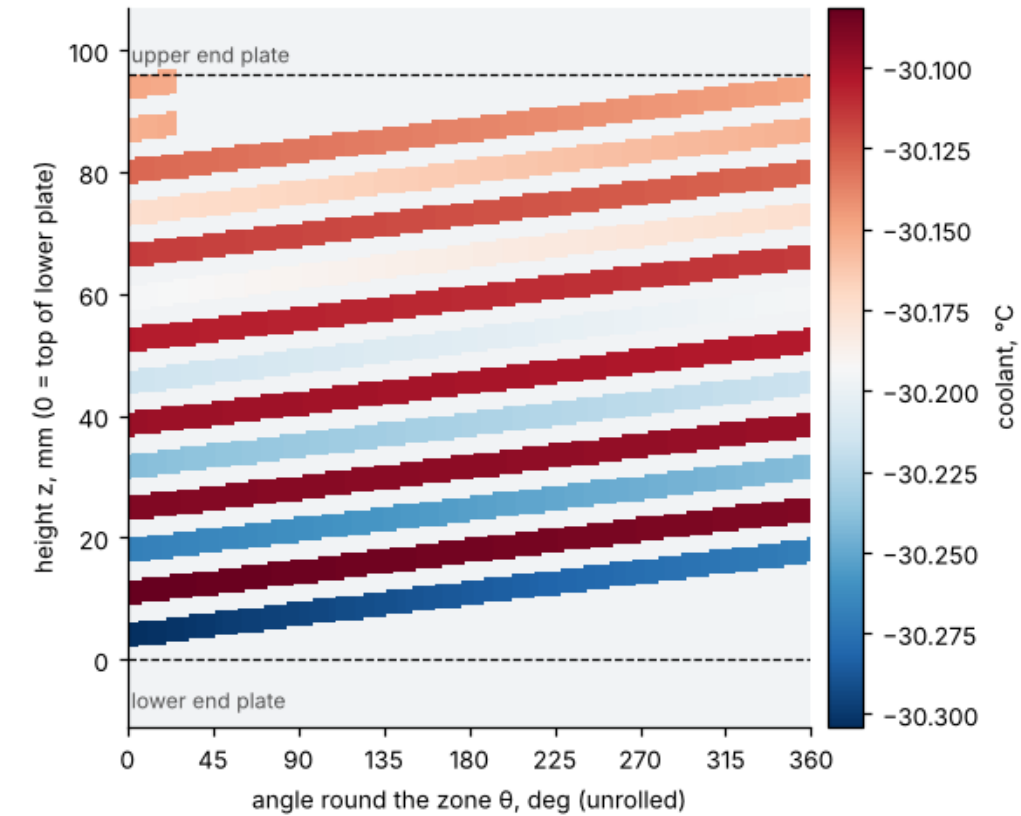


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

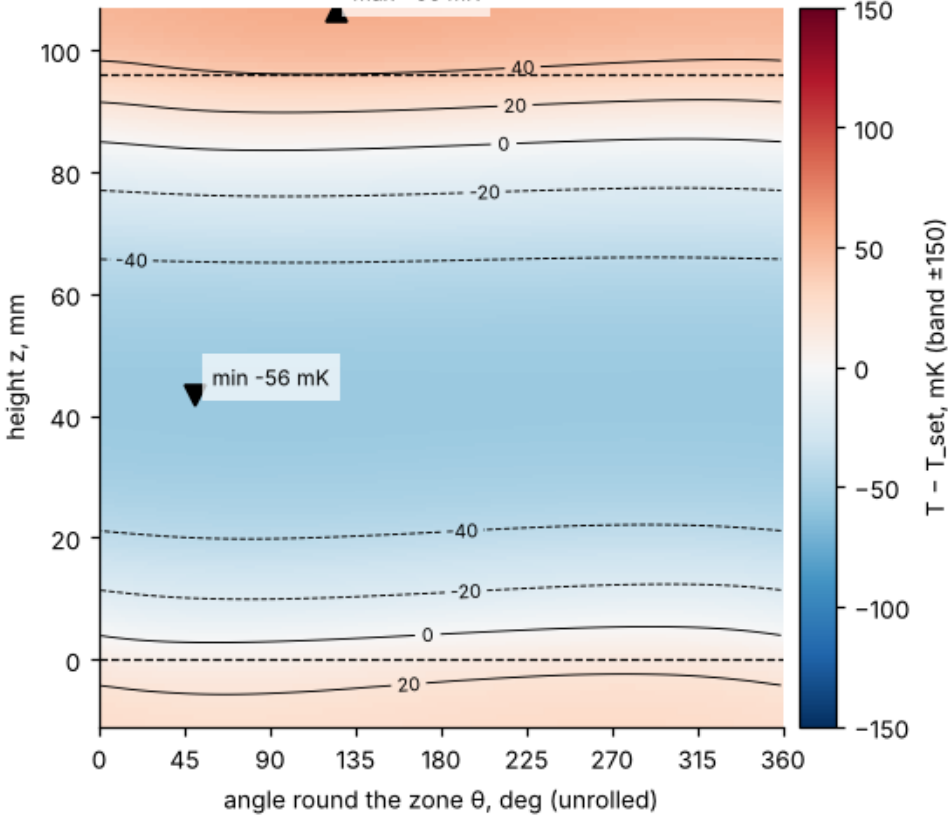
Layout A* — as built: bifilar tube on buffer

Zone 1 · total coolant flow 5 g/s · heat 2.55 W · inlet -30.305 °C · 1,271,160 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



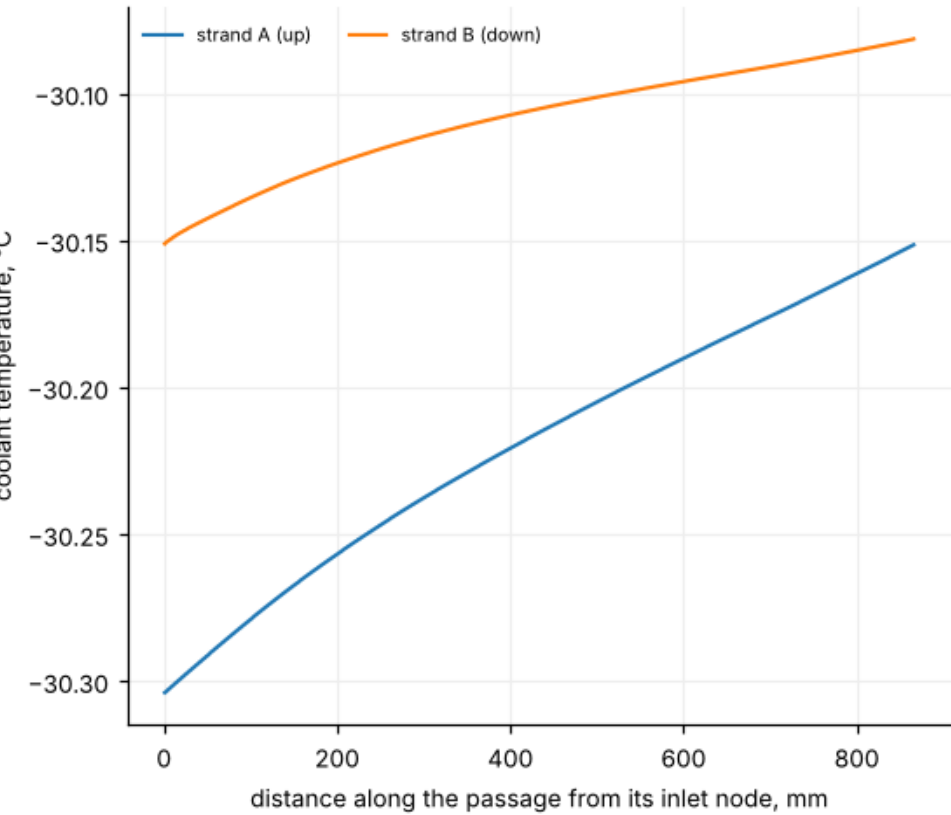
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.296 °C	-30.305 °C
Allowed inlet window, °C	-30.386 to -30.200	-30.399 to -30.210
Inner-wall spread	119.1 mK	111.8 mK
Margin inside ±0.15 K	90.5 mK	94.1 mK
Spread at 5 W (same pattern)	—	219 mK
Coolant rise, mixed outlet	225 mK	223 mK
Pressure drop, worst path	5.38 kPa	5.47 kPa*
Reynolds number, passages	1116 – 1116	1113 – 1113
Coolant-copper UA	23.2 W/K	25.1 W/K
Energy balance error		-4.2e-12 W

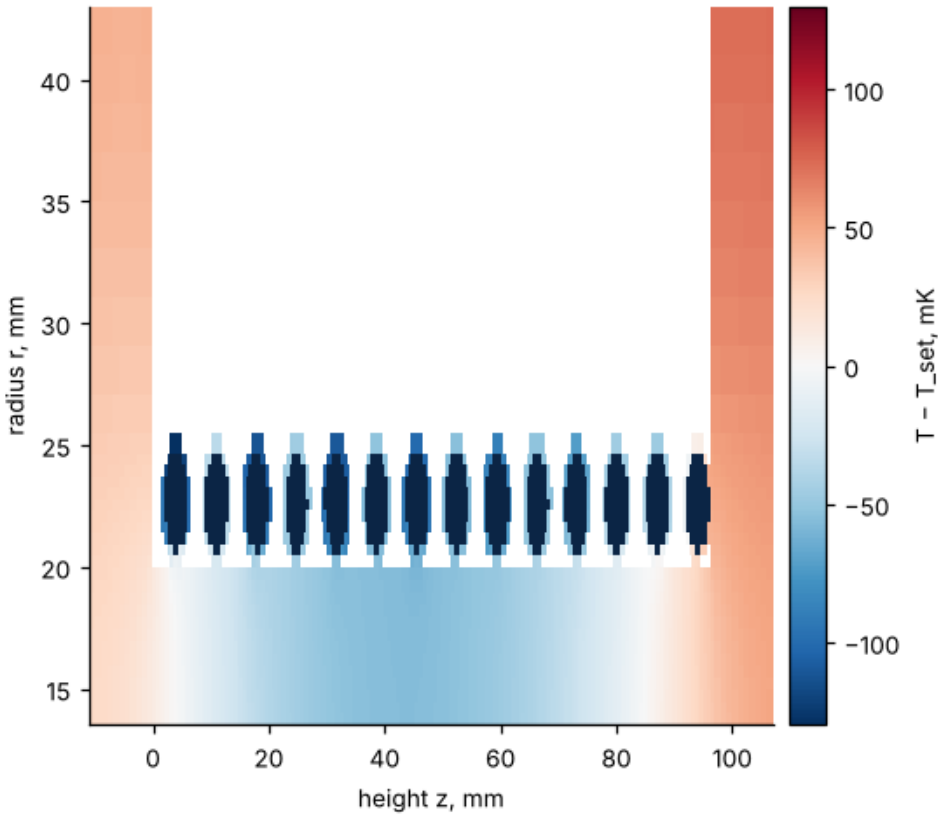
* friction of developed flow only (no bends / fittings)

Δ spread -7.3 mK · Δ inlet -9 mK → CLOSE

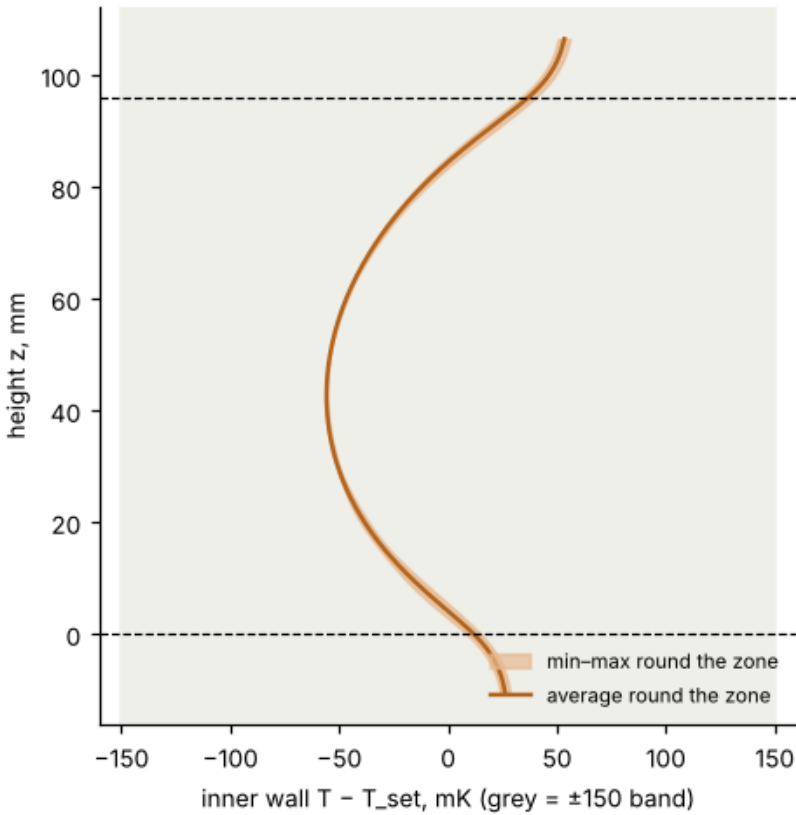
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



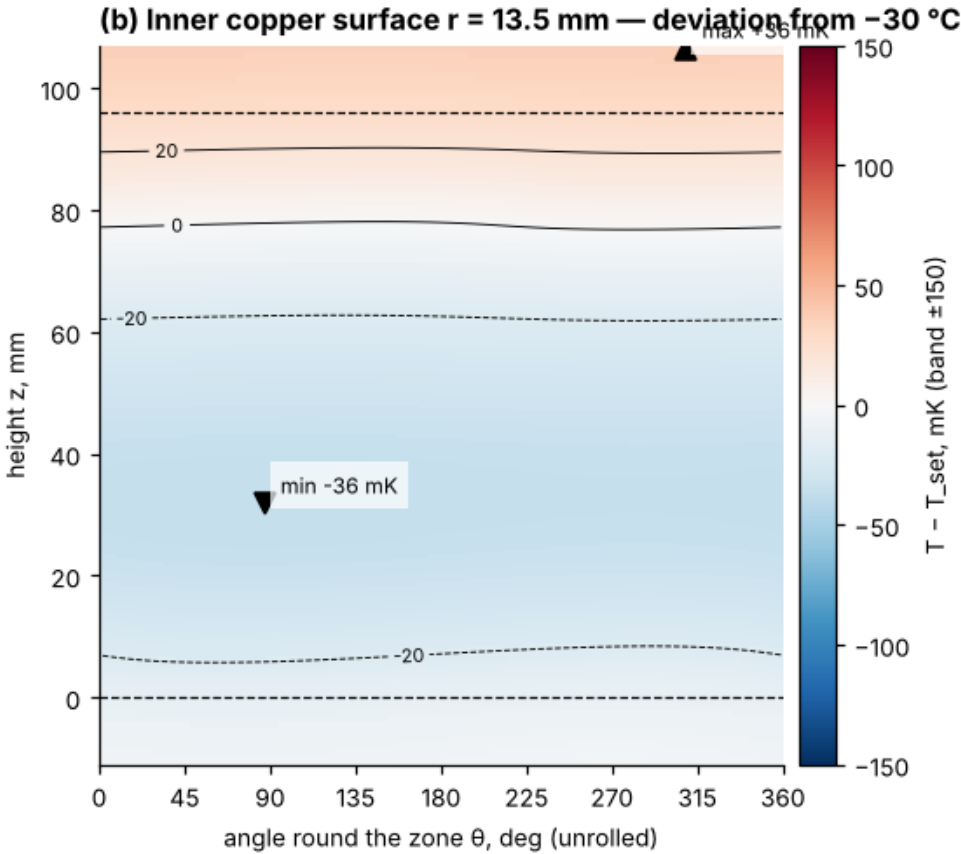
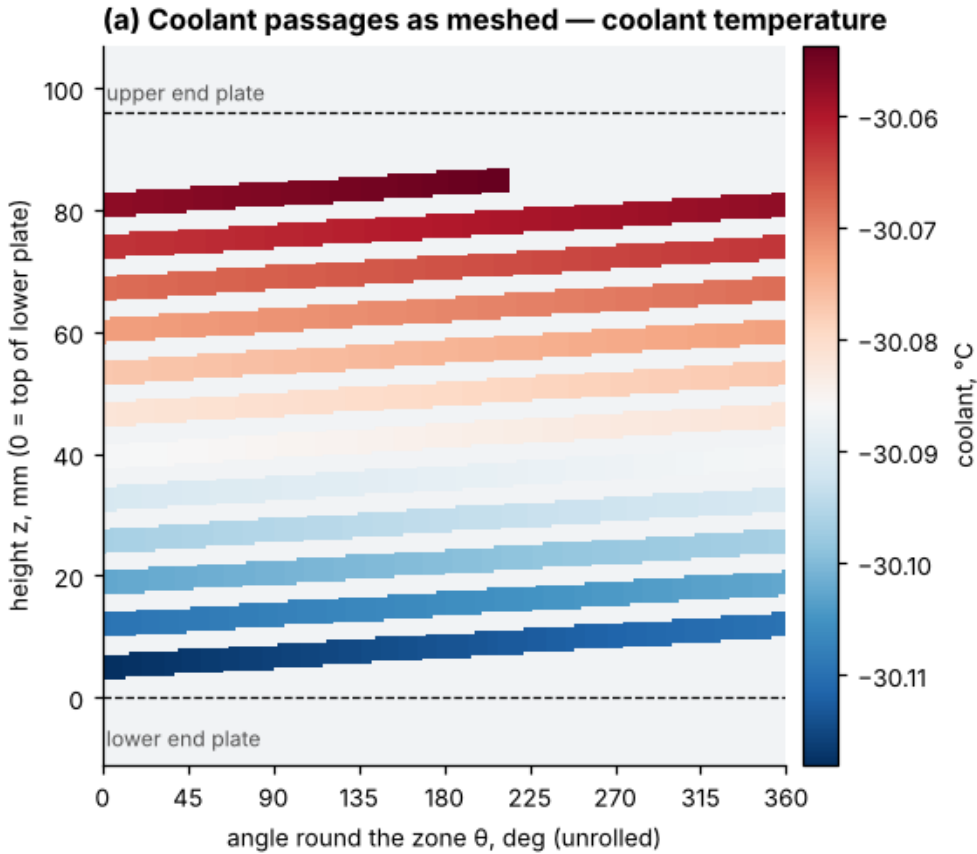
(f) Inner wall along the height



Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

Layout P — plain helix

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.118 °C · 1,348,200 cells · re-validation model

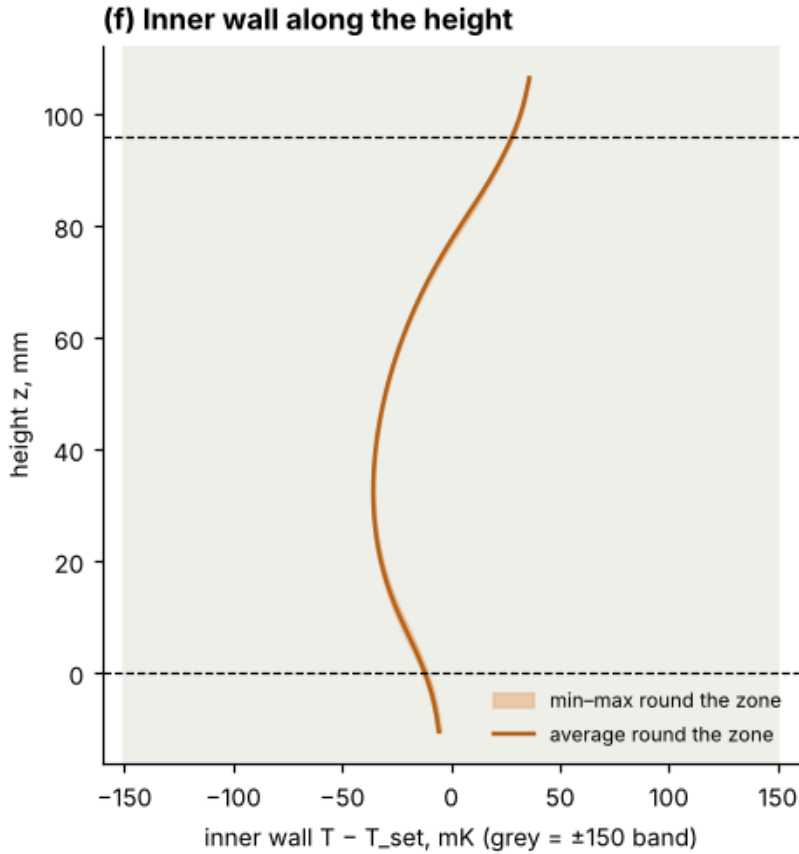
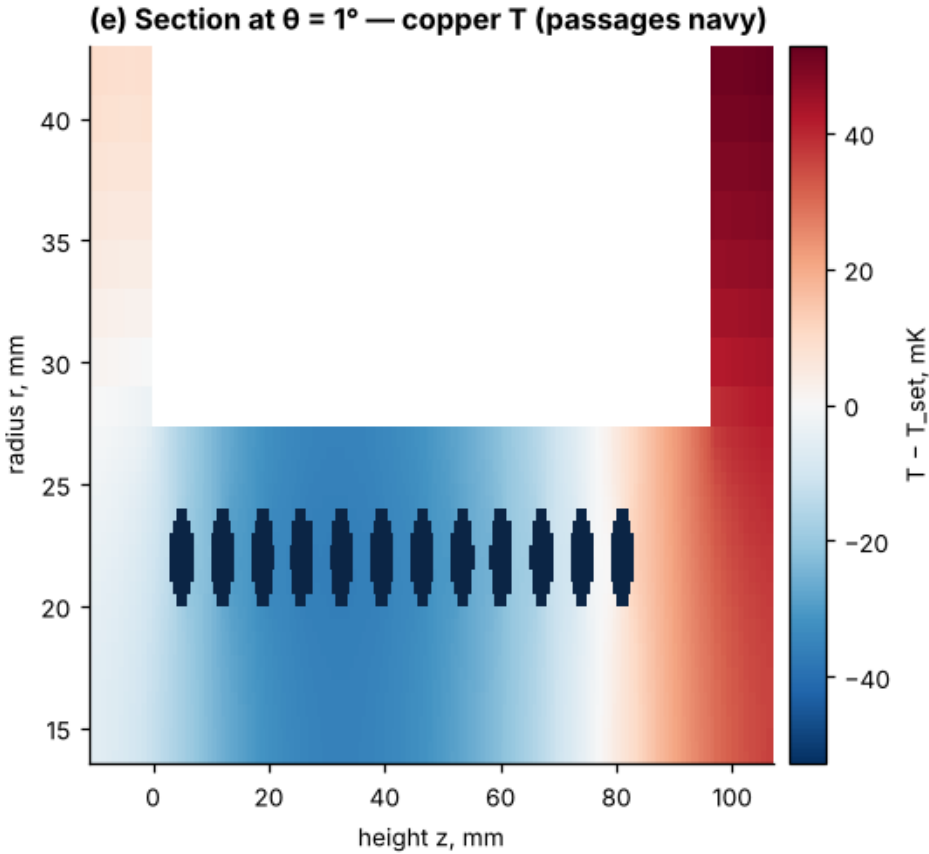
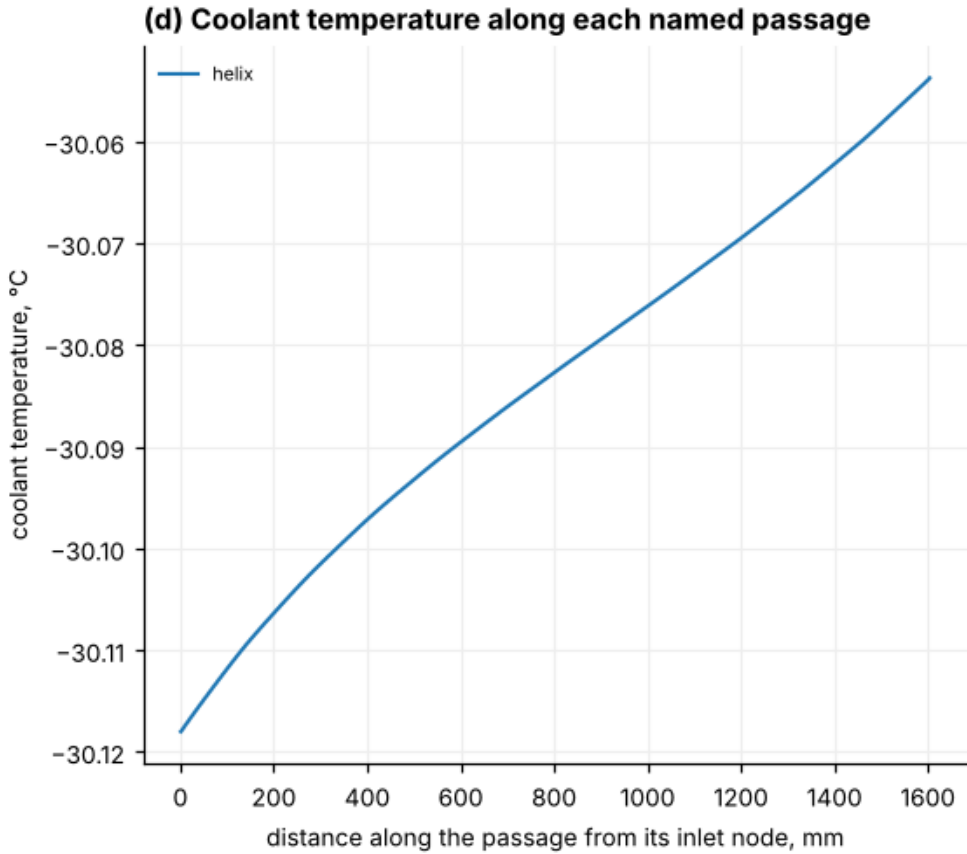


(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.118 °C	-30.118 °C
Allowed inlet window, °C	-30.232 to -30.004	-30.232 to -30.004
Inner-wall spread	71.4 mK	72.5 mK
Margin inside ±0.15 K	114.3 mK	113.8 mK
Spread at 5 W (same pattern)	138 mK	140 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	33.06 kPa	29.91 kPa*
Reynolds number, passages	3907 – 3907	3895 – 3895
Coolant-copper UA	43.0 W/K	43.0 W/K
Energy balance error		1.5e-12 W

* friction of developed flow only (no bends / fittings)

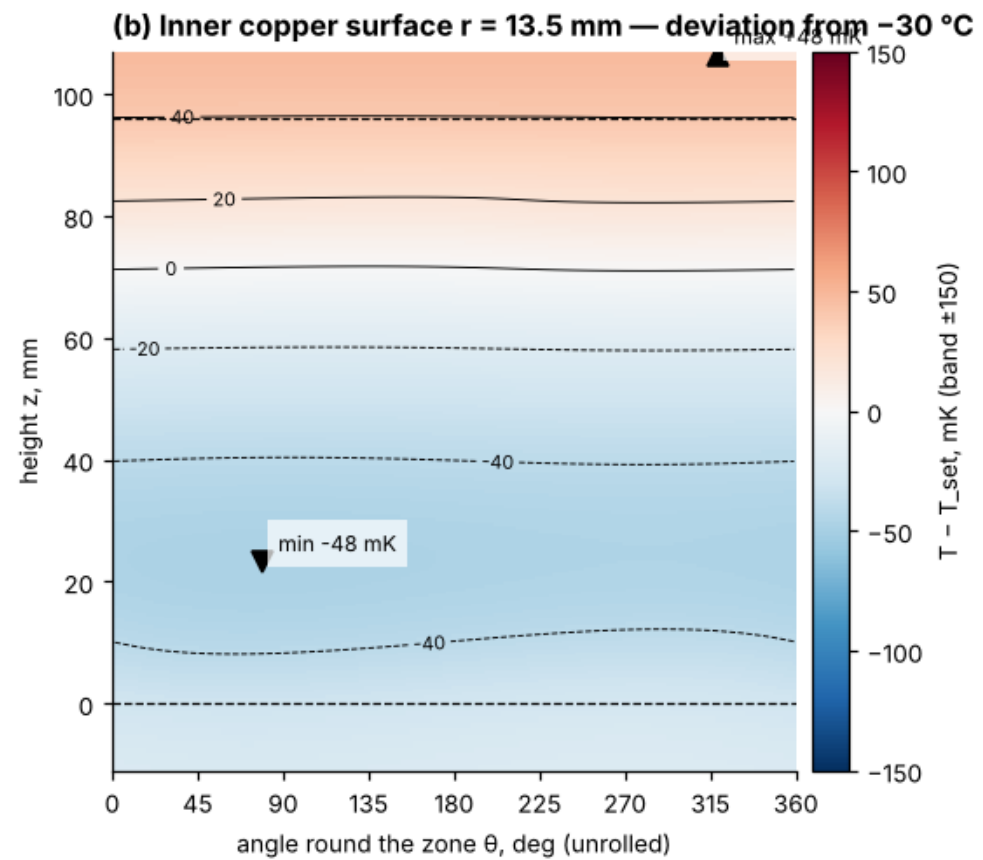
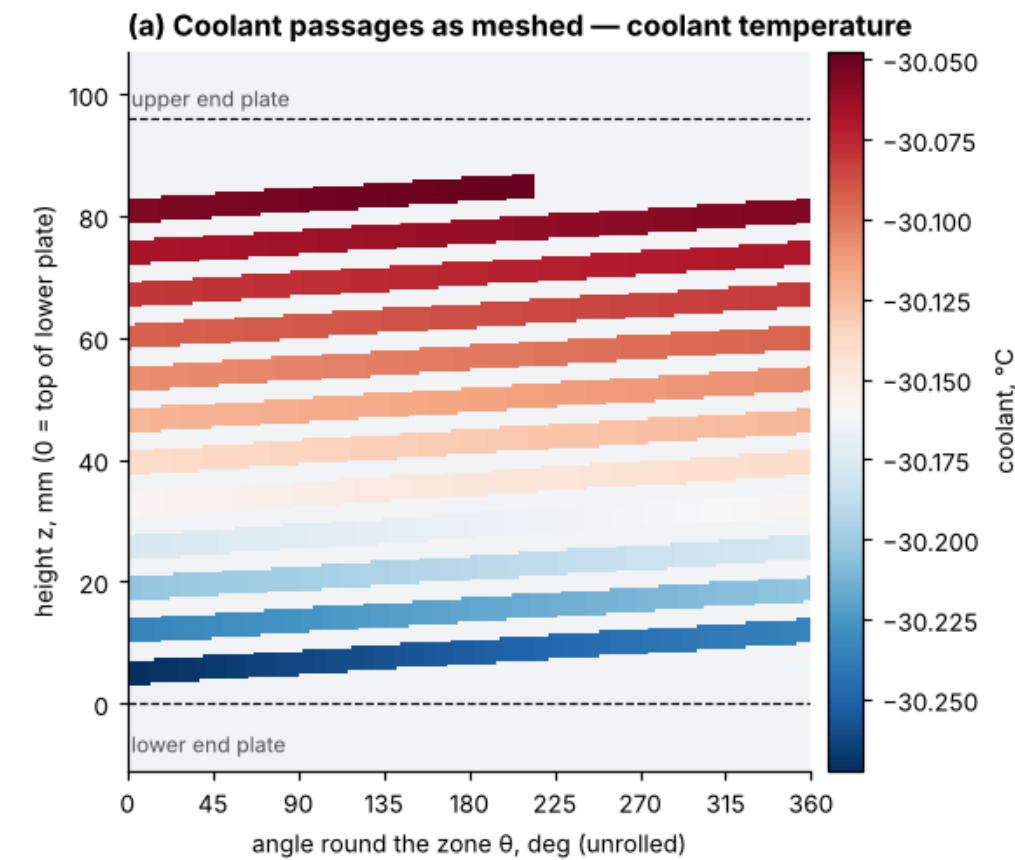
Δ spread +1.1 mK · Δ inlet -0 mK → SAME



Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

Layout P — plain helix

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.273 °C · 1,348,200 cells · re-validation model

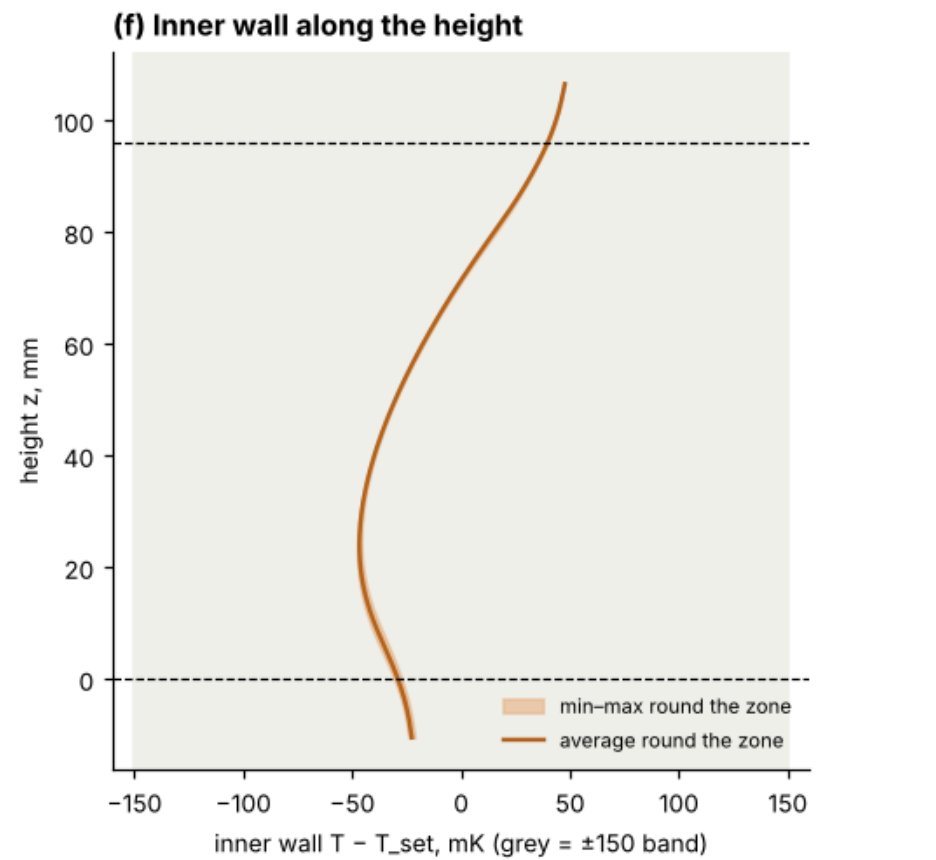
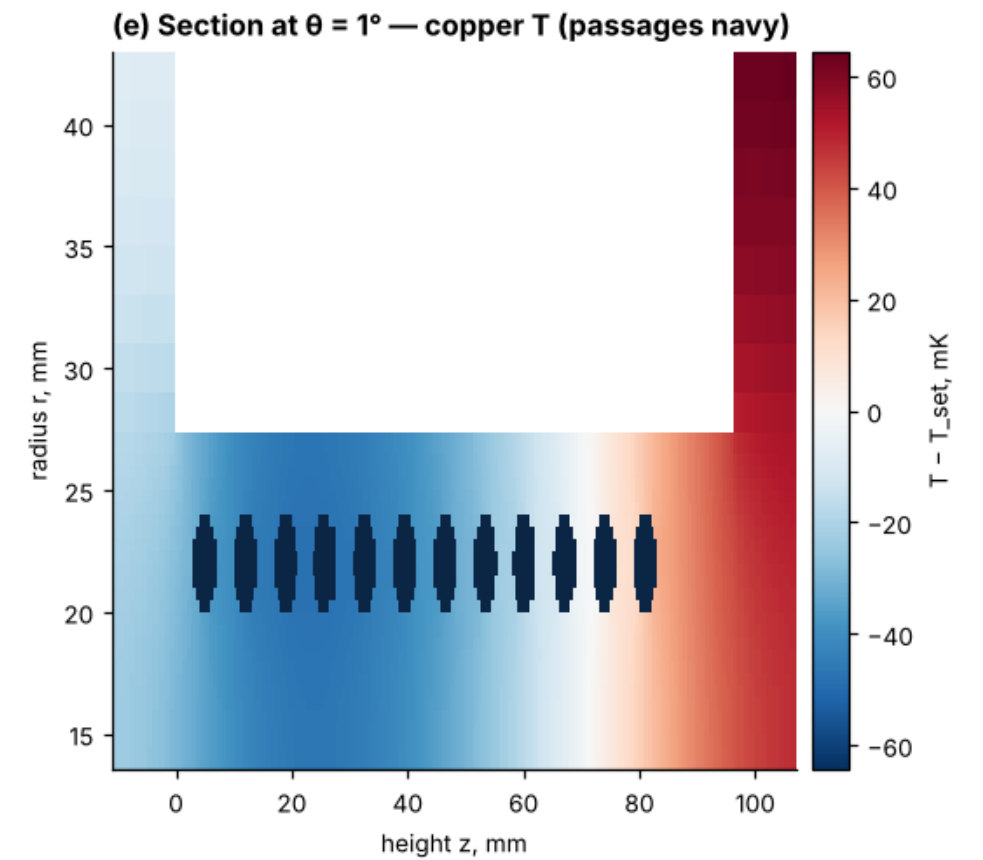
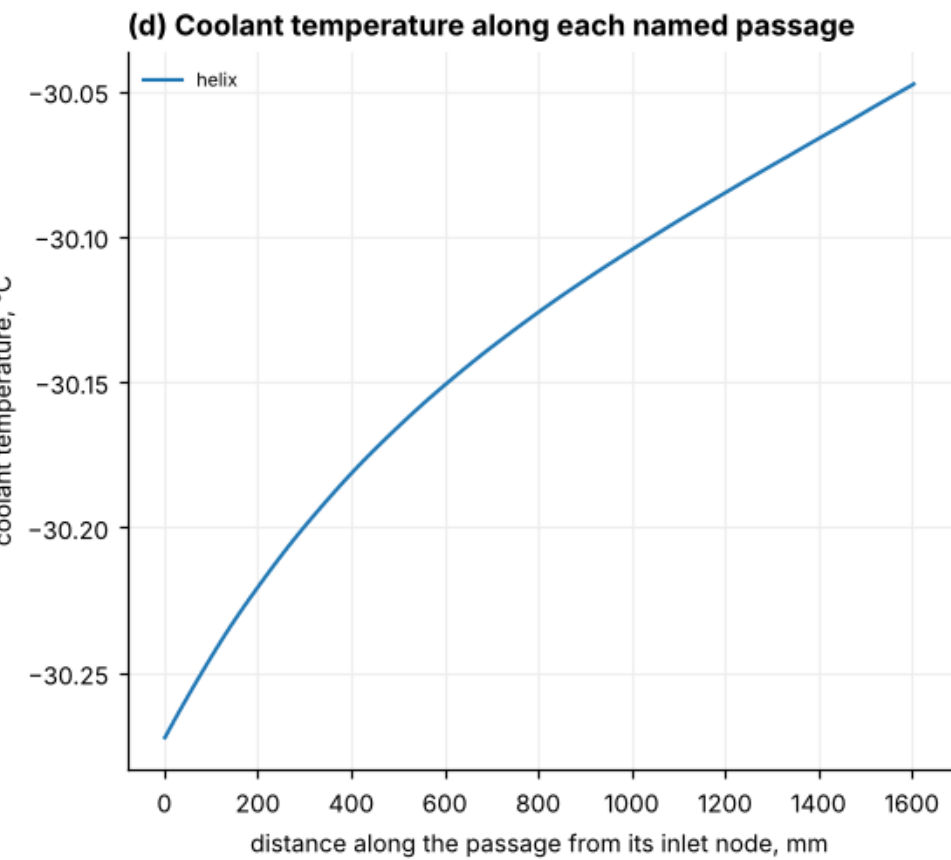


(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.272 °C	-30.273 °C
Allowed inlet window, °C	-30.375 to -30.16	-30.375 to -30.171
Inner-wall spread	93.7 mK	95.5 mK
Margin inside ±0.15 K	103.2 mK	102.3 mK
Spread at 5 W (same pattern)	182 mK	185 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	5.16 kPa	5.09 kPa*
Reynolds number, passages	1116 – 1116	1113 – 1113
Coolant-copper UA	23.5 W/K	23.5 W/K
Energy balance error		1.2e-13 W

* friction of developed flow only (no bends / fittings)

Δ spread +1.8 mK · Δ inlet -1 mK → SAME

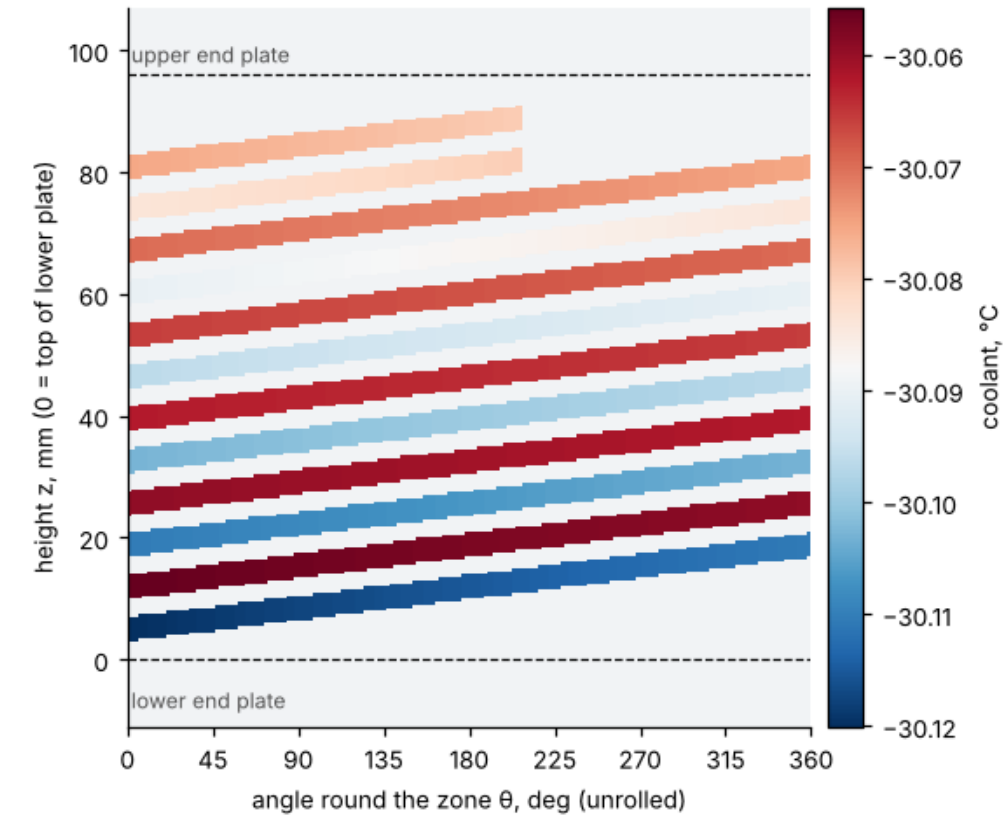


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

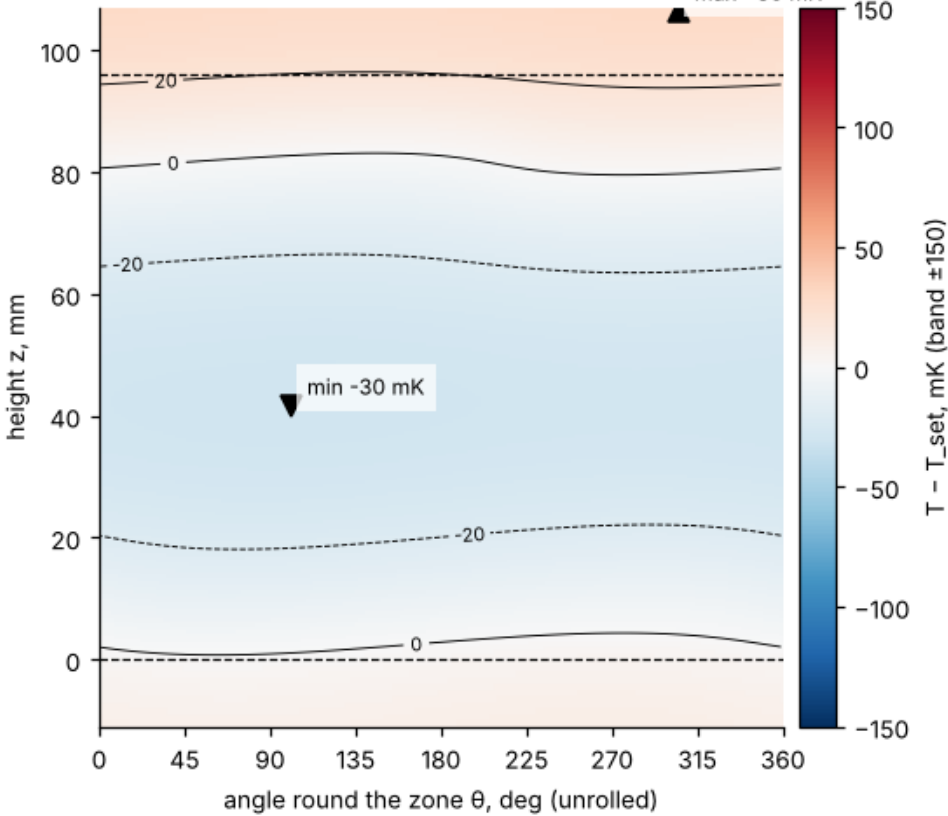
Layout A — folded helix (bifilar)

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.120 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



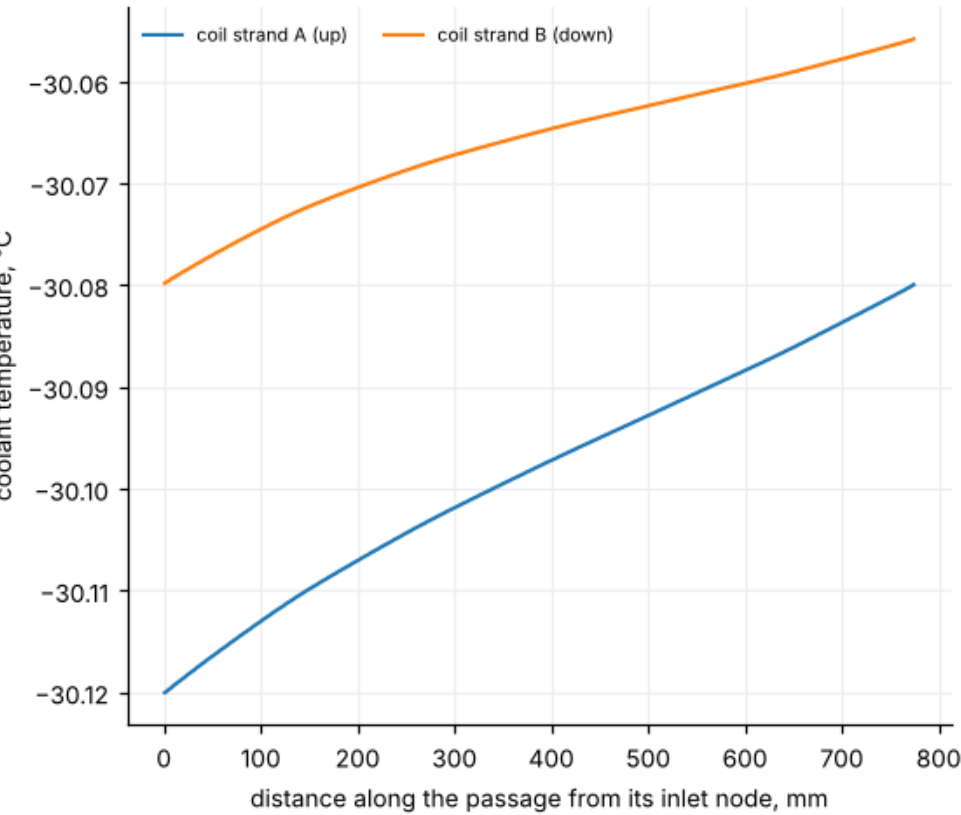
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.120 °C	-30.120 °C
Allowed inlet window, °C	-30.240 to -30.00	-30.240 to -30.000
Inner-wall spread	60.0 mK	60.1 mK
Margin inside ±0.15 K	120.0 mK	119.9 mK
Spread at 5 W (same pattern)	116 mK	117 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	31.97 kPa	28.89 kPa*
Reynolds number, passages	3907 – 3907	3895 – 3895
Coolant-copper UA	41.4 W/K	41.5 W/K
Energy balance error		9.2e-13 W

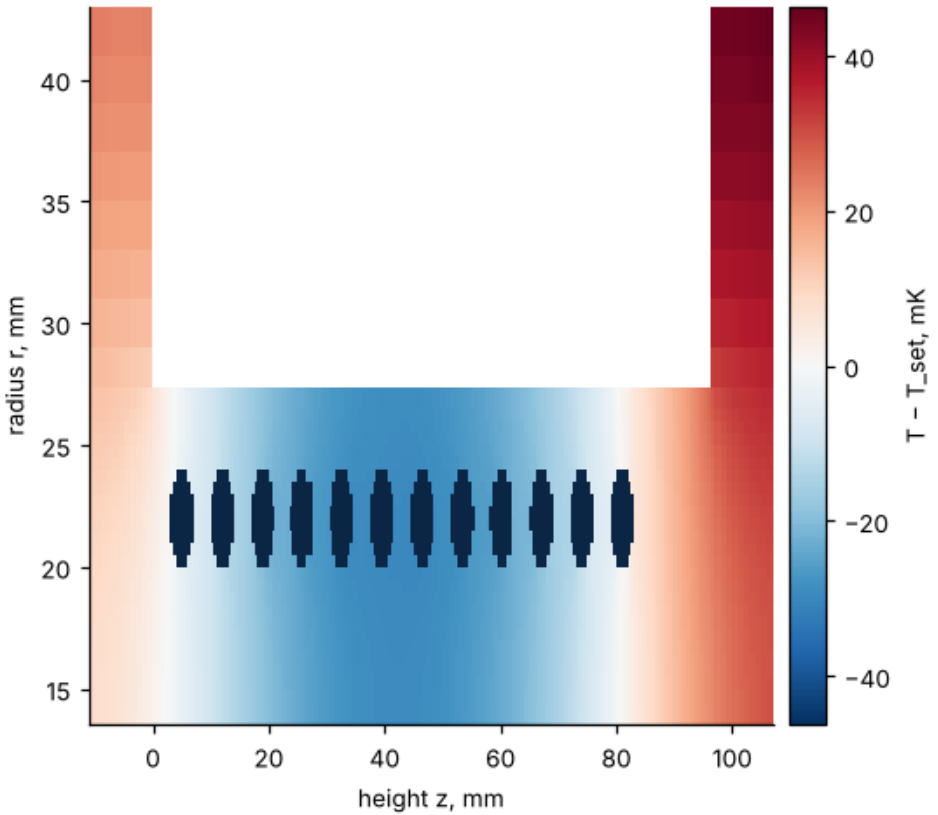
* friction of developed flow only (no bends / fittings)

Δ spread +0.1 mK · Δ inlet -0 mK → SAME

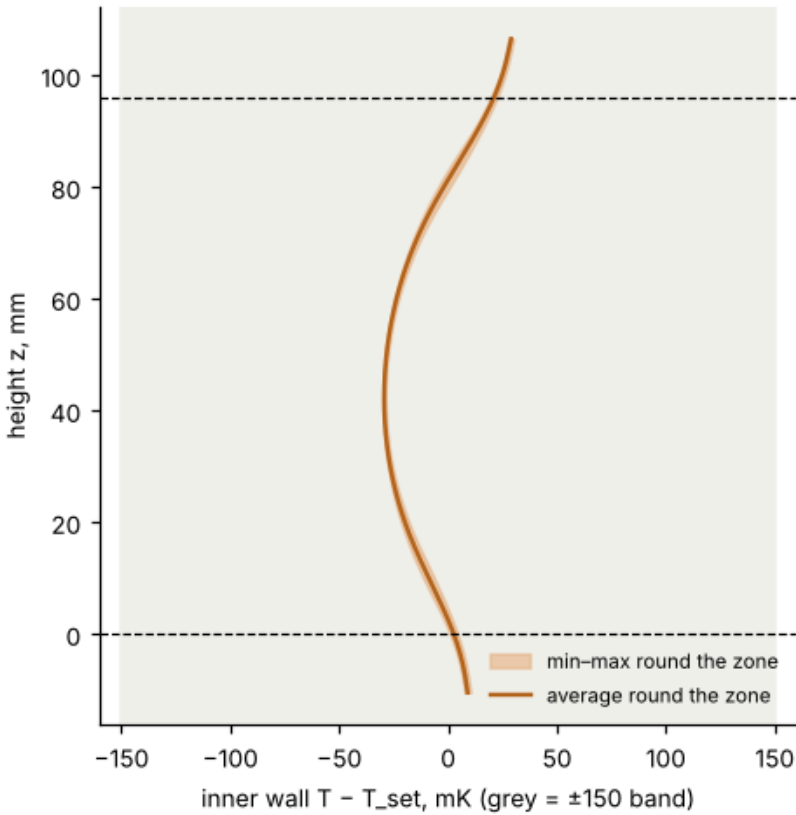
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



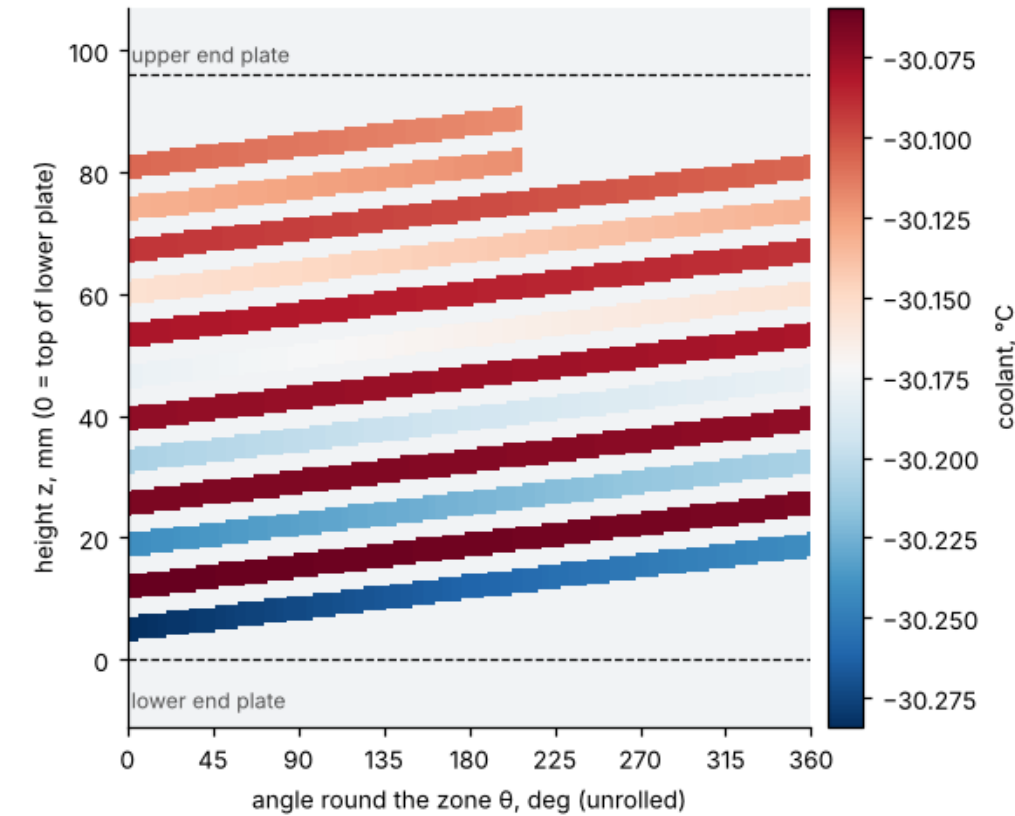
(f) Inner wall along the height



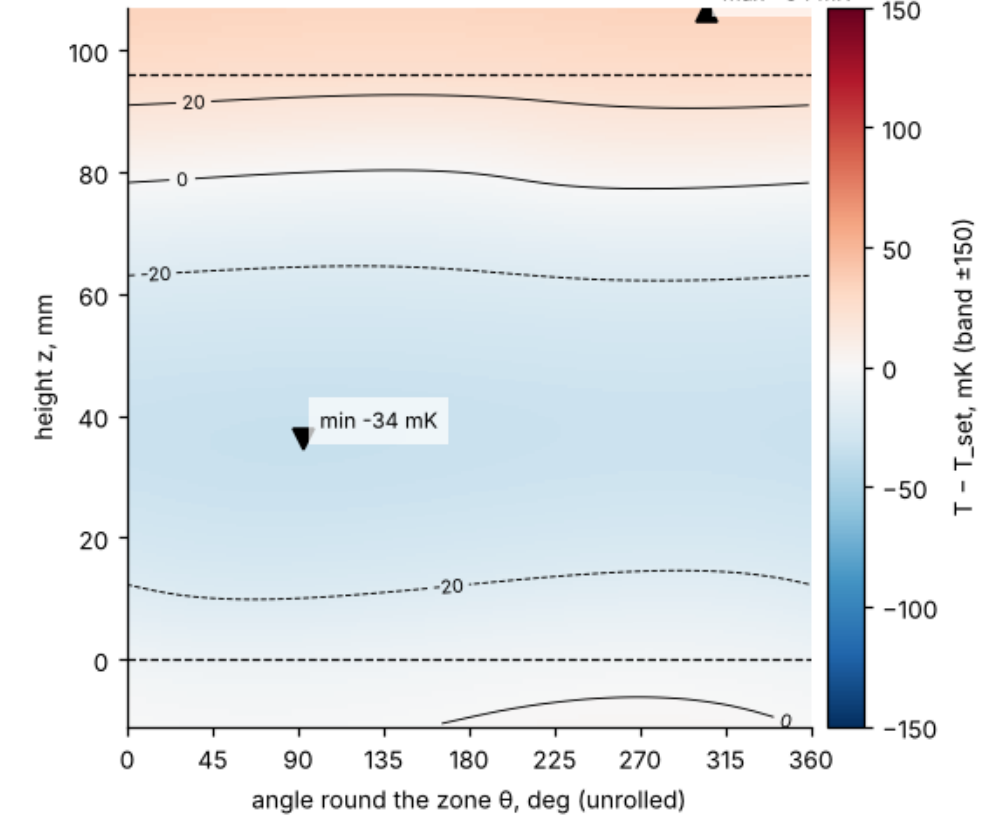
Layout A — folded helix (bifilar)

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.285 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



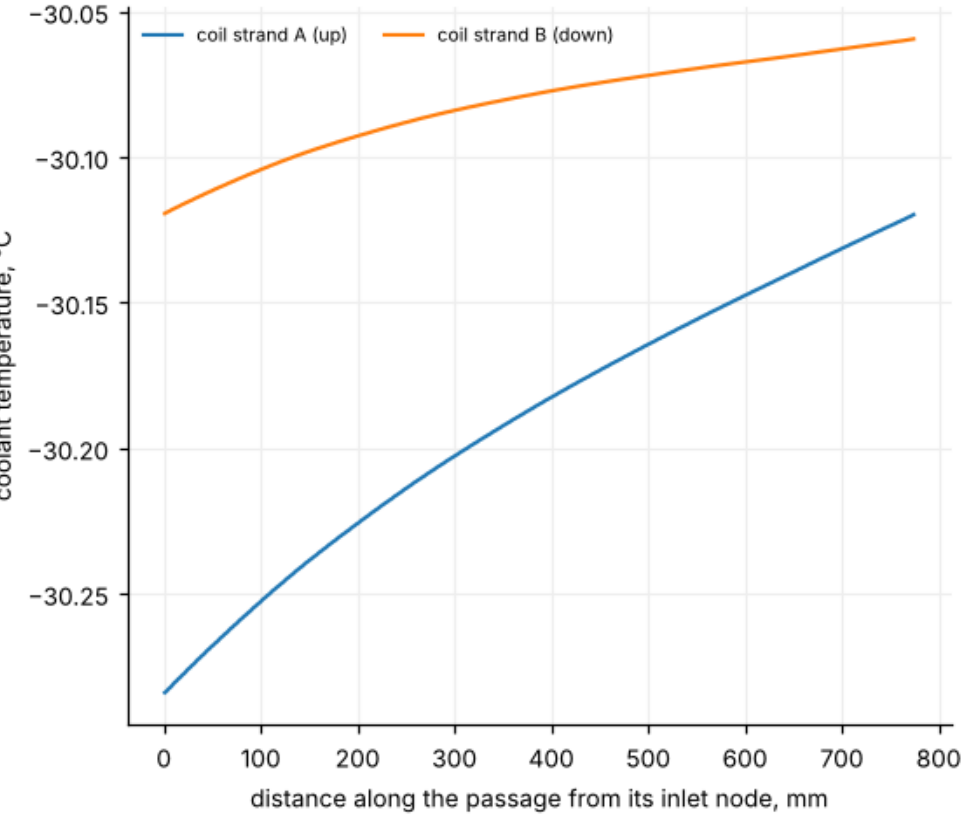
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.284 °C	-30.285 °C
Allowed inlet window, °C	-30.400 to -30.16	-30.401 to -30.169
Inner-wall spread	68.3 mK	68.6 mK
Margin inside ±0.15 K	115.9 mK	115.7 mK
Spread at 5 W (same pattern)	132 mK	133 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	4.98 kPa	4.92 kPa*
Reynolds number, passages	1116 – 1116	1113 – 1113
Coolant-copper UA	22.6 W/K	22.7 W/K
Energy balance error		2.7e-13 W

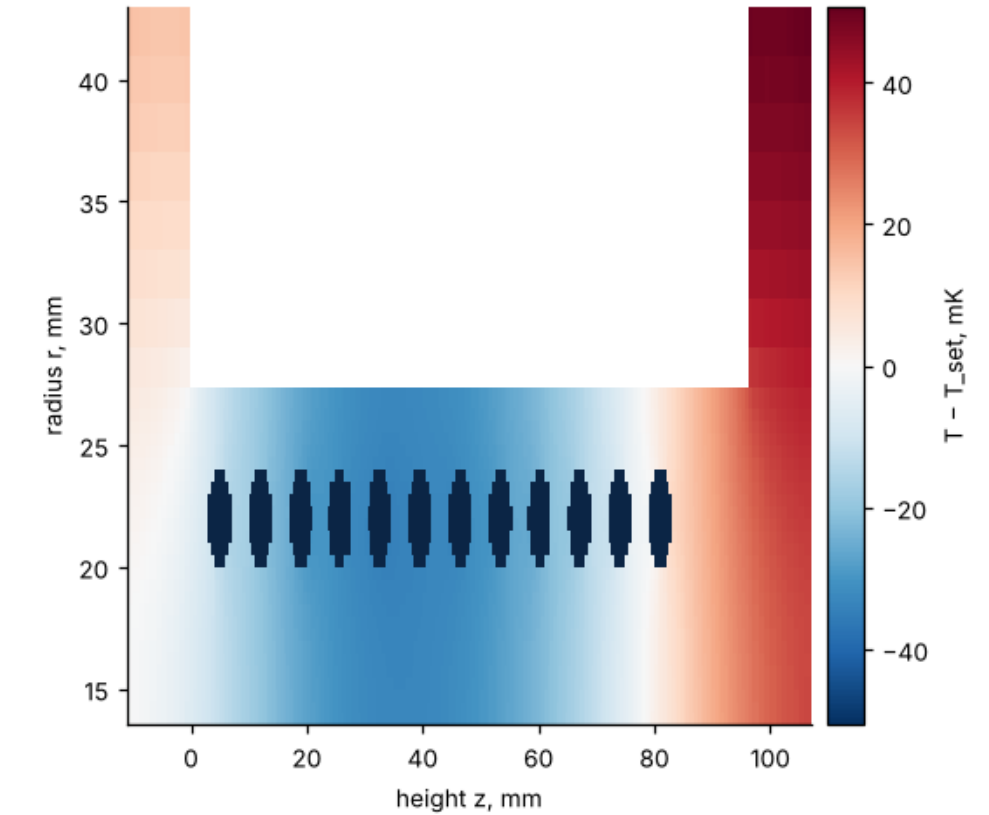
* friction of developed flow only (no bends / fittings)

Δ spread +0.3 mK · Δ inlet -1 mK → SAME

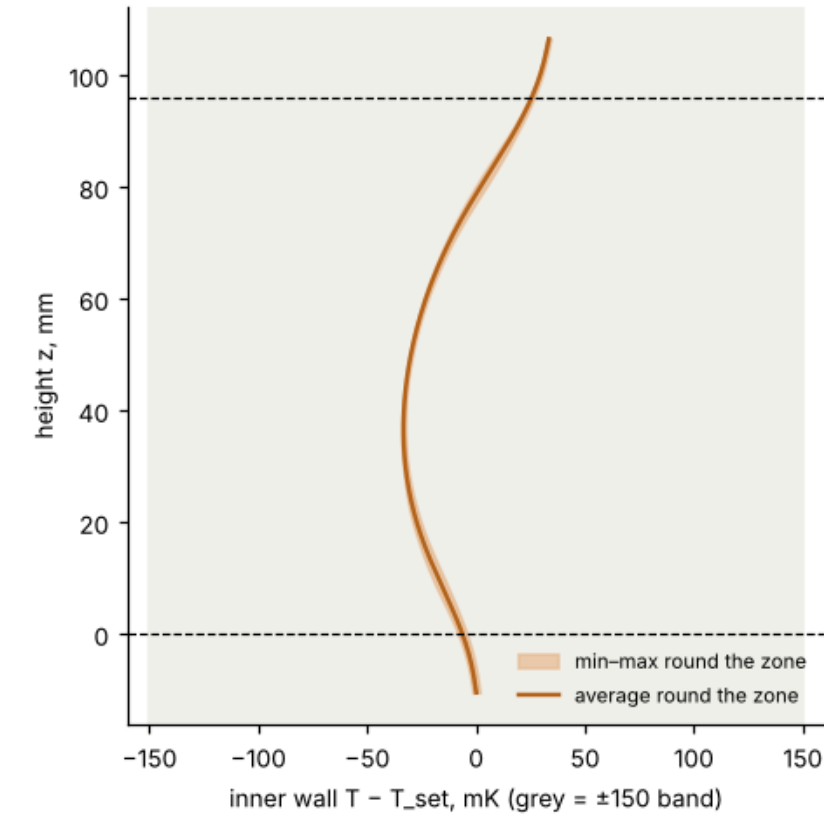
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



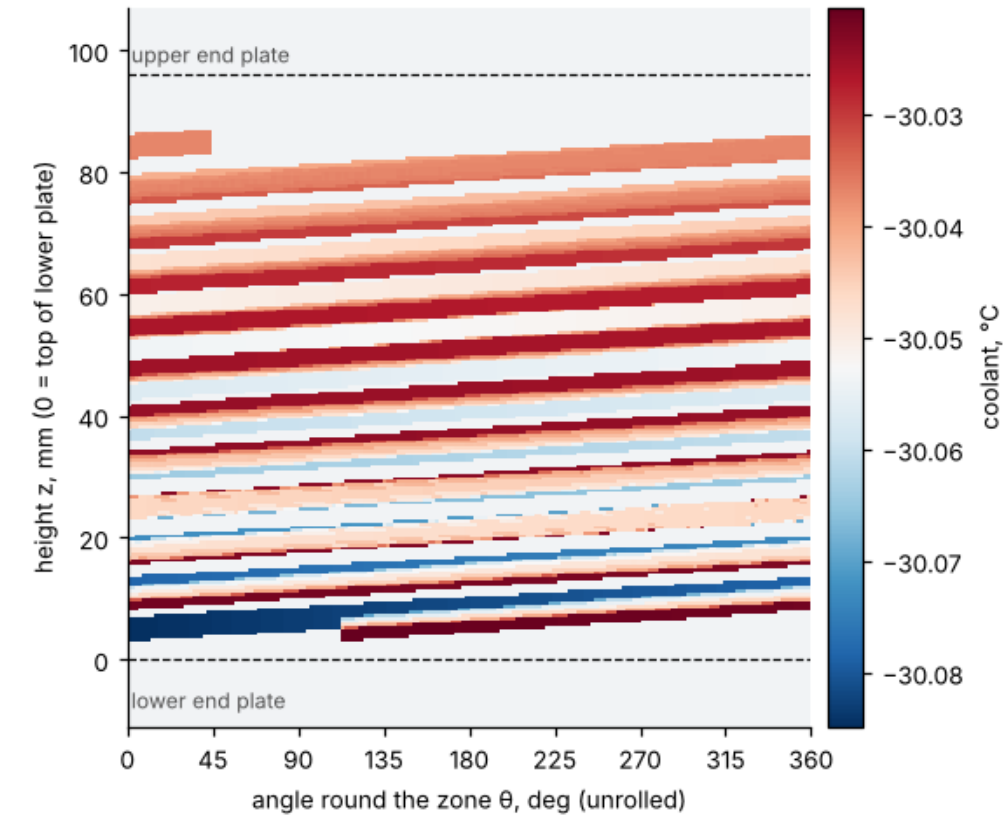
(f) Inner wall along the height



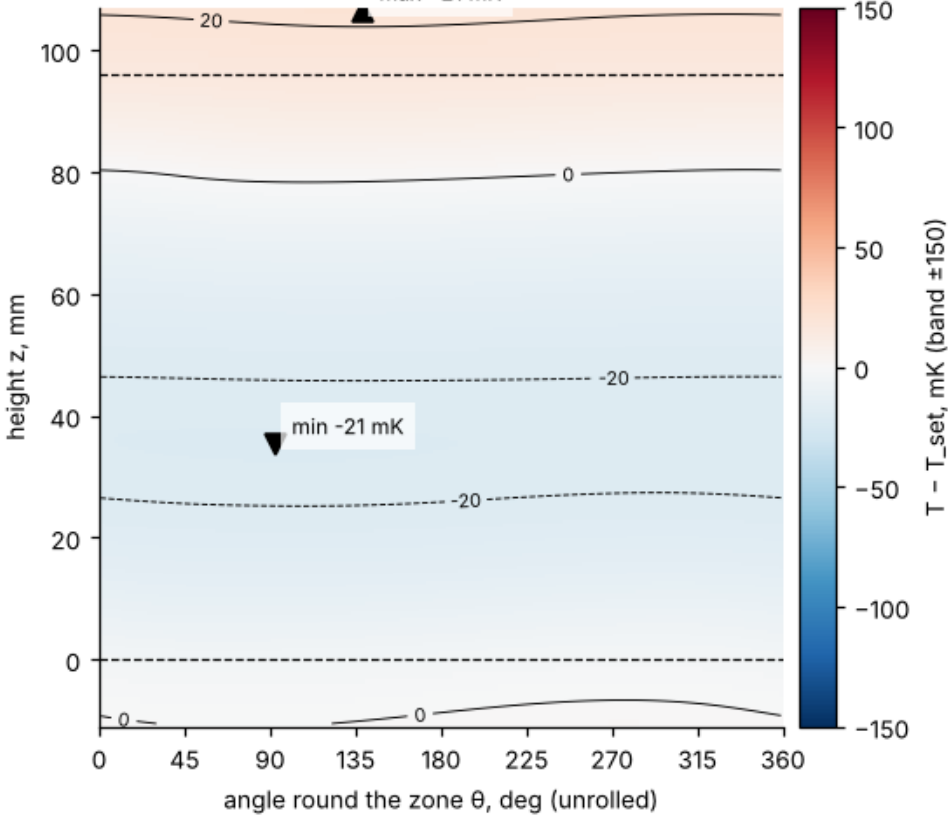
Layout B — two radial layers

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.085 °C · 1,656,360 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface $r = 13.5$ mm — deviation from -30 °C



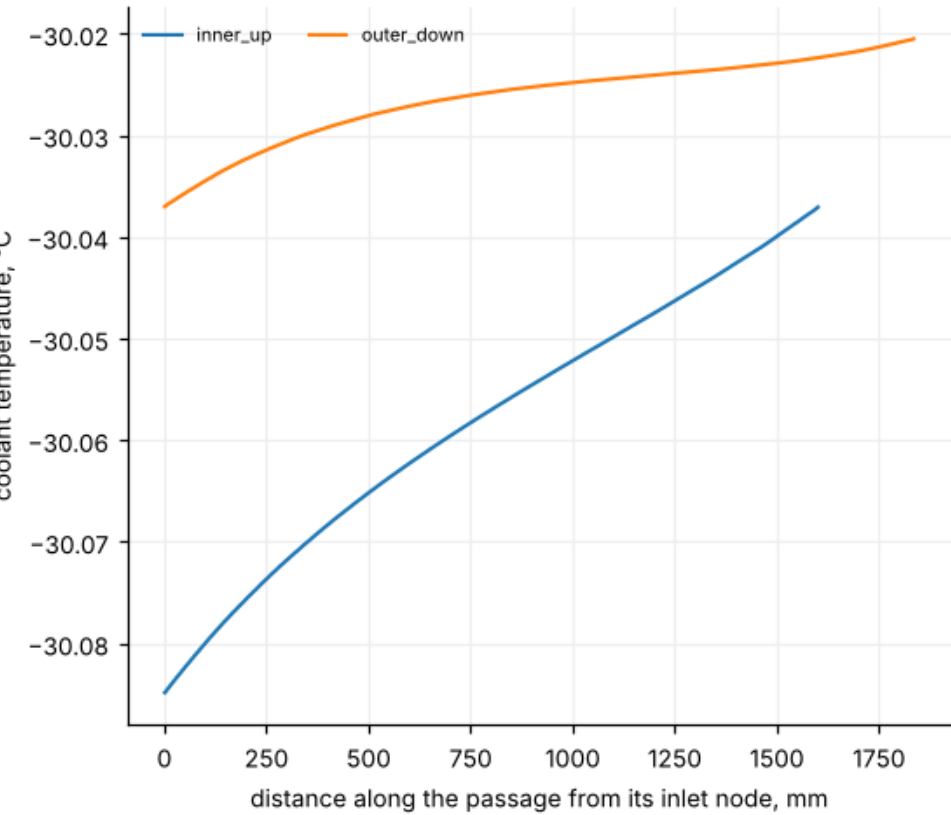
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.086 °C	-30.085 °C
Allowed inlet window, °C	-30.214 to -29.95	-30.214 to -29.956
Inner-wall spread	42.9 mK	42.3 mK
Margin inside ±0.15 K	128.5 mK	128.8 mK
Spread at 5 W (same pattern)	83 mK	82 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	66.91 kPa	62.91 kPa*
Reynolds number, passages	3907 – 3907	3895 – 3895
Coolant-copper UA	89.8 W/K	90.2 W/K
Energy balance error		3.6e-13 W

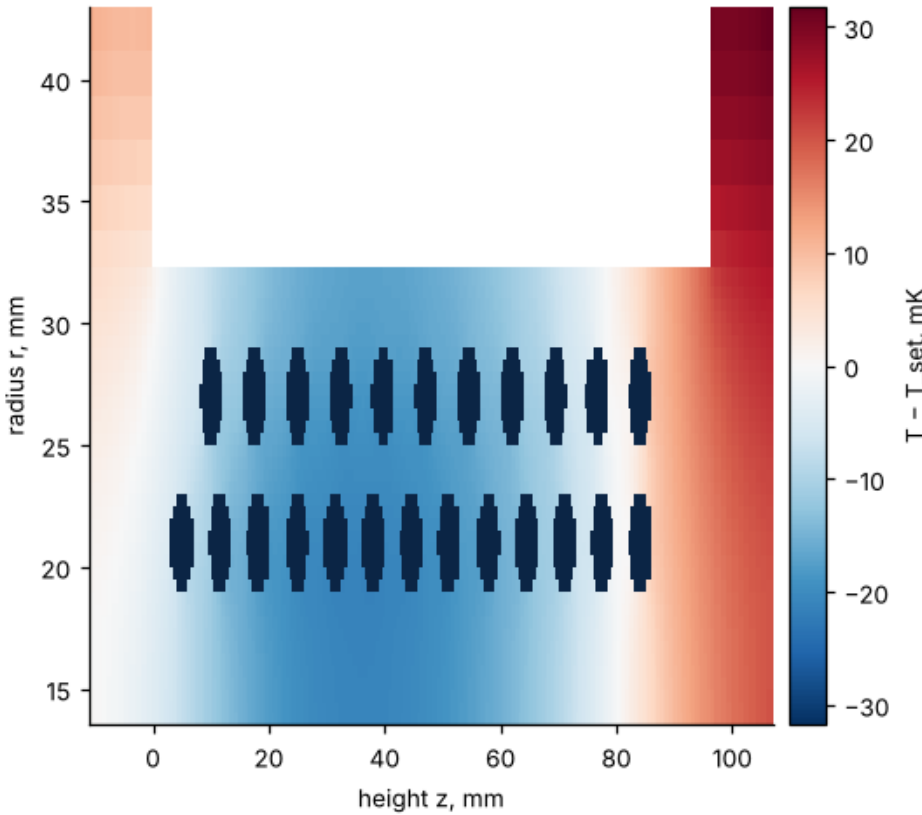
* friction of developed flow only (no bends / fittings)

Δ spread -0.6 mK · Δ inlet +1 mK → SAME

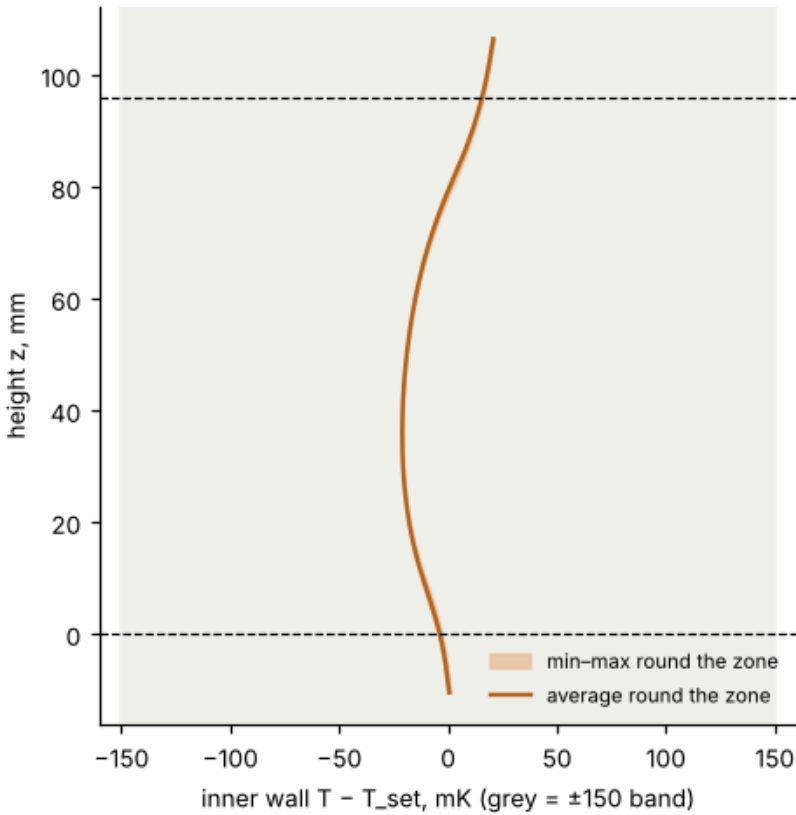
(d) Coolant temperature along each named passage



(e) Section at $\theta = 1^\circ$ — copper T (passages navy)



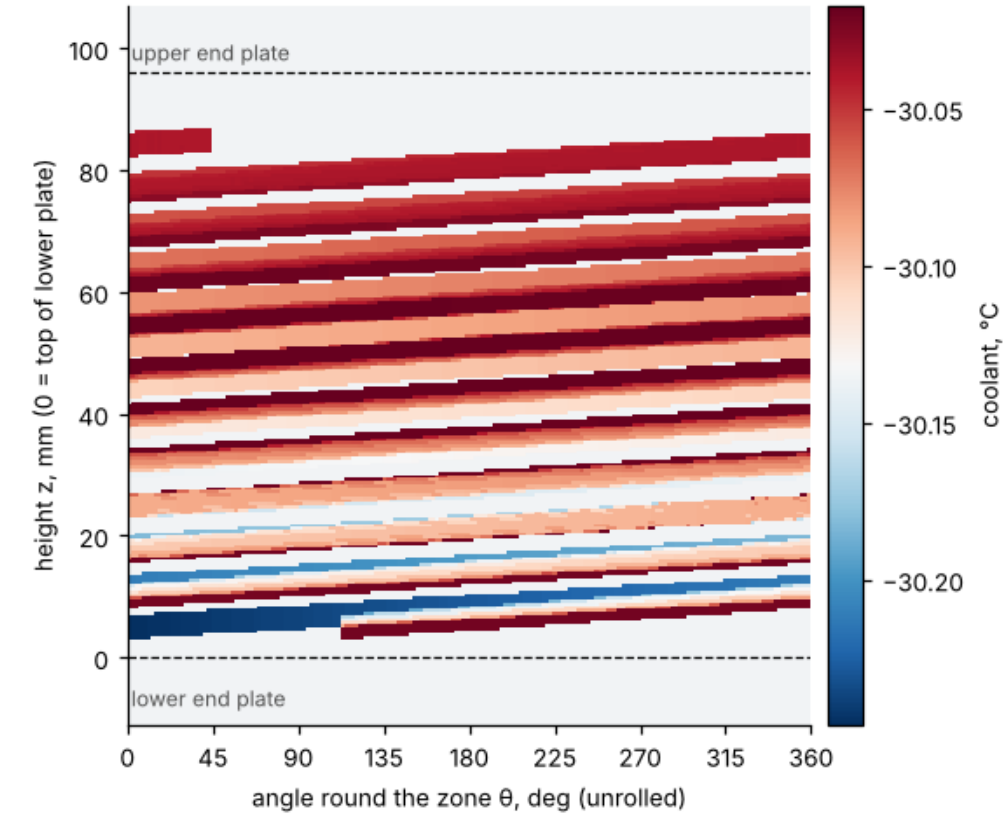
(f) Inner wall along the height



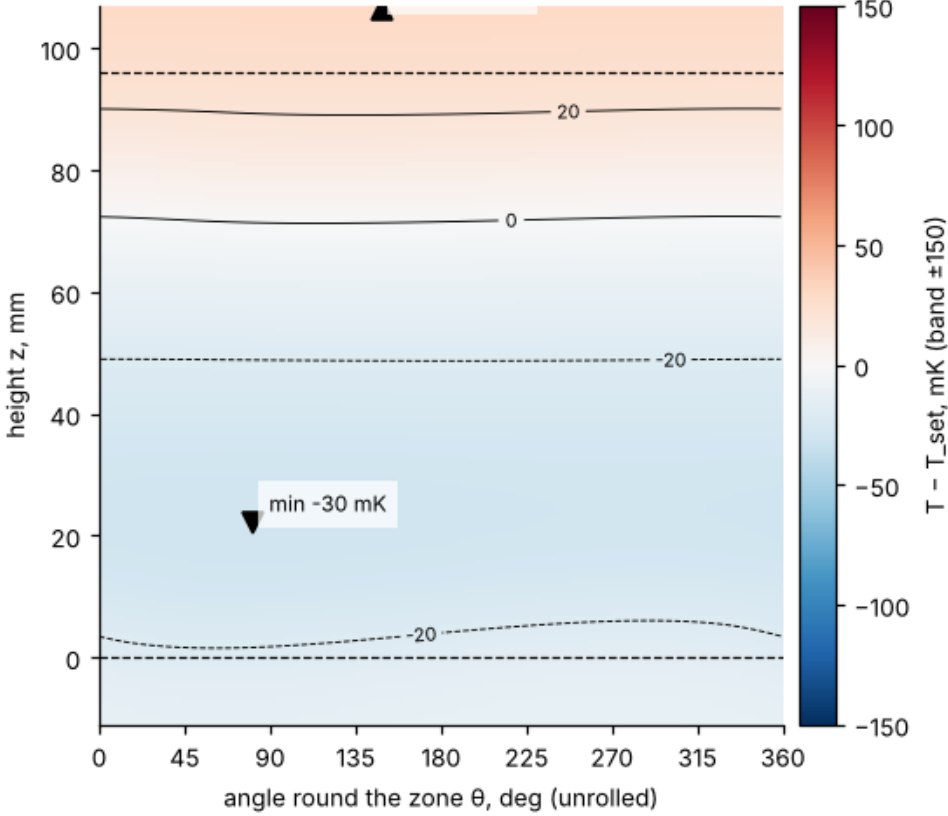
Layout B — two radial layers

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.247 °C · 1,656,360 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



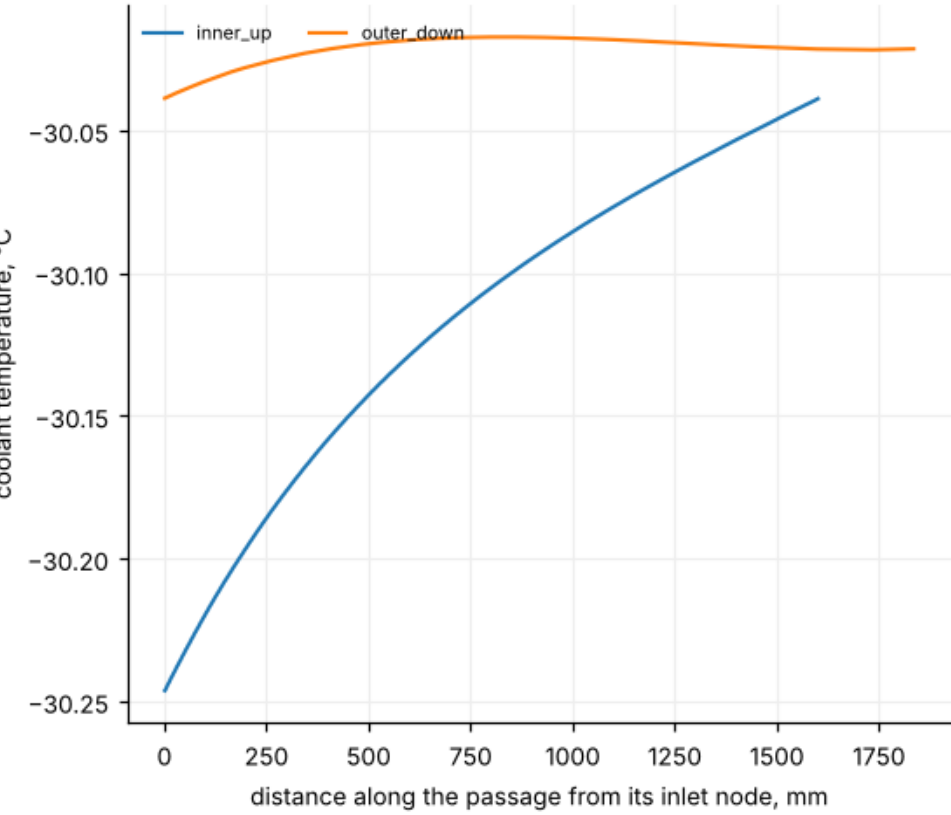
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.248 °C	-30.247 °C
Allowed inlet window, °C	-30.367 to -30.12	-30.367 to -30.127
Inner-wall spread	61.0 mK	60.8 mK
Margin inside ±0.15 K	119.5 mK	119.6 mK
Spread at 5 W (same pattern)	118 mK	118 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	10.64 kPa	10.73 kPa*
Reynolds number, passages	1116 – 1116	1113 – 1113
Coolant-copper UA	49.1 W/K	49.2 W/K
Energy balance error		-2.8e-12 W

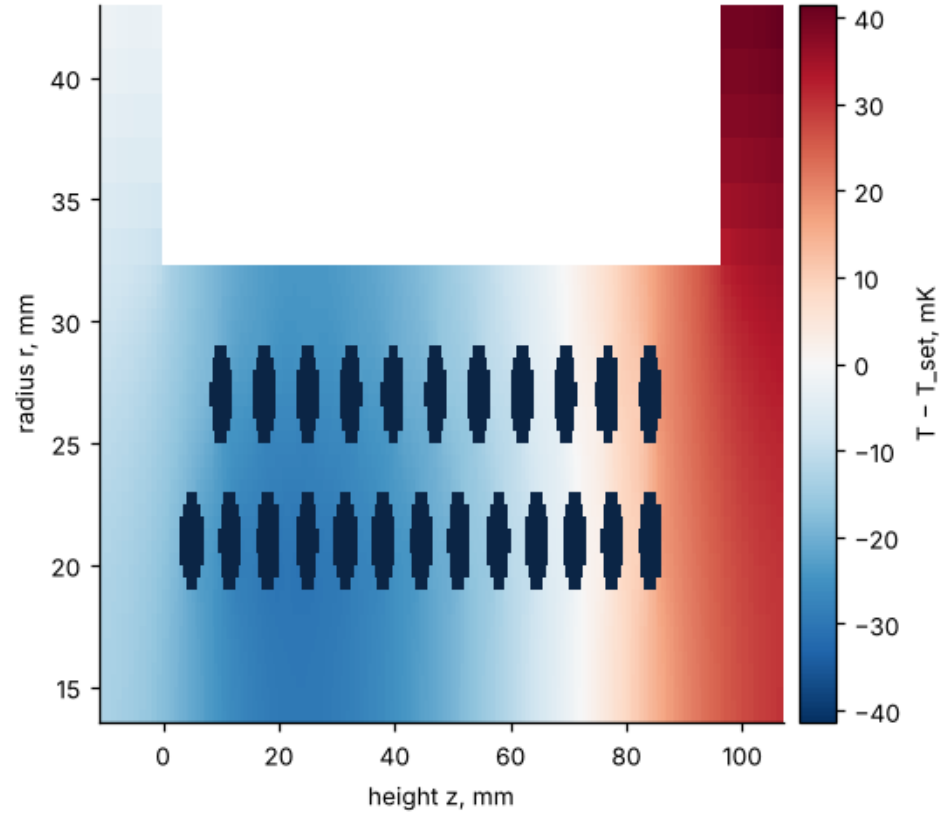
* friction of developed flow only (no bends / fittings)

Δ spread -0.2 mK · Δ inlet +1 mK → SAME

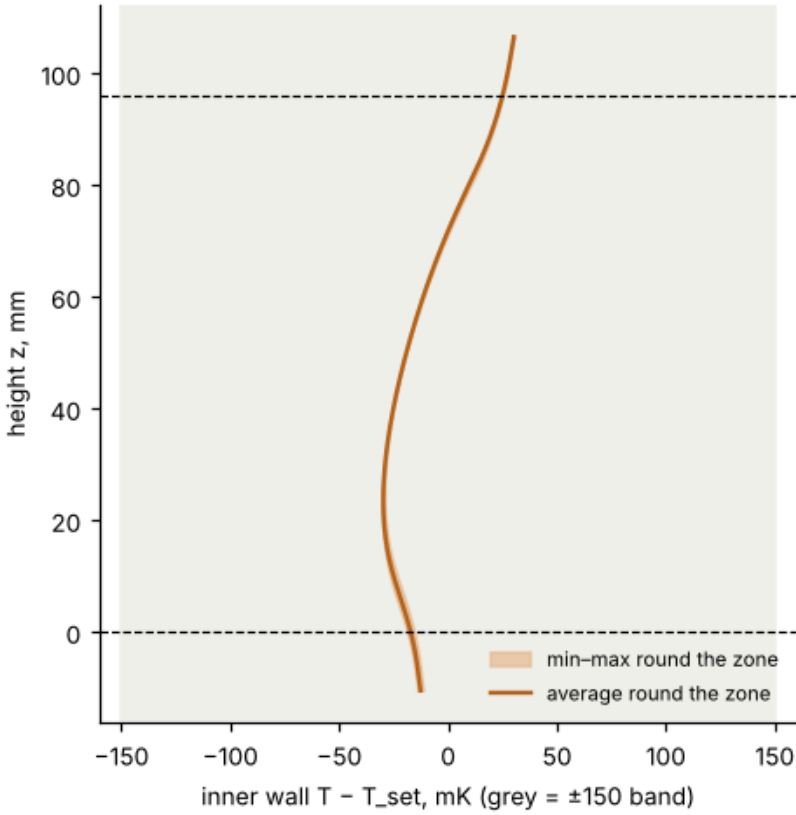
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

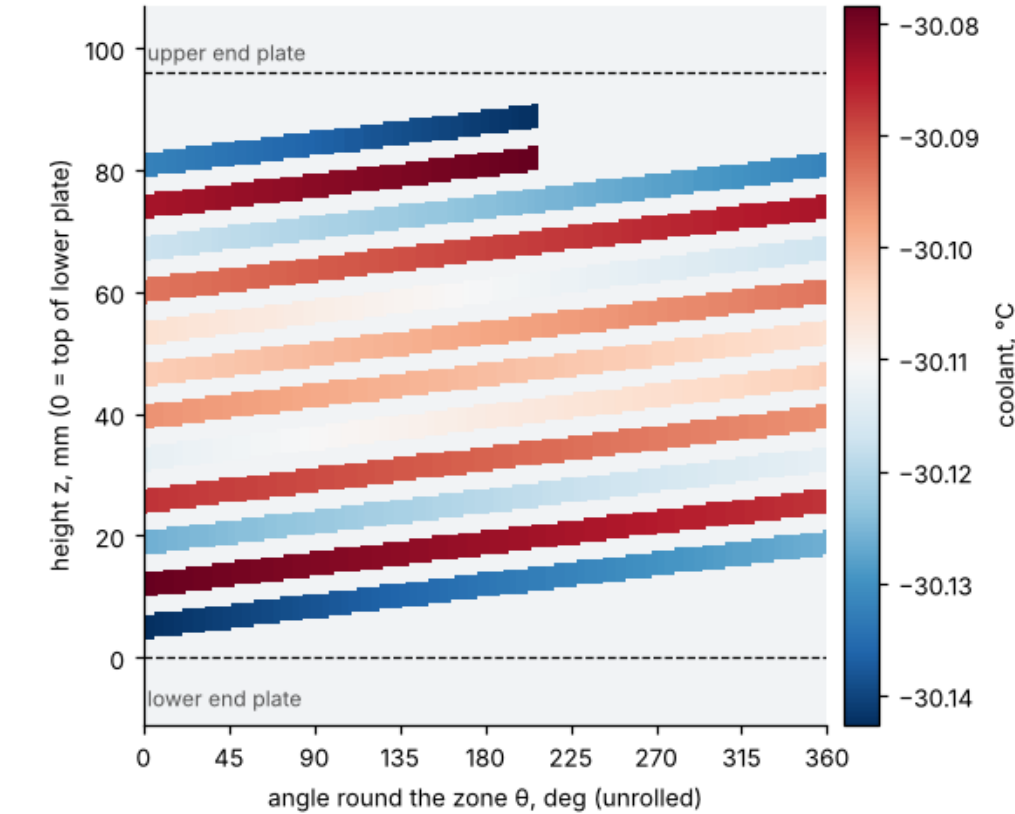


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

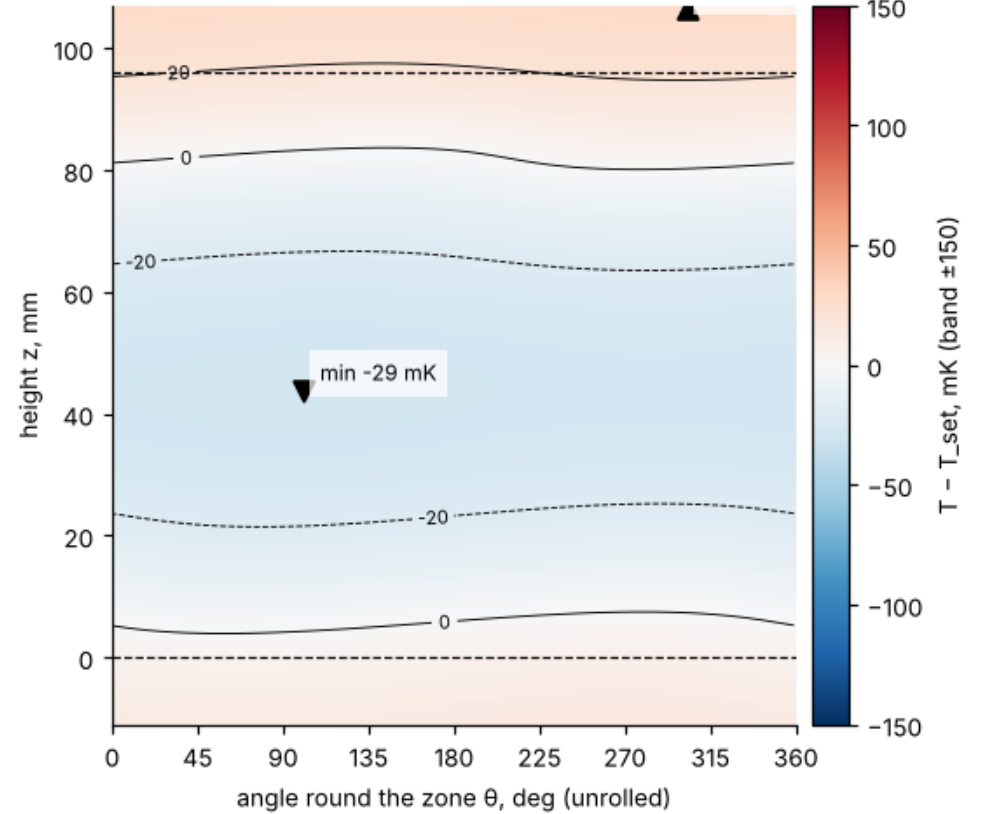
Layout C — opposite-end feeds

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.143 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



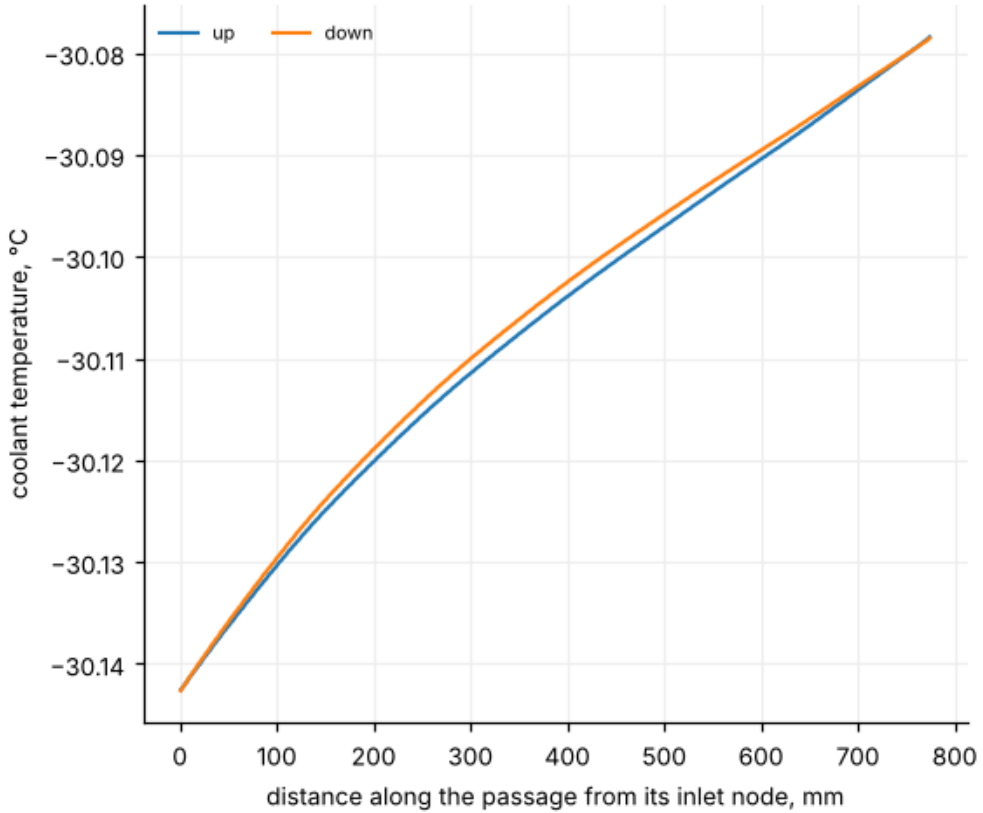
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.142 °C	-30.143 °C
Allowed inlet window, °C	-30.263 to -30.02	-30.264 to -30.022
Inner-wall spread	57.7 mK	57.9 mK
Margin inside ±0.15 K	121.1 mK	121.0 mK
Spread at 5 W (same pattern)	112 mK	112 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	6.09 kPa	5.40 kPa*
Reynolds number, passages	1954 – 1954	1948 – 1948
Coolant-copper UA	29.6 W/K	29.6 W/K
Energy balance error		3.3e-13 W

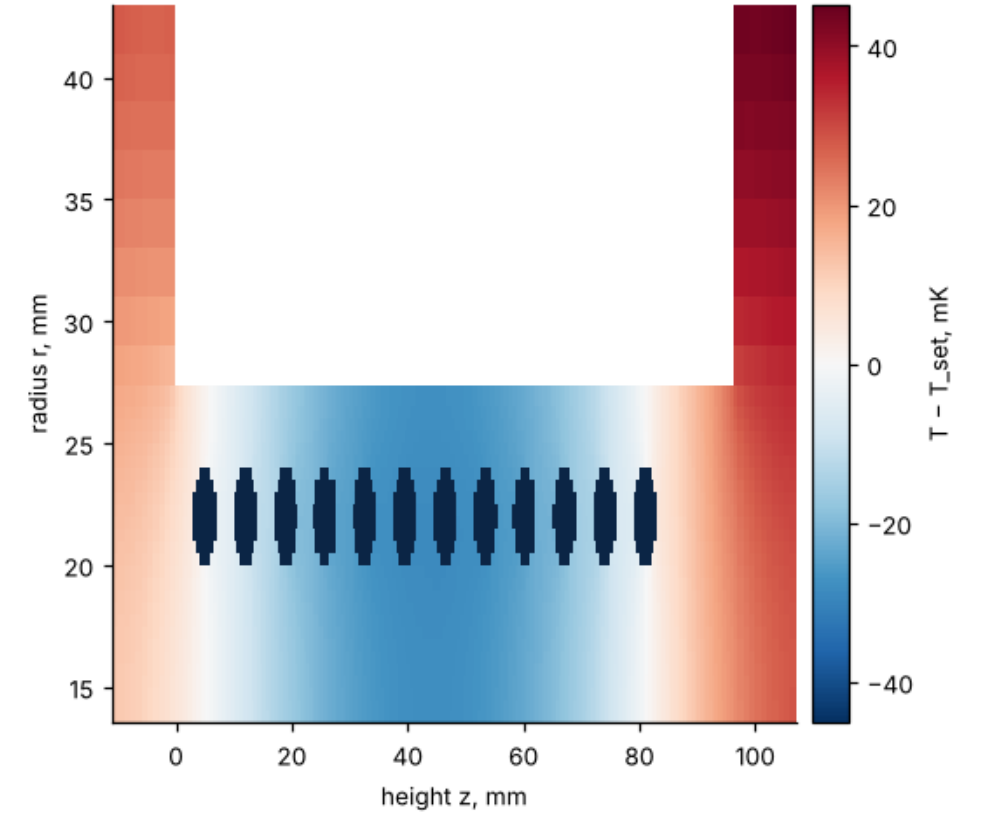
* friction of developed flow only (no bends / fittings)

Δ spread +0.2 mK · Δ inlet -1 mK → SAME

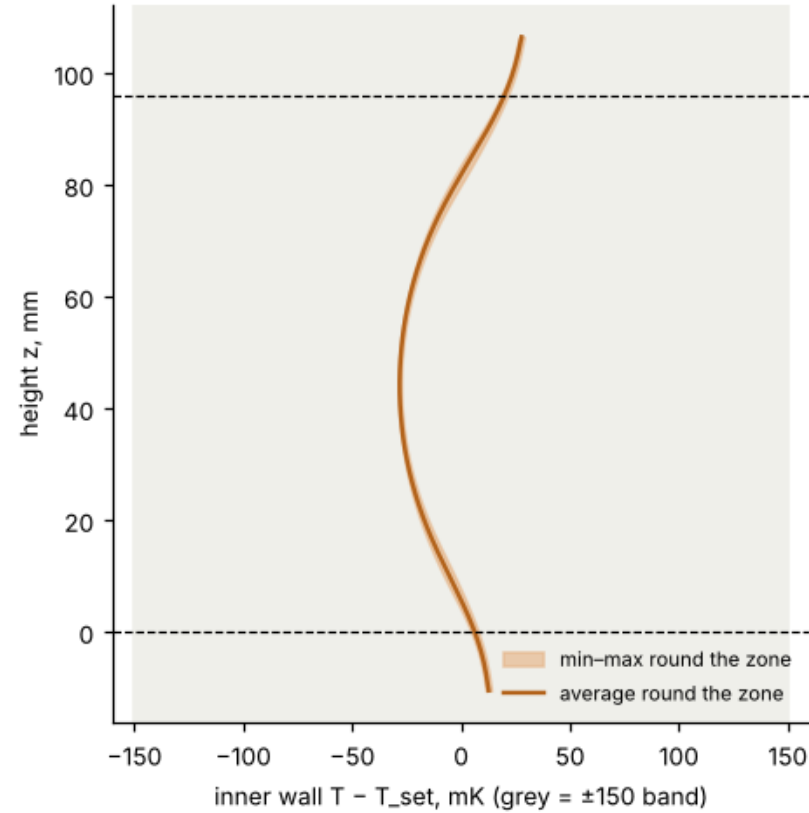
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

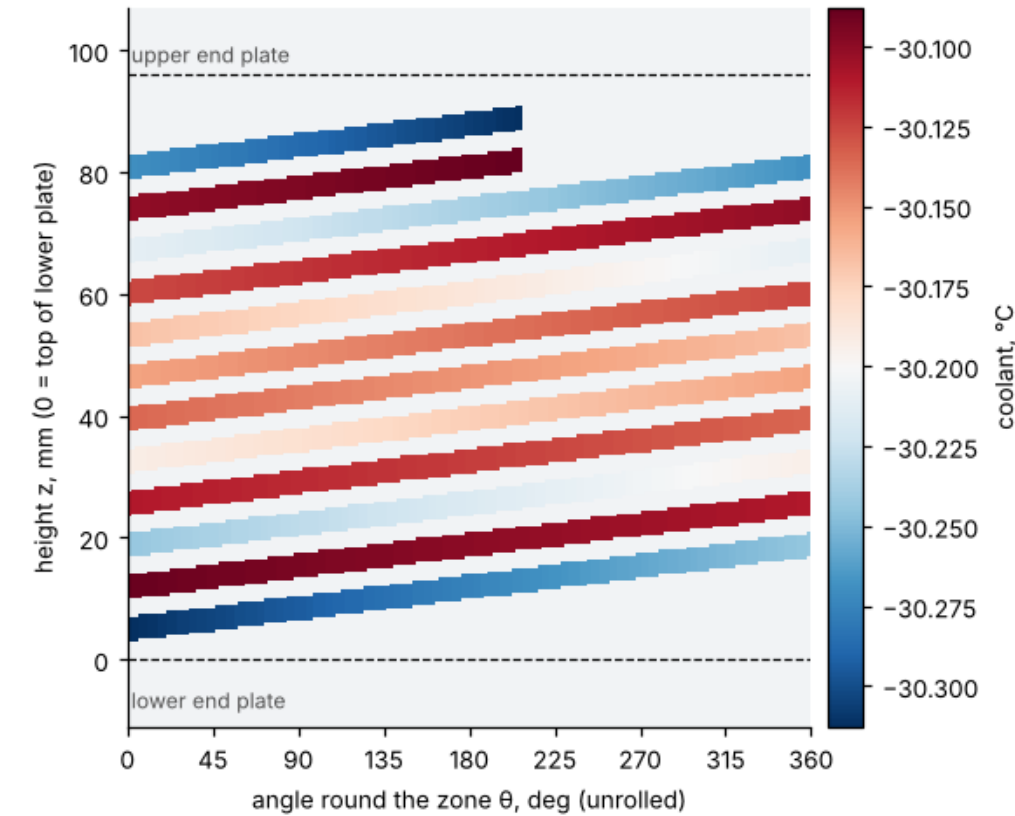


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

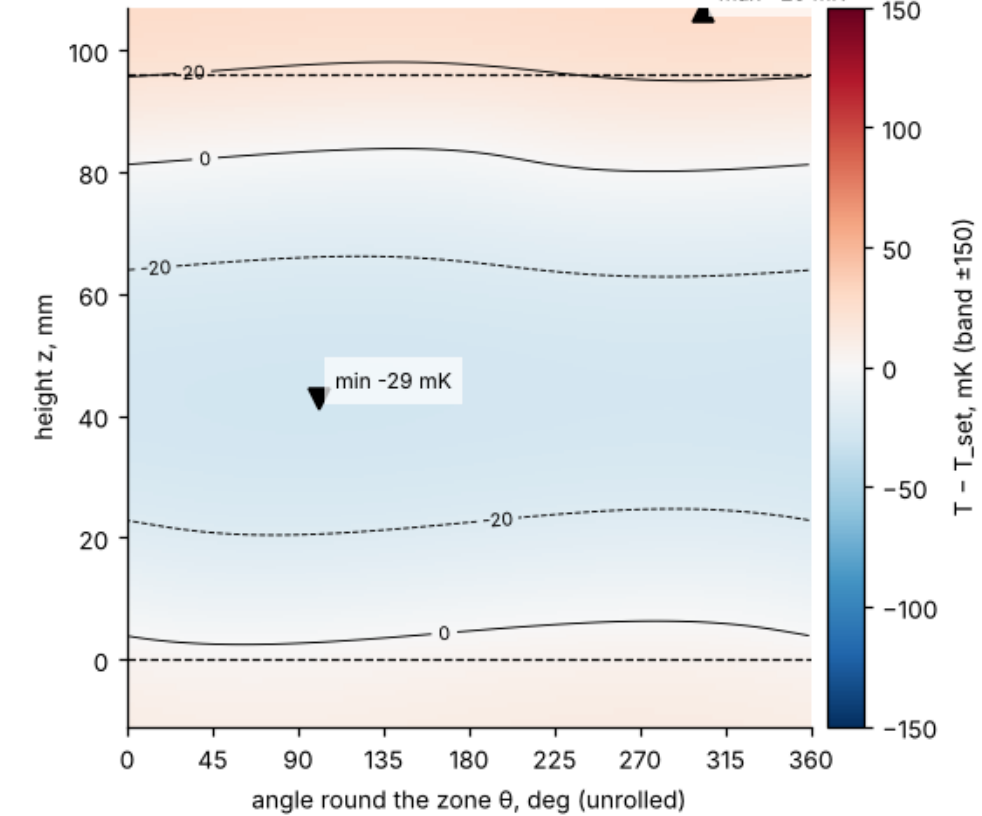
Layout C — opposite-end feeds

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.314 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



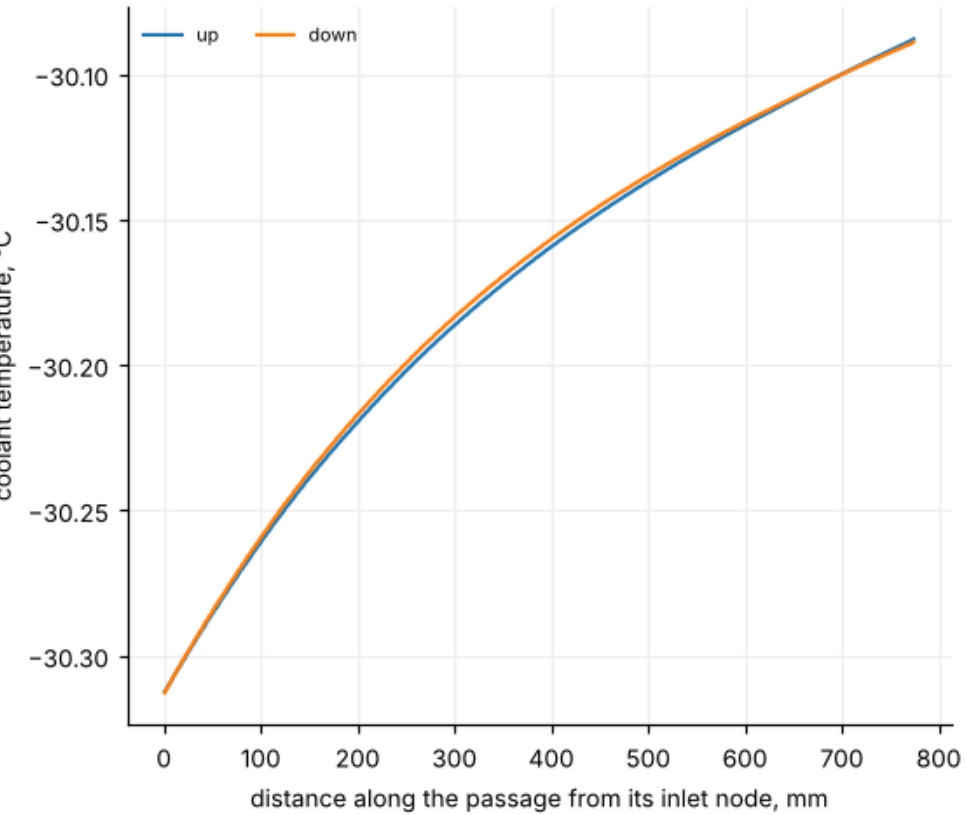
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.313 °C	-30.314 °C
Allowed inlet window, °C	-30.434 to -30.19	-30.435 to -30.193
Inner-wall spread	57.2 mK	57.4 mK
Margin inside ±0.15 K	121.4 mK	121.3 mK
Spread at 5 W (same pattern)	111 mK	111 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	0.98 kPa	0.95 kPa*
Reynolds number, passages	558 – 558	556 – 556
Coolant-copper UA	16.7 W/K	16.7 W/K
Energy balance error		8.1e-13 W

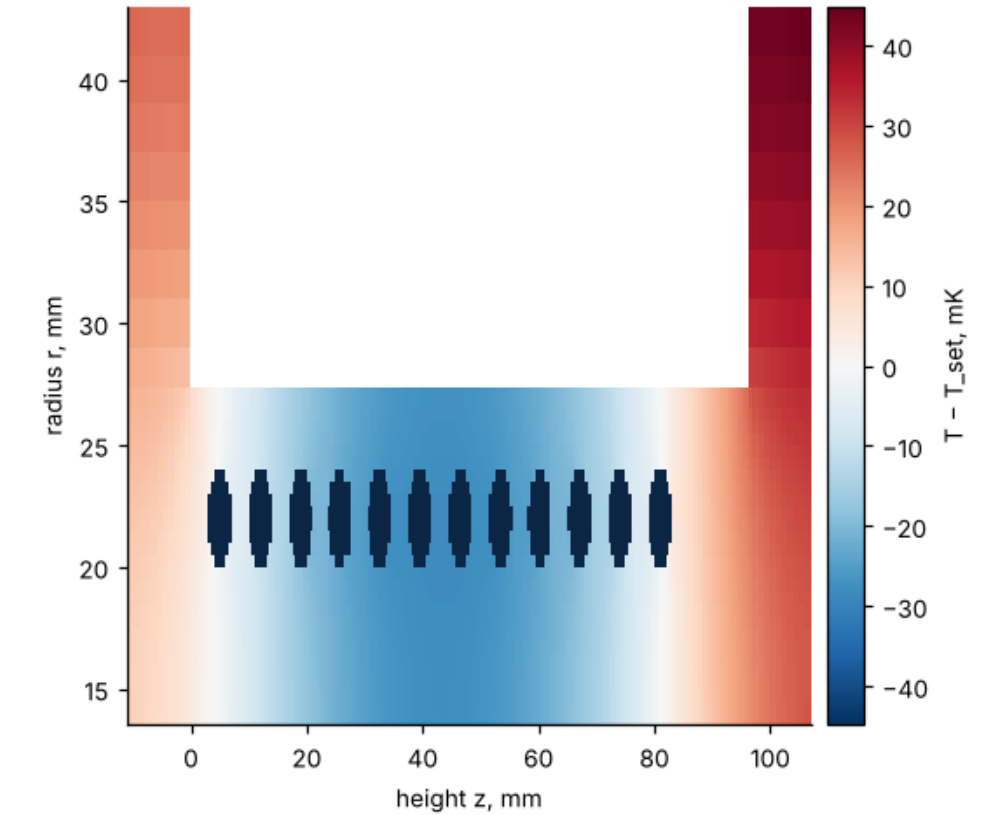
* friction of developed flow only (no bends / fittings)

Δ spread +0.2 mK · Δ inlet -1 mK → SAME

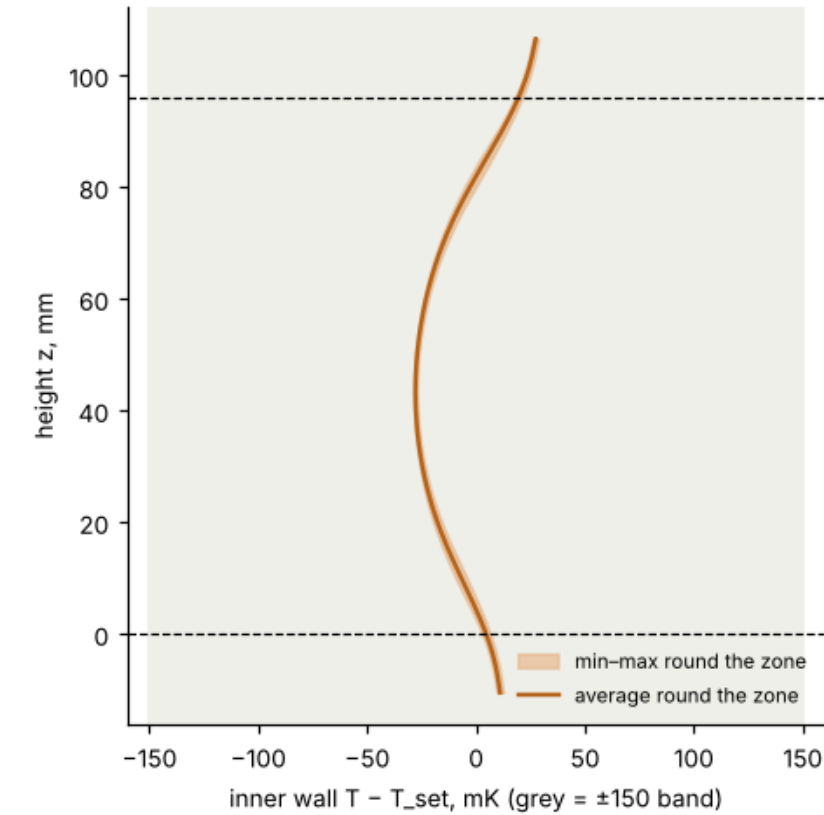
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

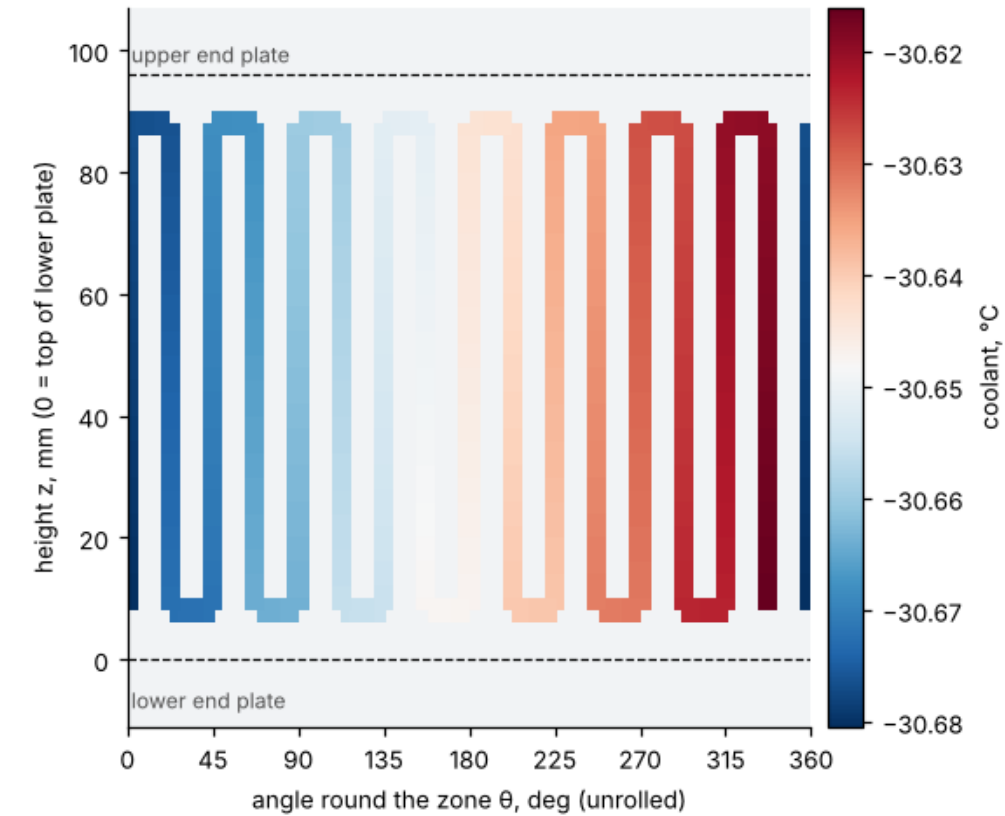


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

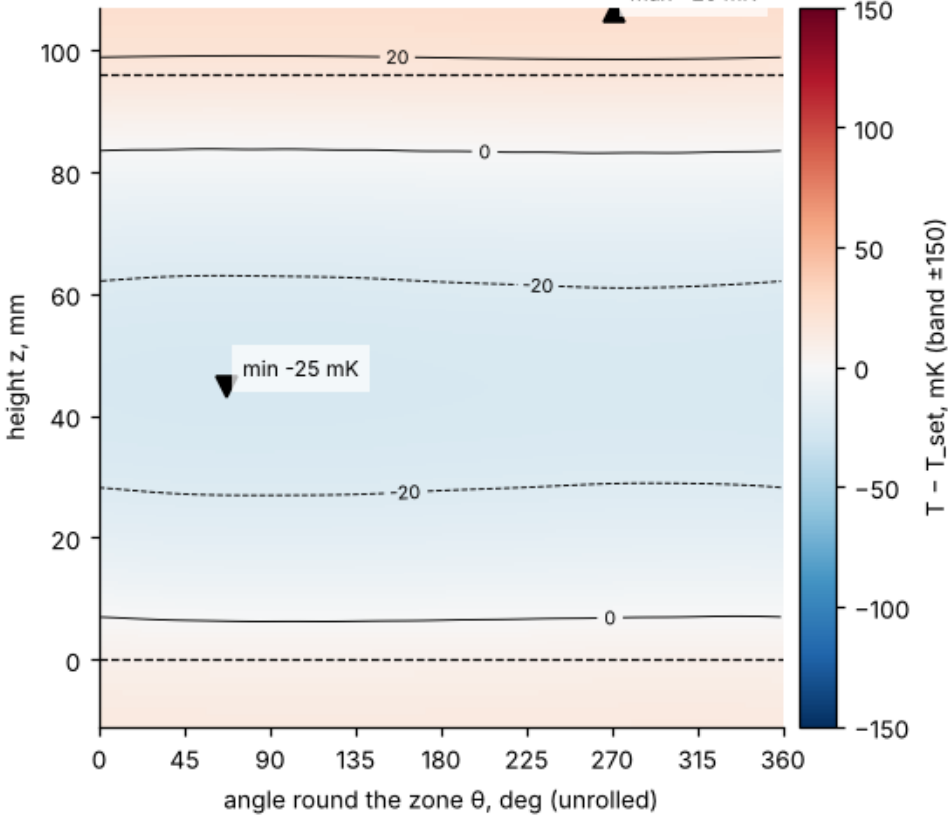
Layout D — axial serpentine

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.681 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



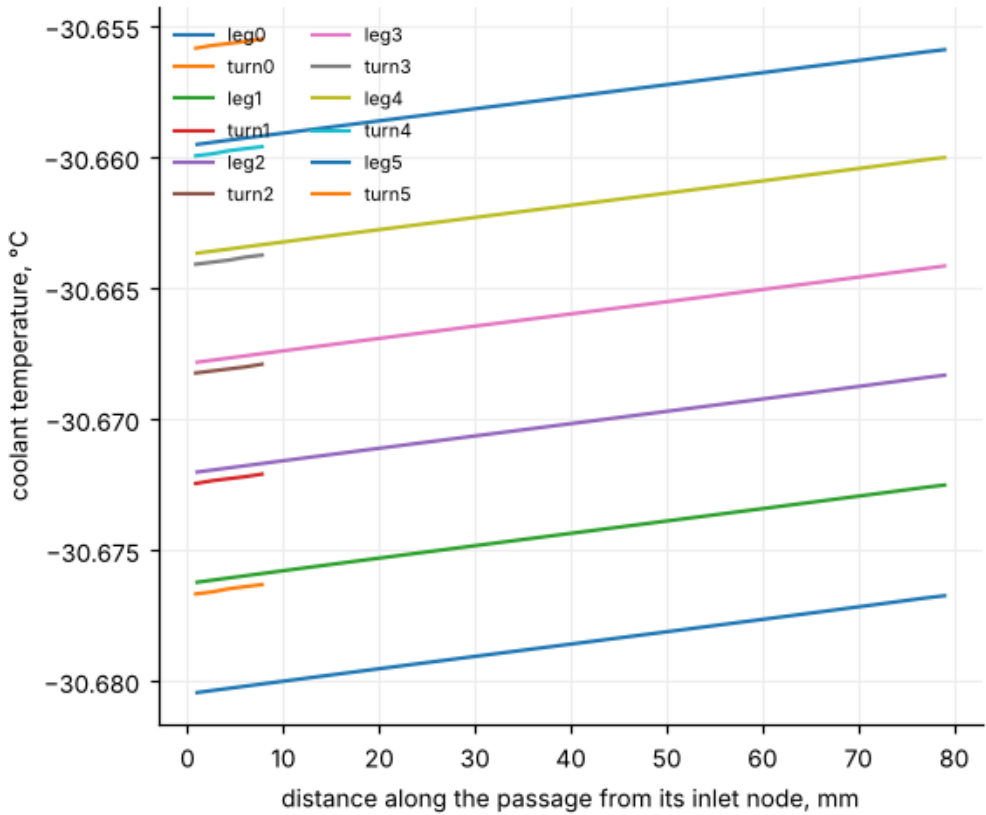
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.676 °C	-30.681 °C
Allowed inlet window, °C	-30.799 to -30.55	-30.805 to -30.556
Inner-wall spread	53.0 mK	50.7 mK
Margin inside ±0.15 K	123.5 mK	124.7 mK
Spread at 5 W (same pattern)	103 mK	98 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	50.12 kPa	6.68 kPa*
Reynolds number, passages	3907 – 3907	3895 – 3895
Coolant-copper UA	4.1 W/K	4.1 W/K
Energy balance error		1.3e-12 W

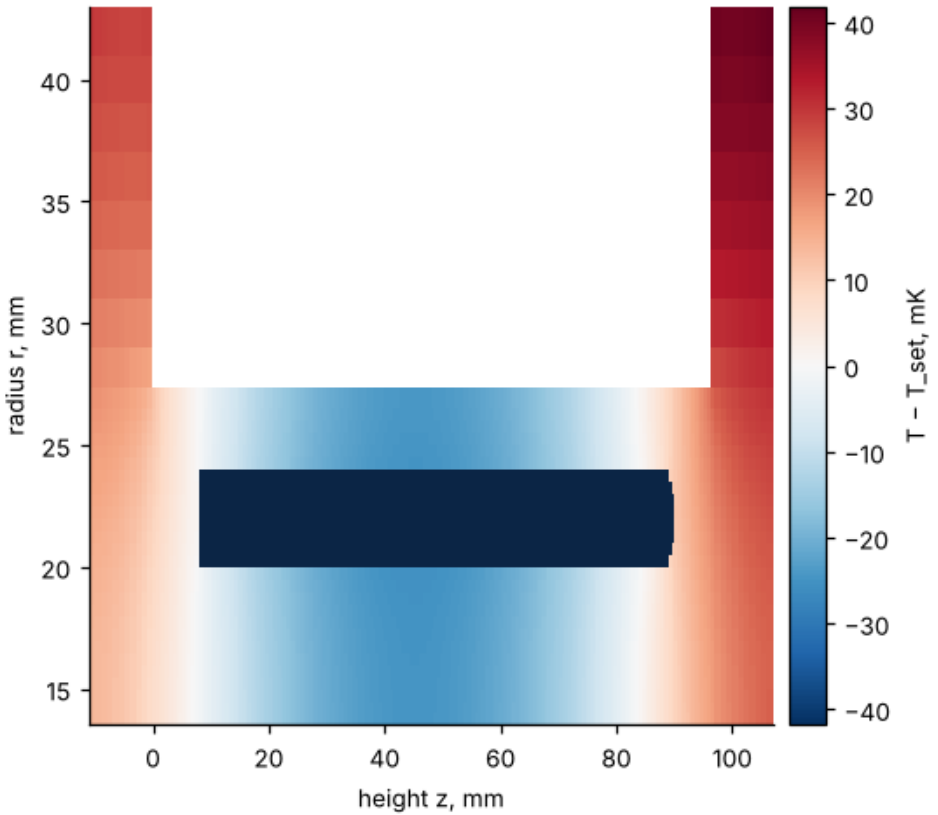
* friction of developed flow only (no bends / fittings)

Δ spread -2.3 mK · Δ inlet -5 mK → SAME

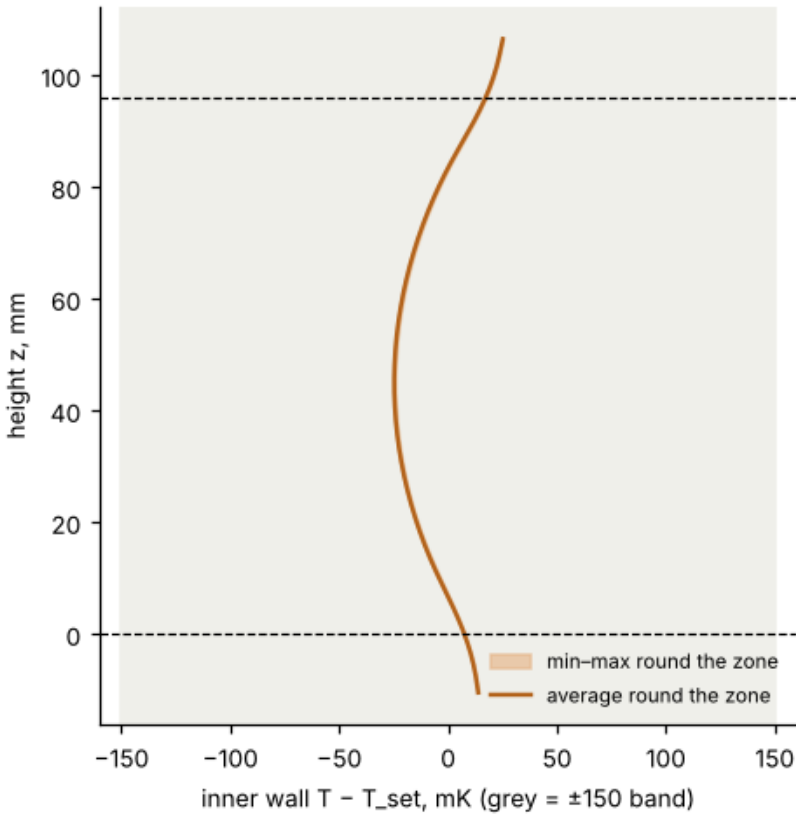
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



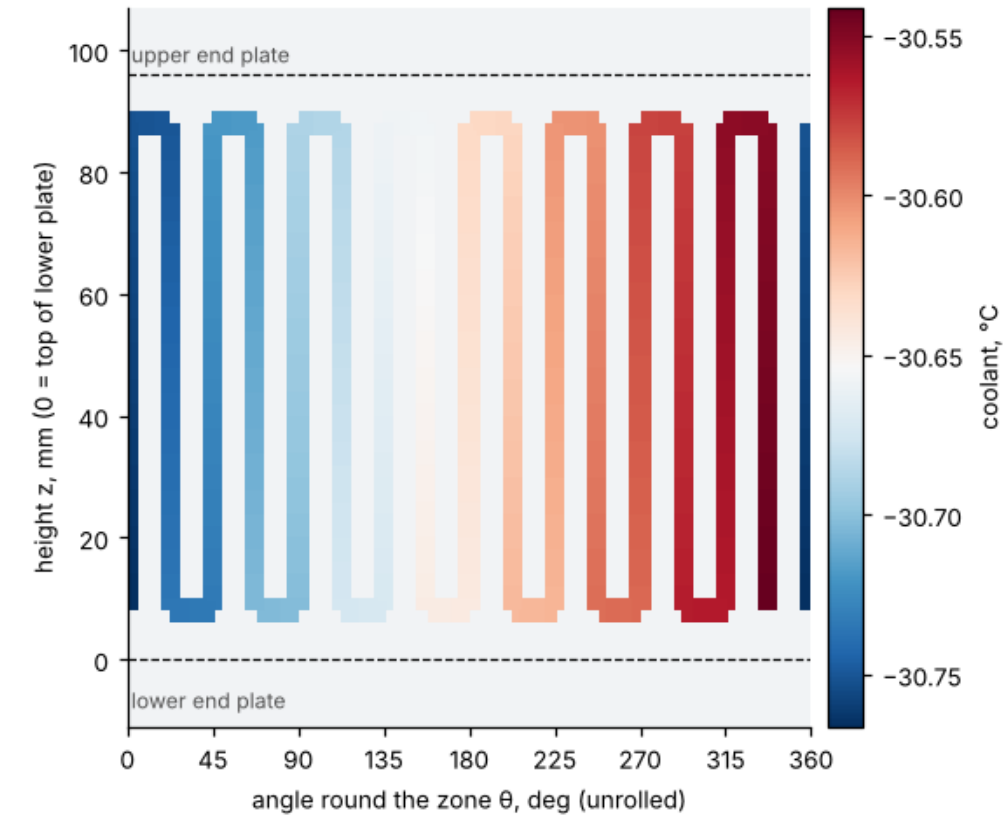
(f) Inner wall along the height



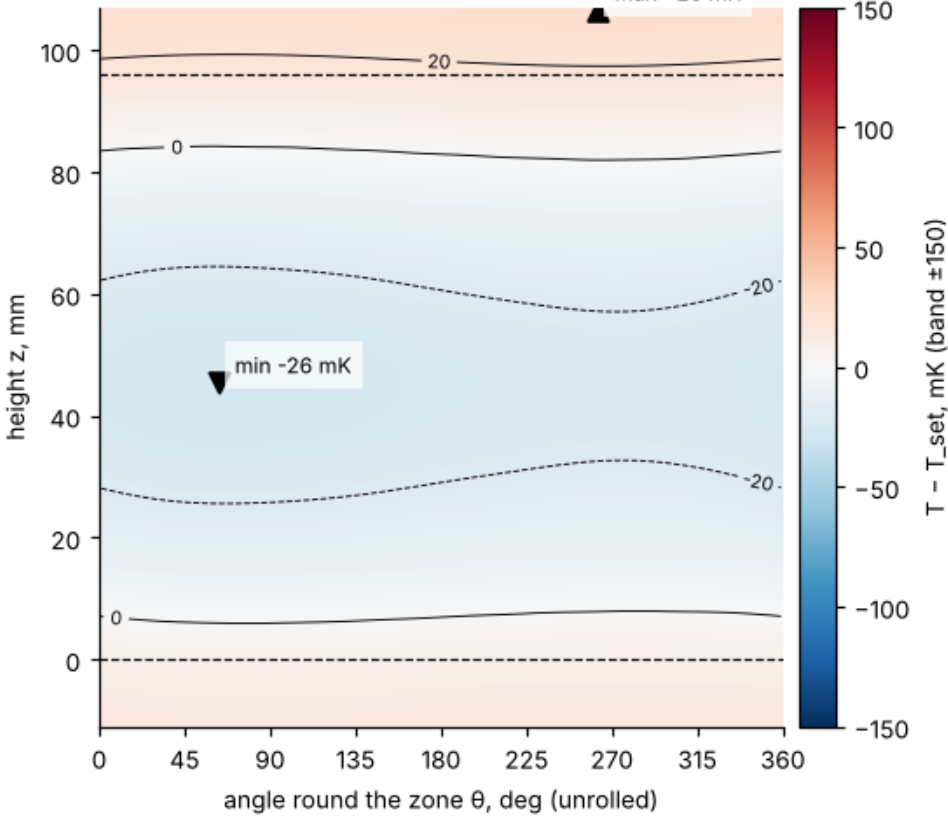
Layout D — axial serpentine

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.767 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



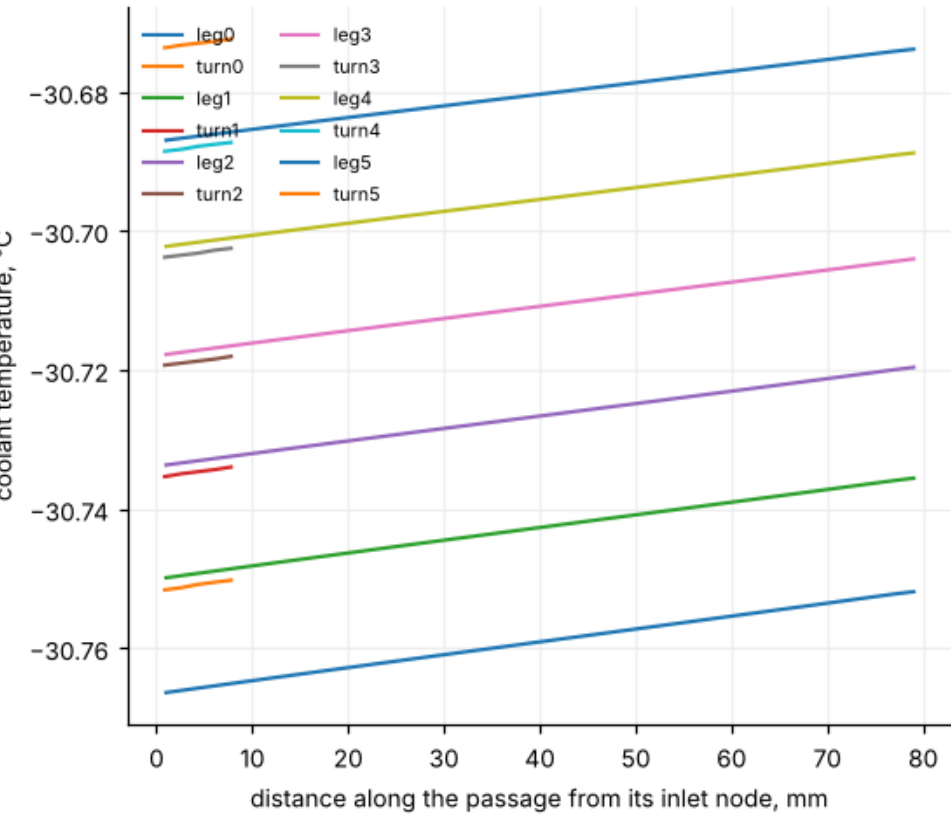
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.762 °C	-30.767 °C
Allowed inlet window, °C	-30.884 to -30.64	-30.891 to -30.643
Inner-wall spread	54.9 mK	52.6 mK
Margin inside ±0.15 K	122.5 mK	123.7 mK
Spread at 5 W (same pattern)	107 mK	102 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	5.47 kPa	1.91 kPa*
Reynolds number, passages	1116 – 1116	1113 – 1113
Coolant-copper UA	4.1 W/K	4.1 W/K
Energy balance error		1.2e-12 W

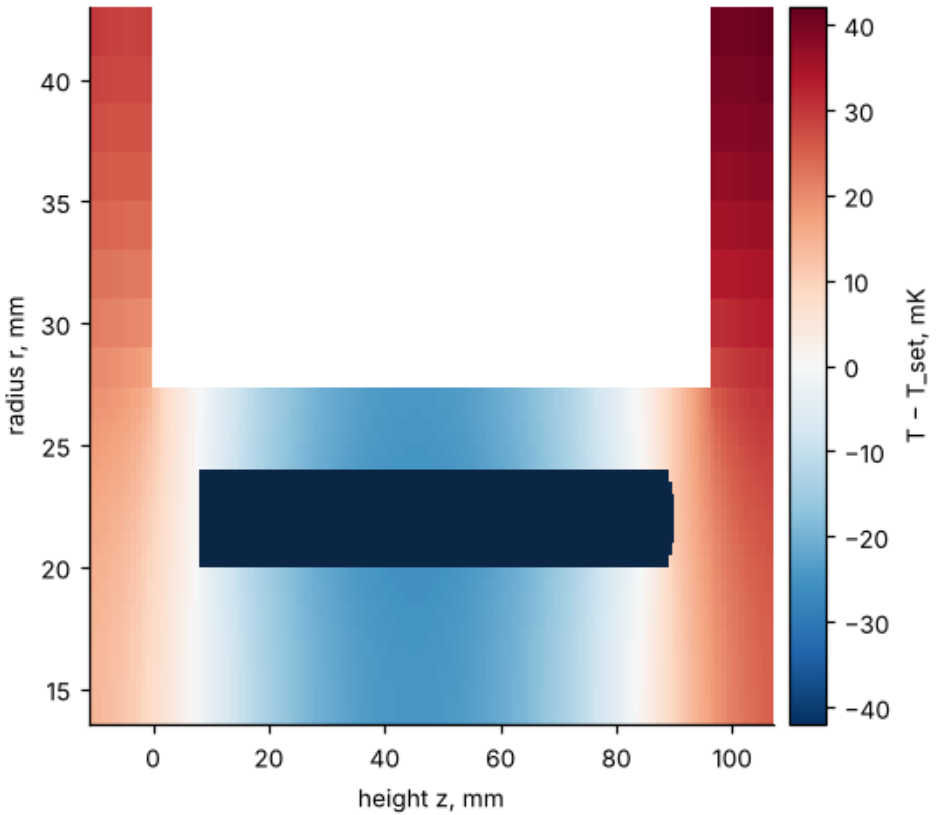
* friction of developed flow only (no bends / fittings)

Δ spread -2.3 mK · Δ inlet -5 mK → SAME

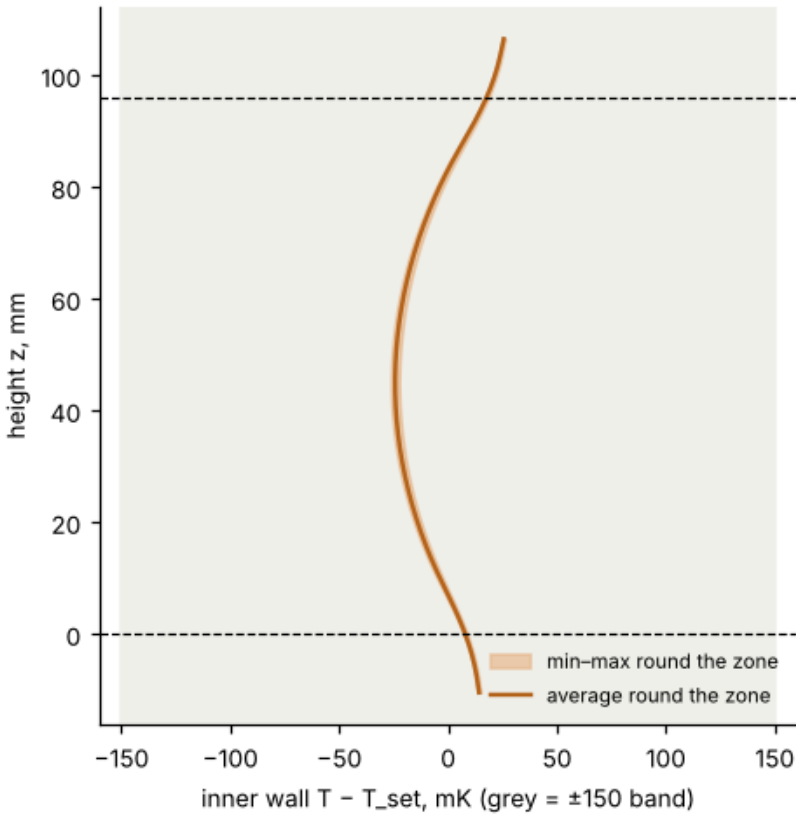
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)

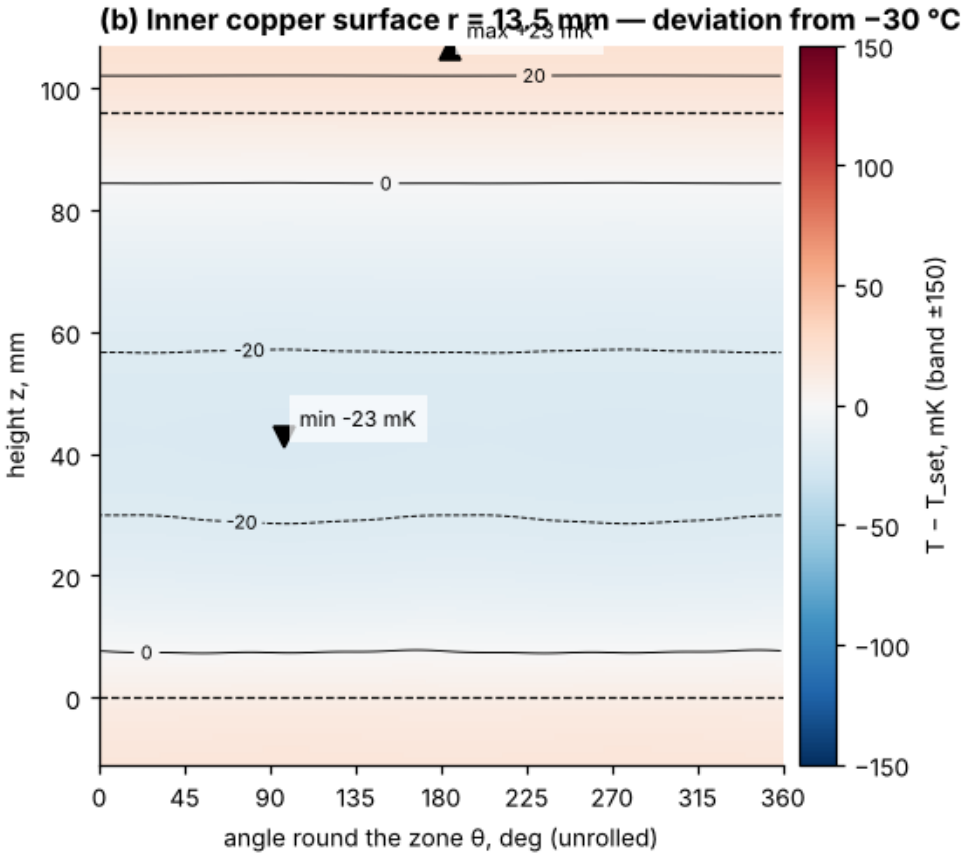
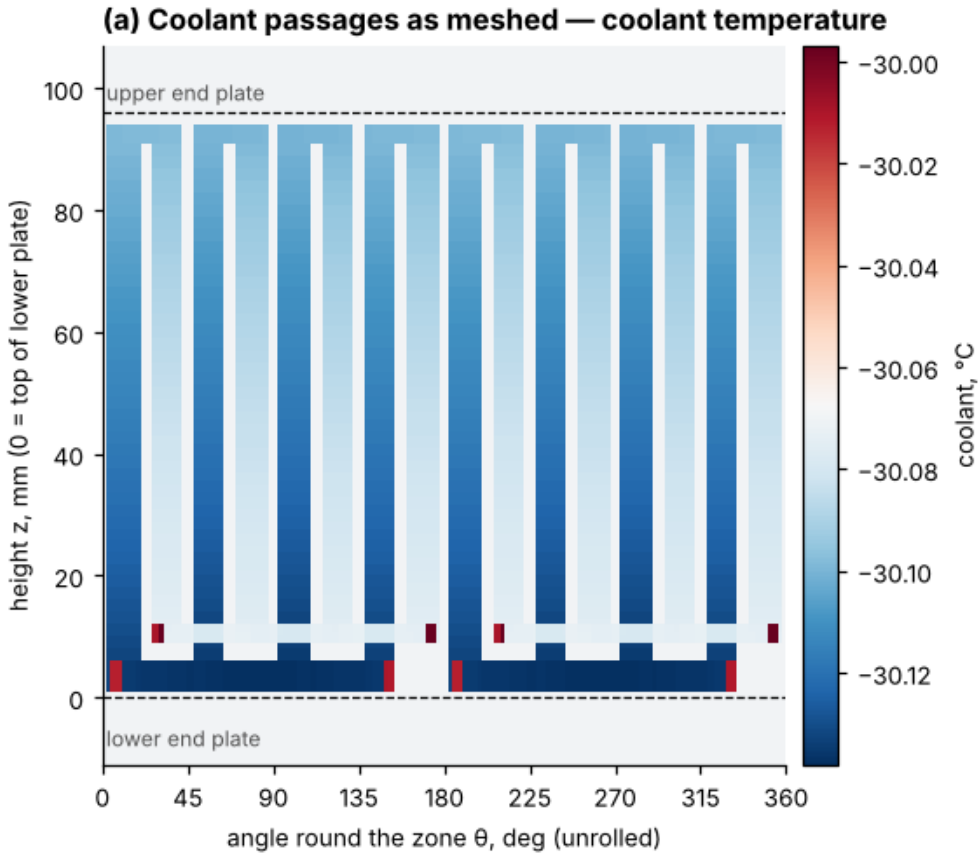


(f) Inner wall along the height



Layout E — paired axial sleeve

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.138 °C · 1,348,200 cells · re-validation model

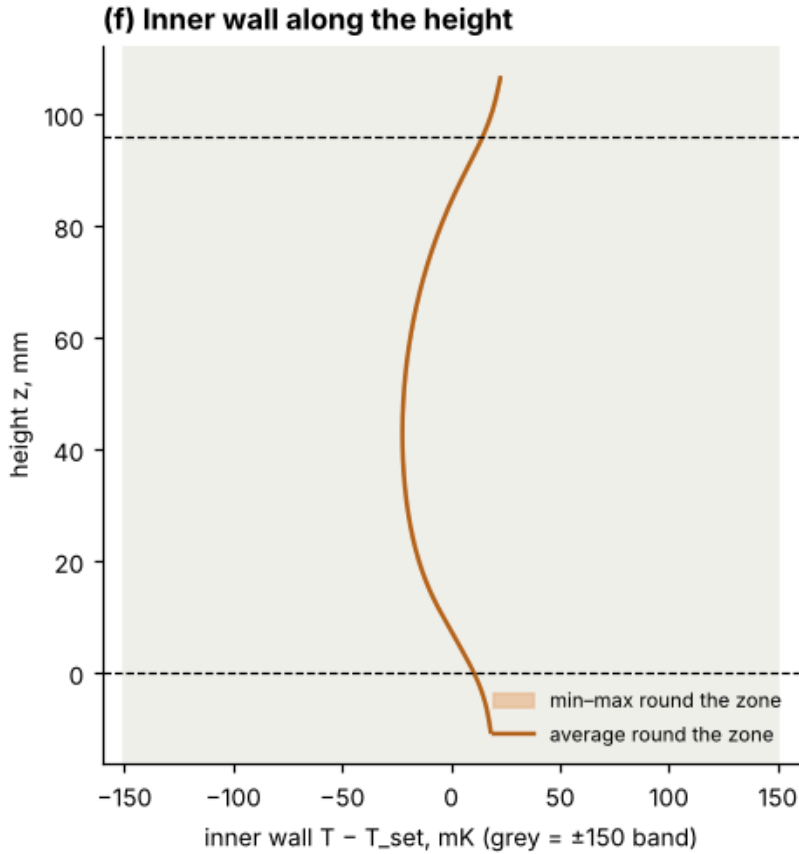
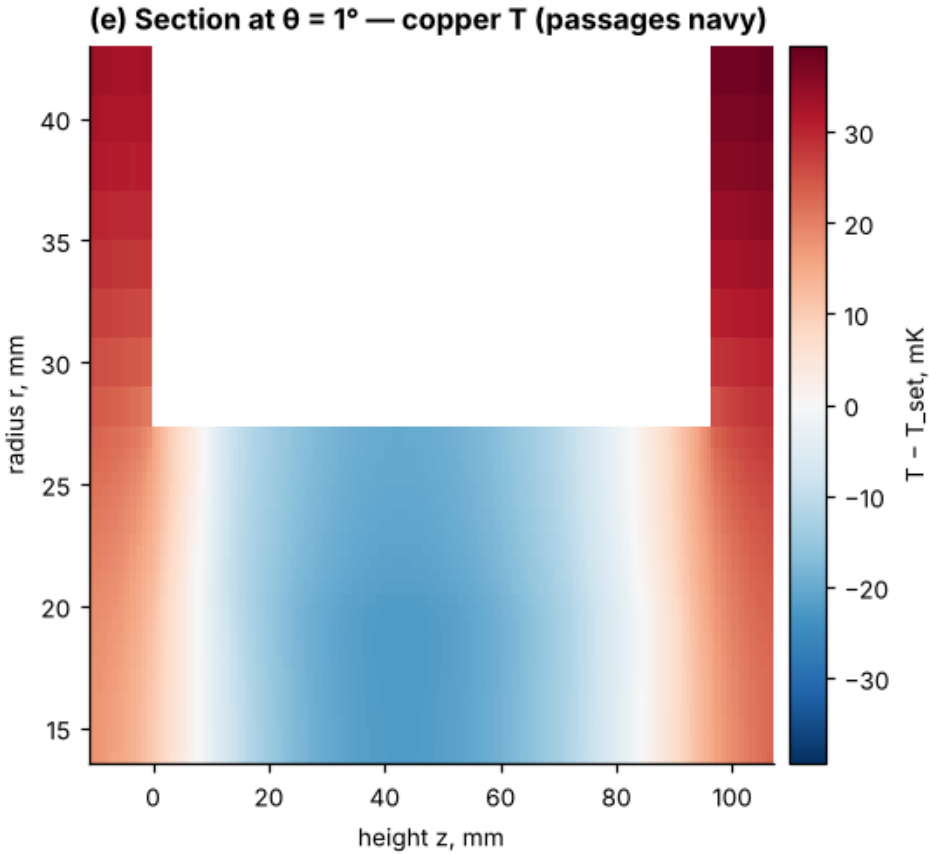
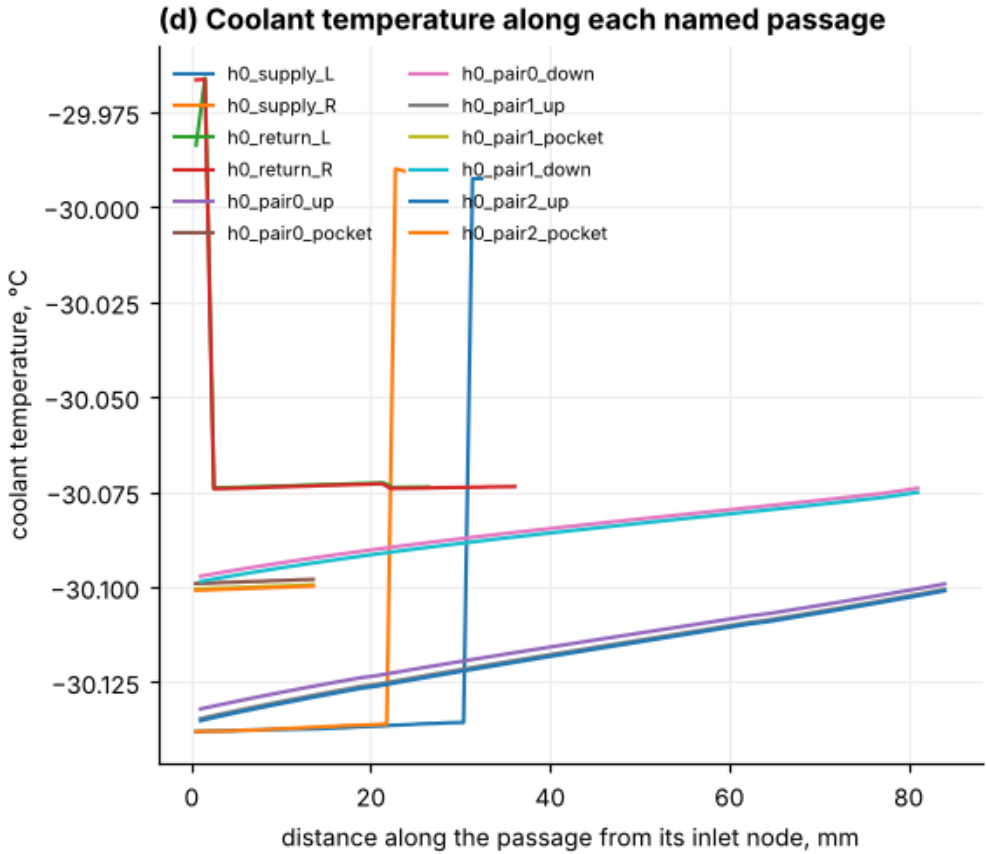


(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.131 °C	-30.138 °C
Allowed inlet window, °C	-30.258 to -30.00	-30.266 to -30.011
Inner-wall spread	45.6 mK	45.1 mK
Margin inside $\pm 0.15\text{ K}$	127.2 mK	127.4 mK
Spread at 5 W (same pattern)	88 mK	87 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	13.78 kPa	10.55 kPa*
Reynolds number, passages	472 – 1534	340 – 765
Coolant-copper UA	31.8 W/K	29.0 W/K
Energy balance error		-2.5e-13 W

* friction of developed flow only (no bends / fittings)

Δ spread -0.5 mK · Δ inlet -7 mK → SAME

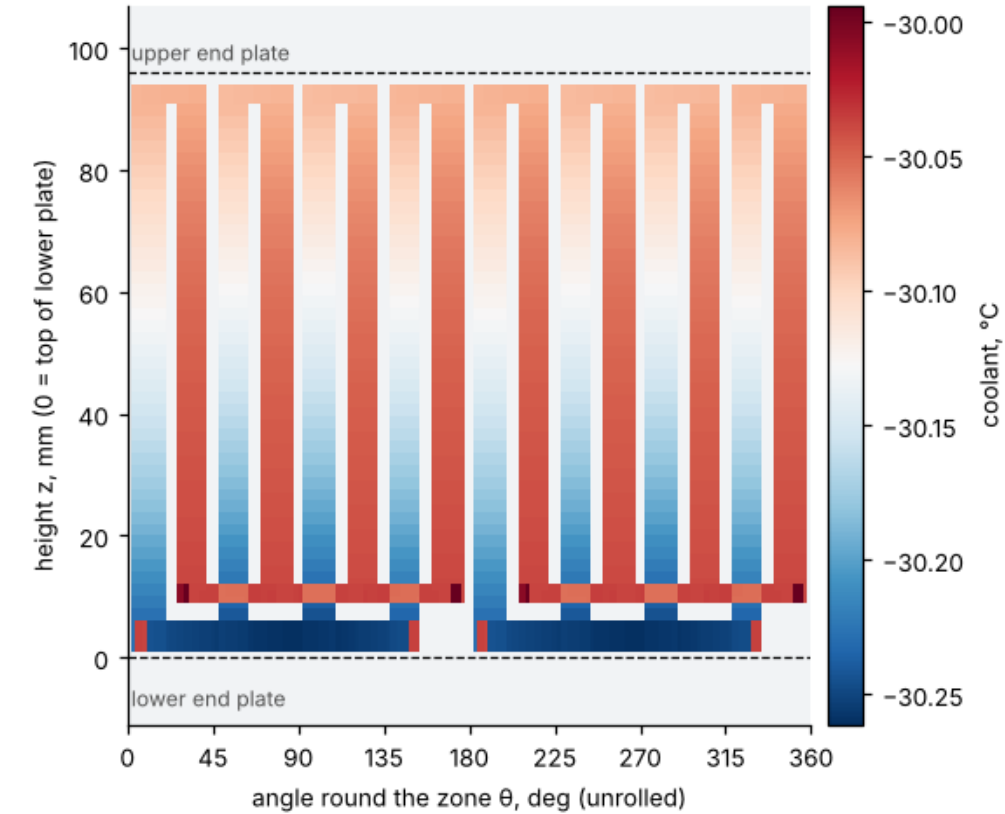


Model: 3-D finite-volume conduction (copper wall $r\ 13.5\text{--}15$, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C , heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C .

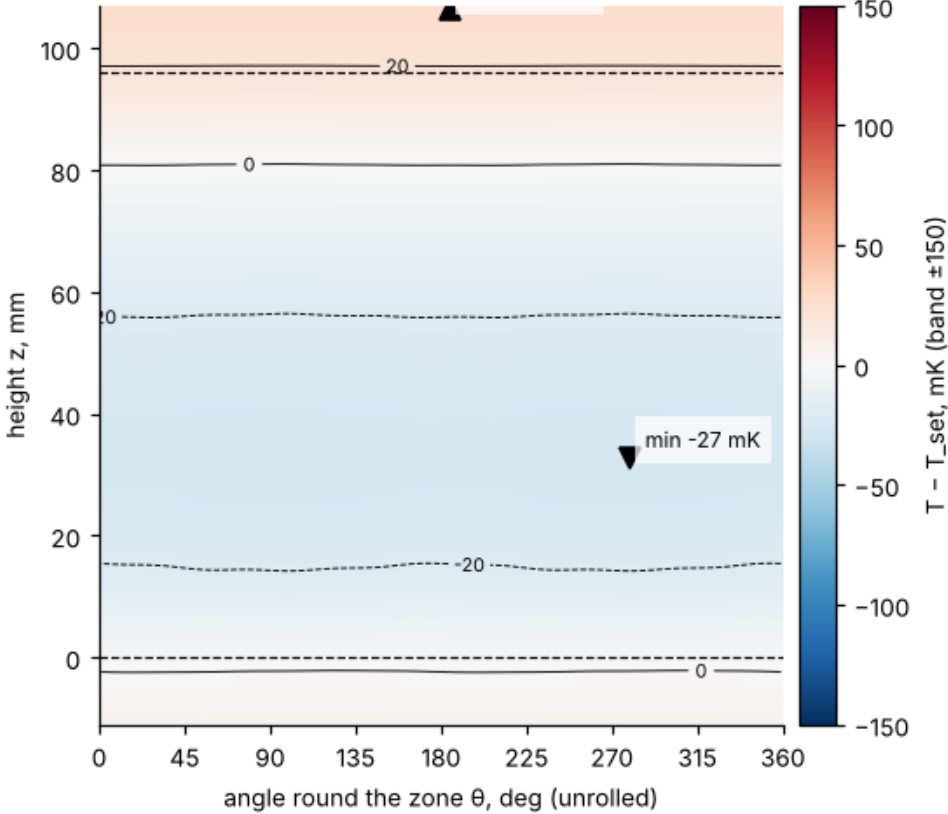
Layout E — paired axial sleeve

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.262 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface $r = 13.5$ mm — deviation from -30 °C



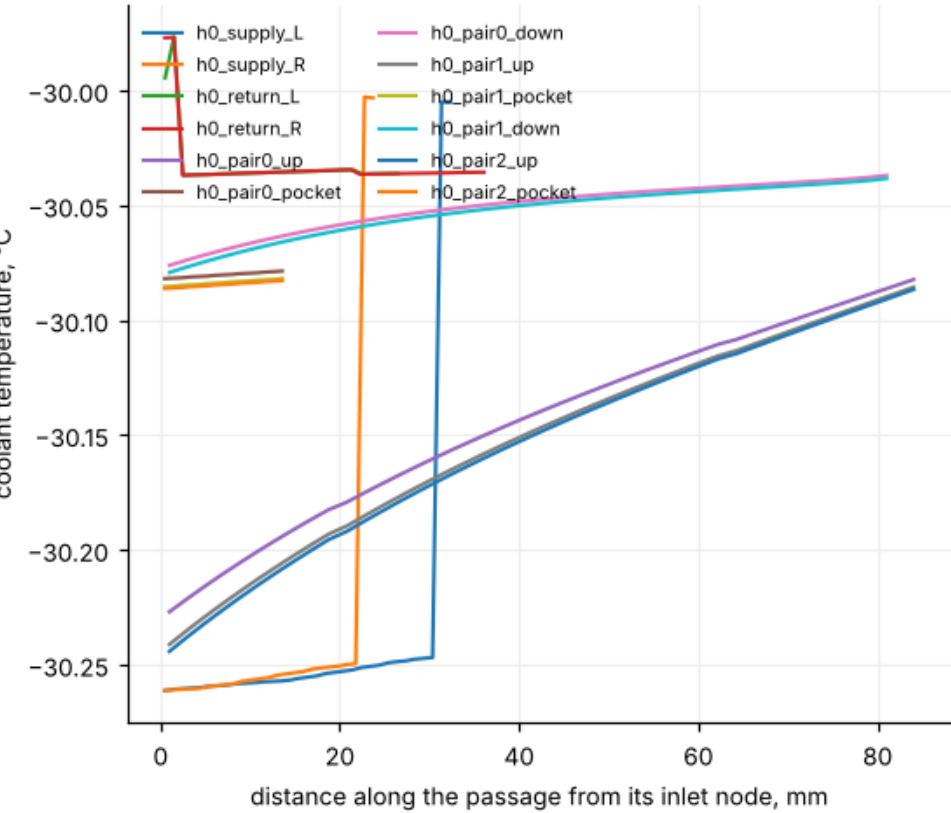
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.257 °C	-30.262 °C
Allowed inlet window, °C	-30.380 to -30.13	-30.385 to -30.138
Inner-wall spread	54.4 mK	53.8 mK
Margin inside ±0.15 K	122.8 mK	123.1 mK
Spread at 5 W (same pattern)	106 mK	104 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	3.57 kPa	3.01 kPa*
Reynolds number, passages	135 – 438	97 – 219
Coolant-copper UA	31.8 W/K	29.0 W/K
Energy balance error		-5.8e-13 W

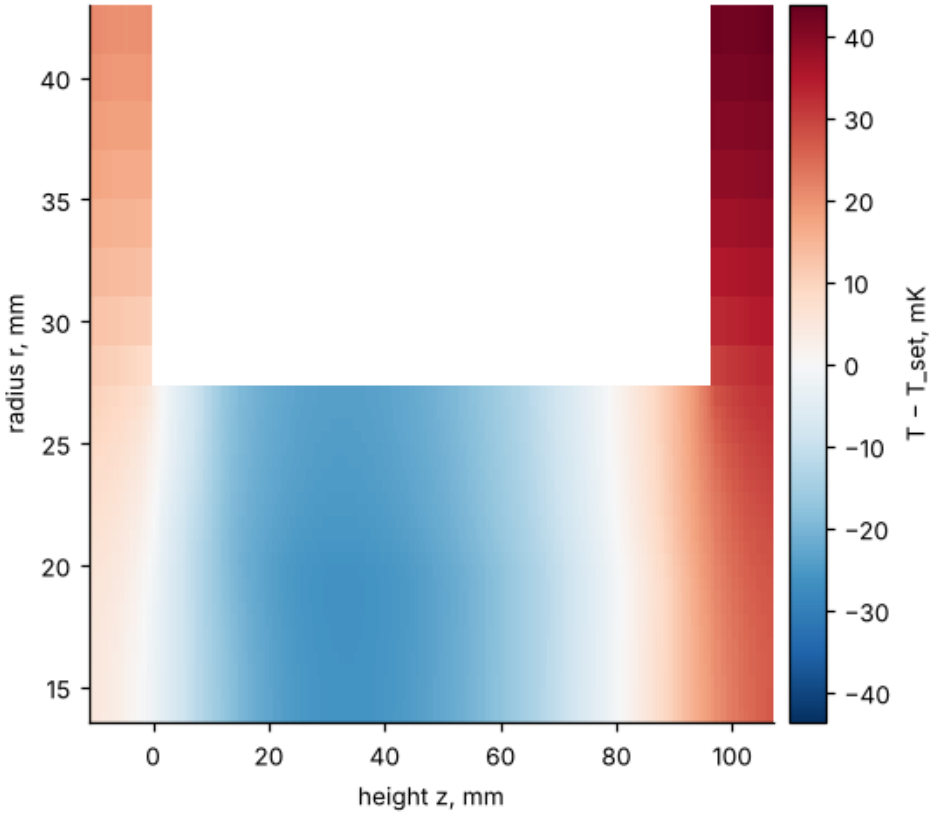
* friction of developed flow only (no bends / fittings)

Δ spread -0.6 mK · Δ inlet -5 mK → SAME

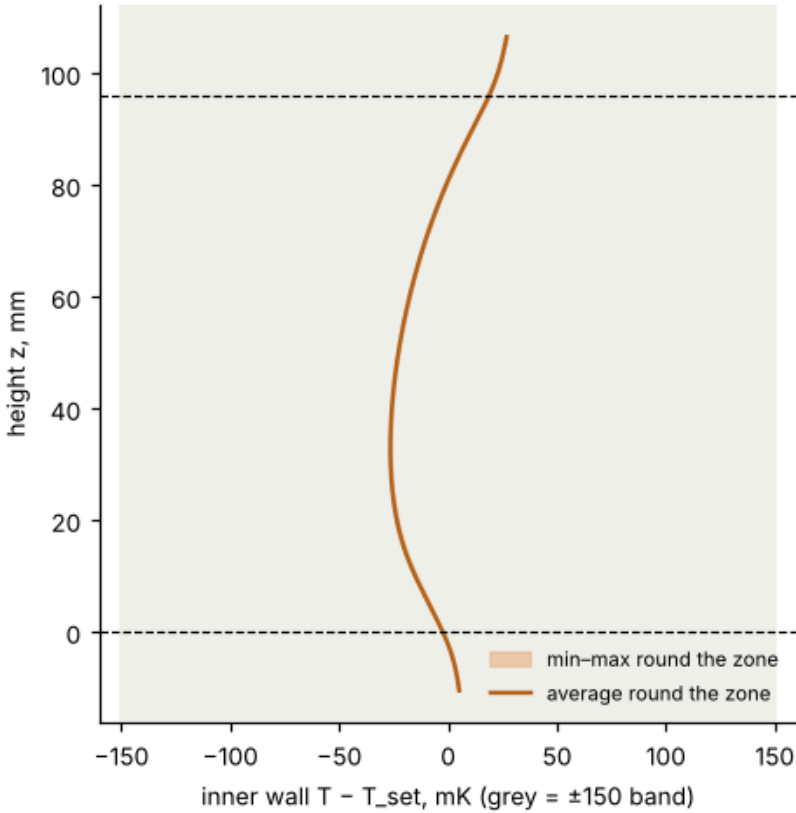
(d) Coolant temperature along each named passage



(e) Section at $\theta = 1^\circ$ — copper T (passages navy)



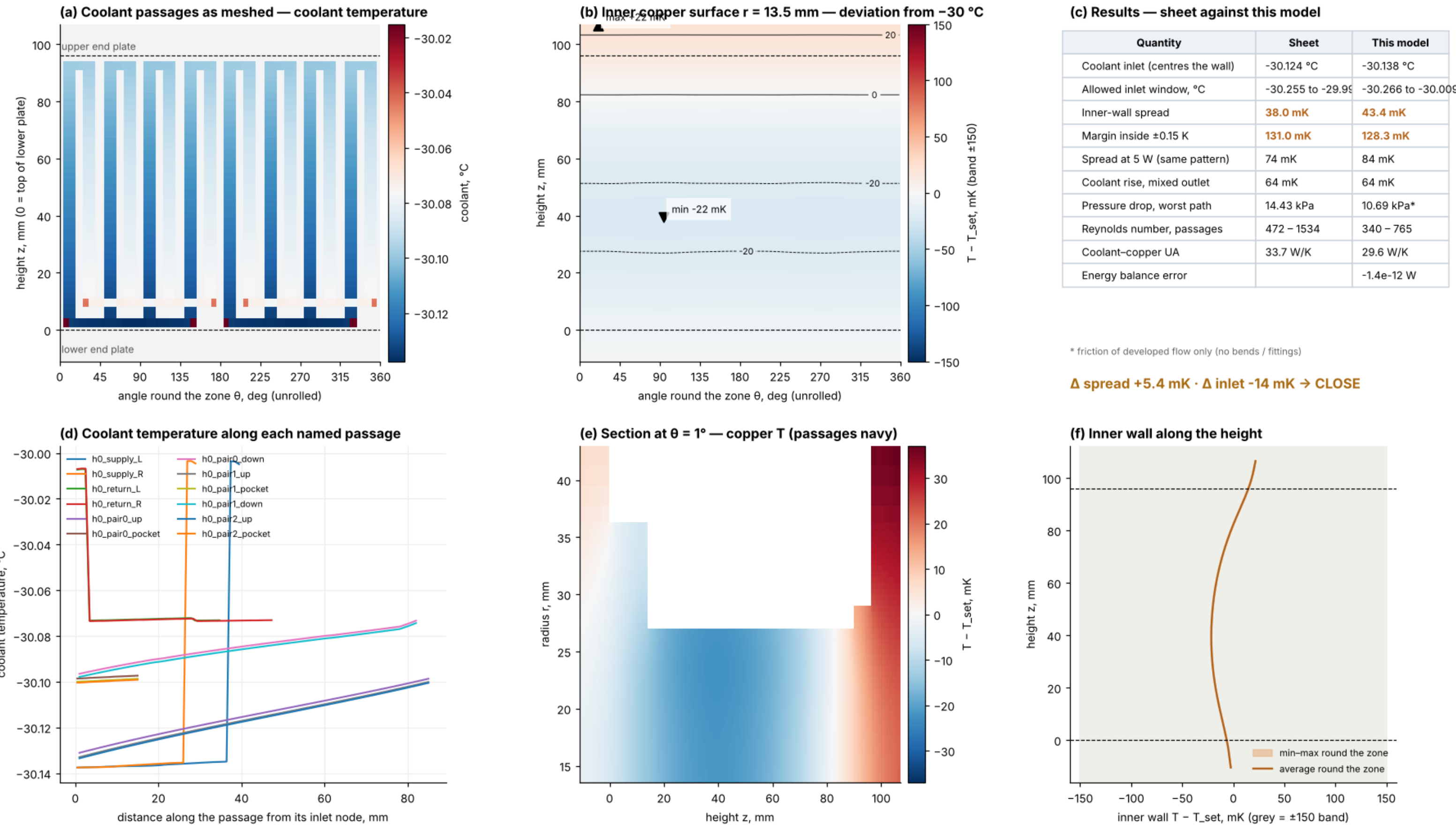
(f) Inner wall along the height



Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

Layout E_PDF — paired sleeve, PDF dims

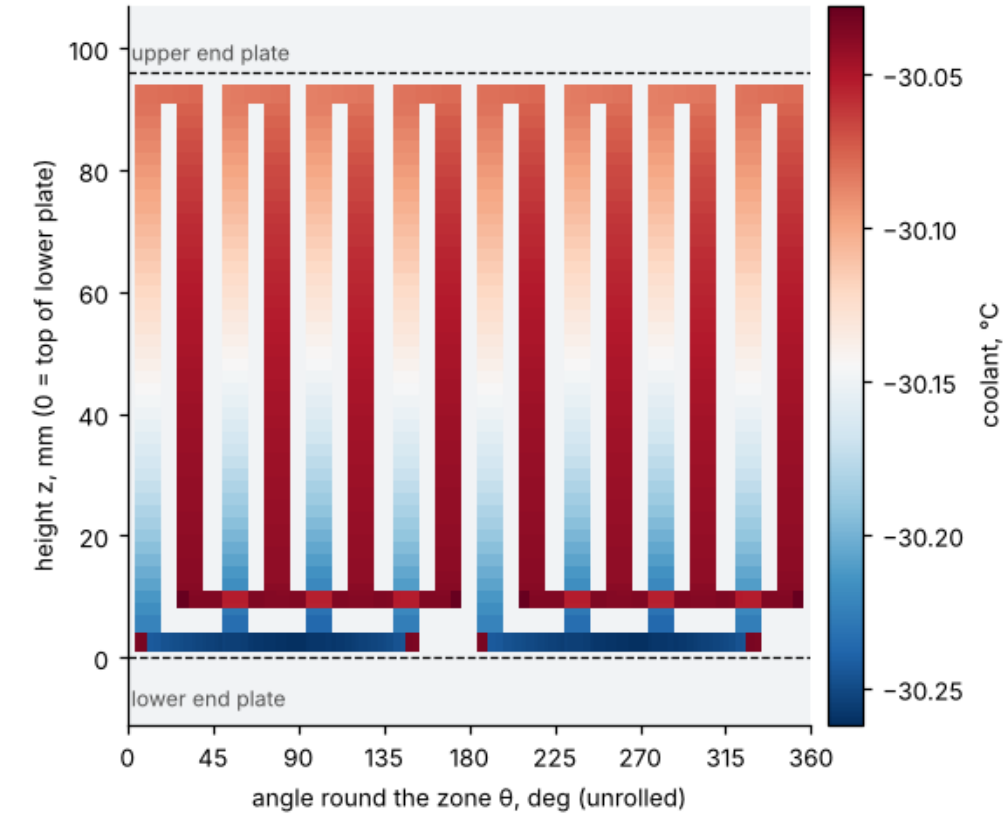
Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.138 °C · 1,926,000 cells · re-validation model



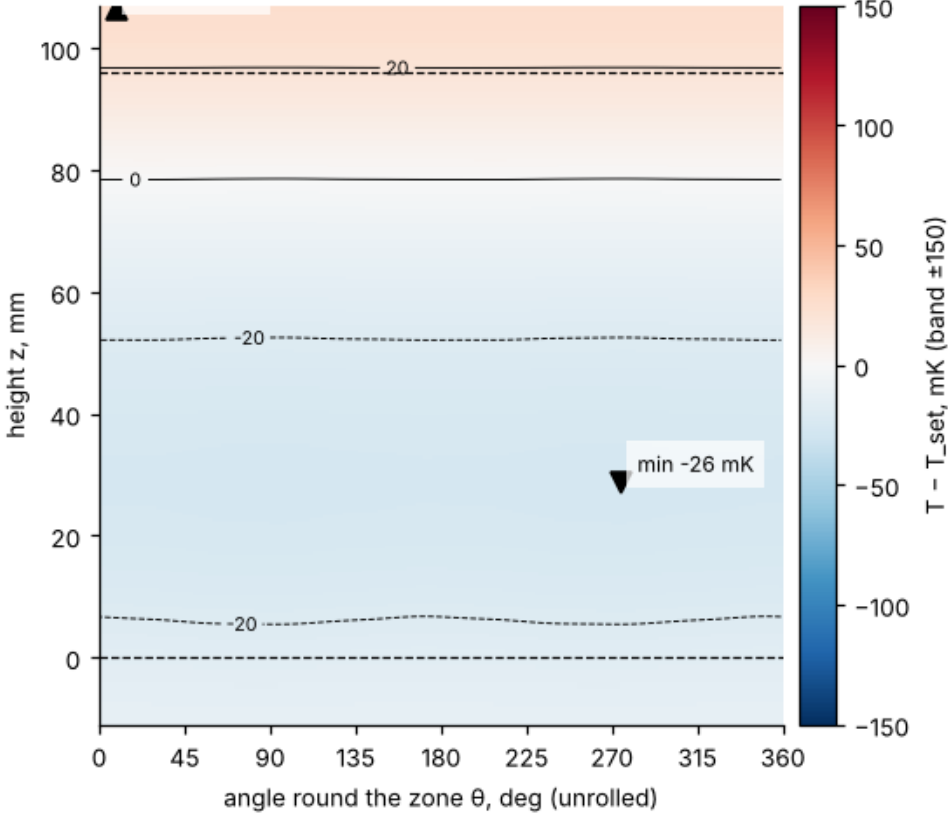
Layout E_PDF — paired sleeve, PDF dims

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.262 °C · 1,926,000 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



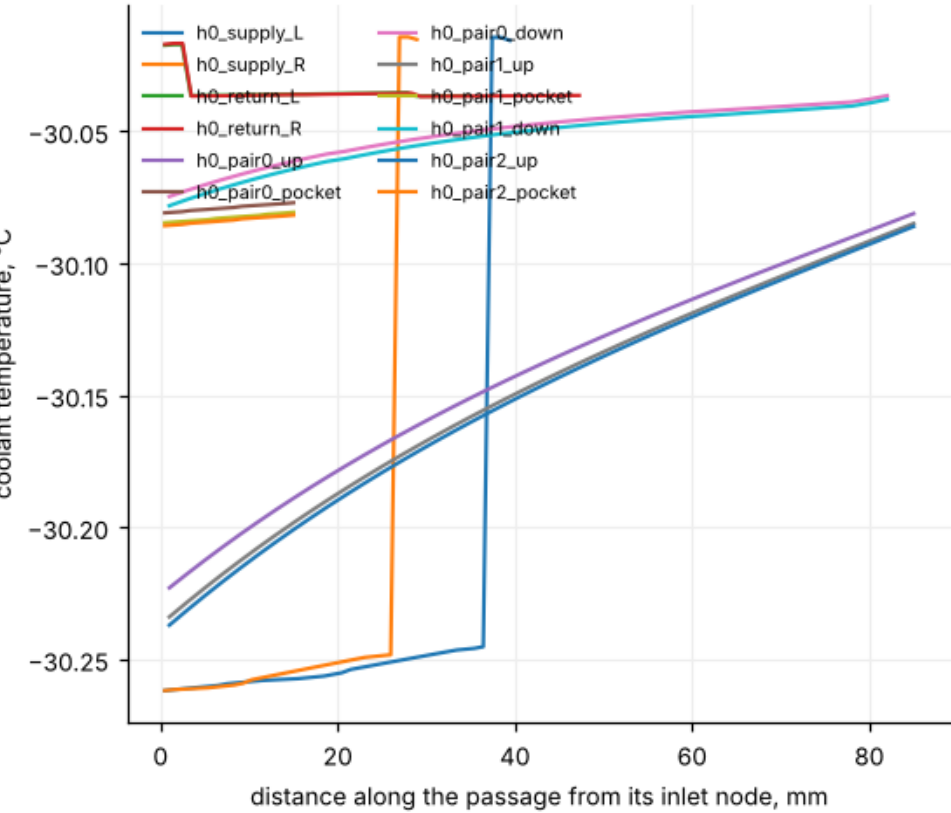
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.253 °C	-30.262 °C
Allowed inlet window, °C	-30.378 to -30.12	-30.386 to -30.138
Inner-wall spread	49.1 mK	52.6 mK
Margin inside ±0.15 K	125.4 mK	123.7 mK
Spread at 5 W (same pattern)	95 mK	102 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	3.76 kPa	3.05 kPa*
Reynolds number, passages	135 – 438	97 – 219
Coolant-copper UA	33.7 W/K	29.6 W/K
Energy balance error		-2.6e-12 W

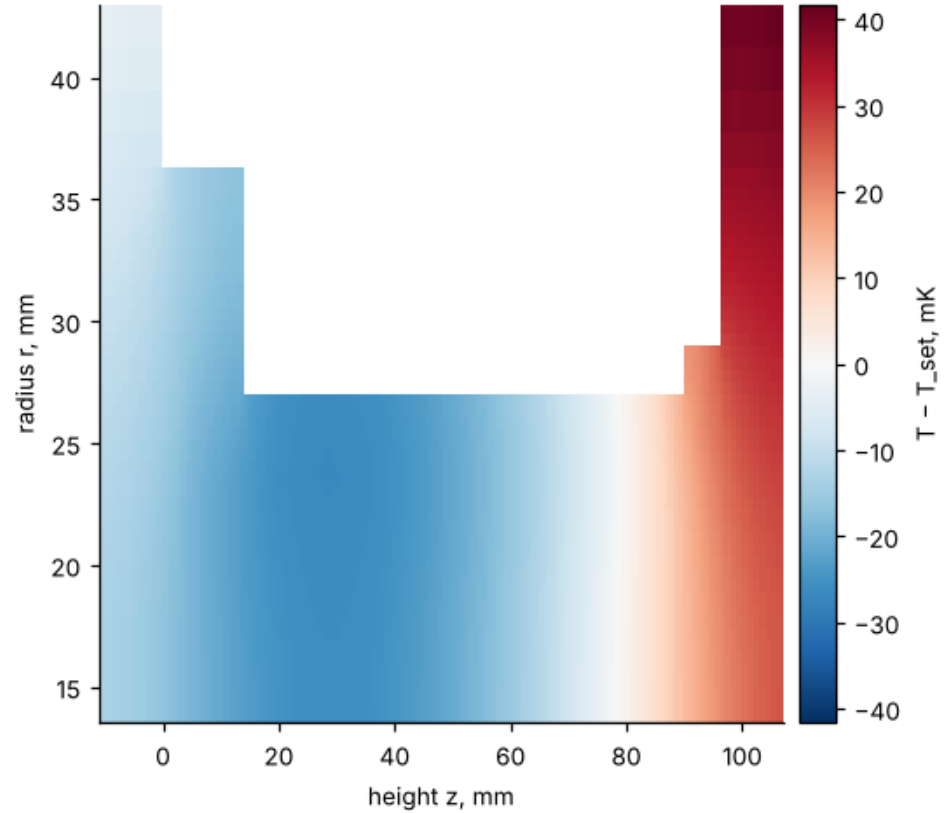
* friction of developed flow only (no bends / fittings)

Δ spread +3.5 mK · Δ inlet -9 mK → SAME

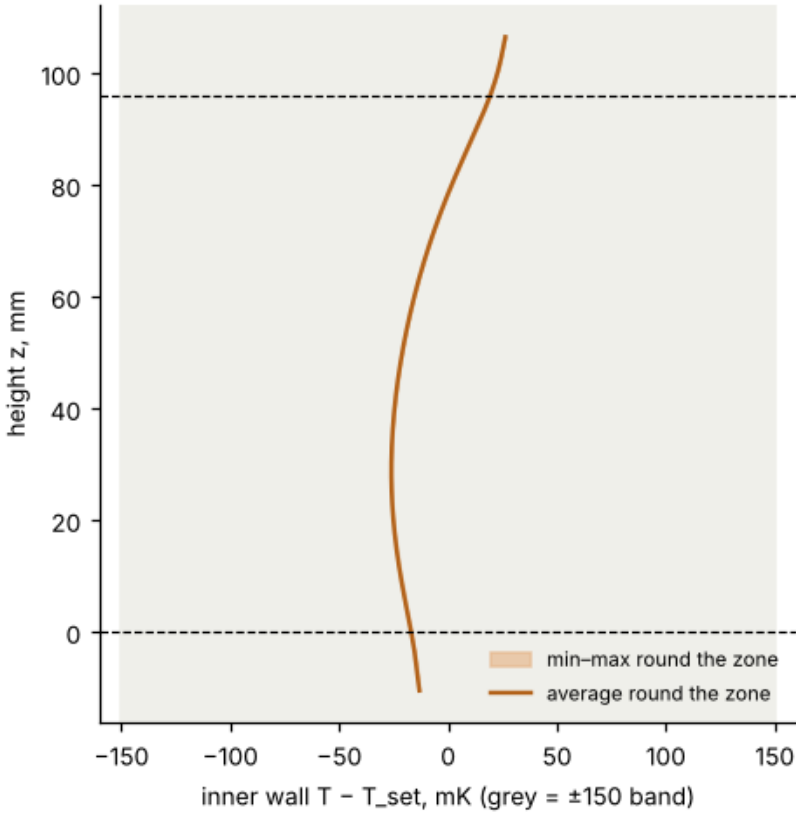
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



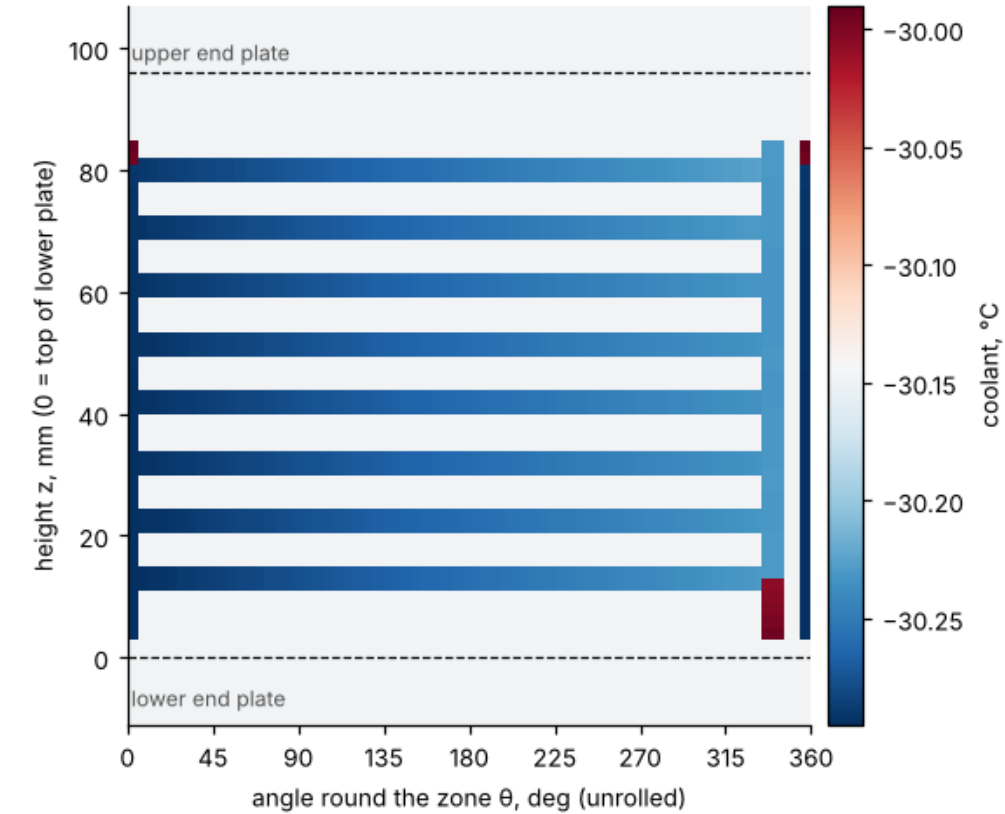
(f) Inner wall along the height



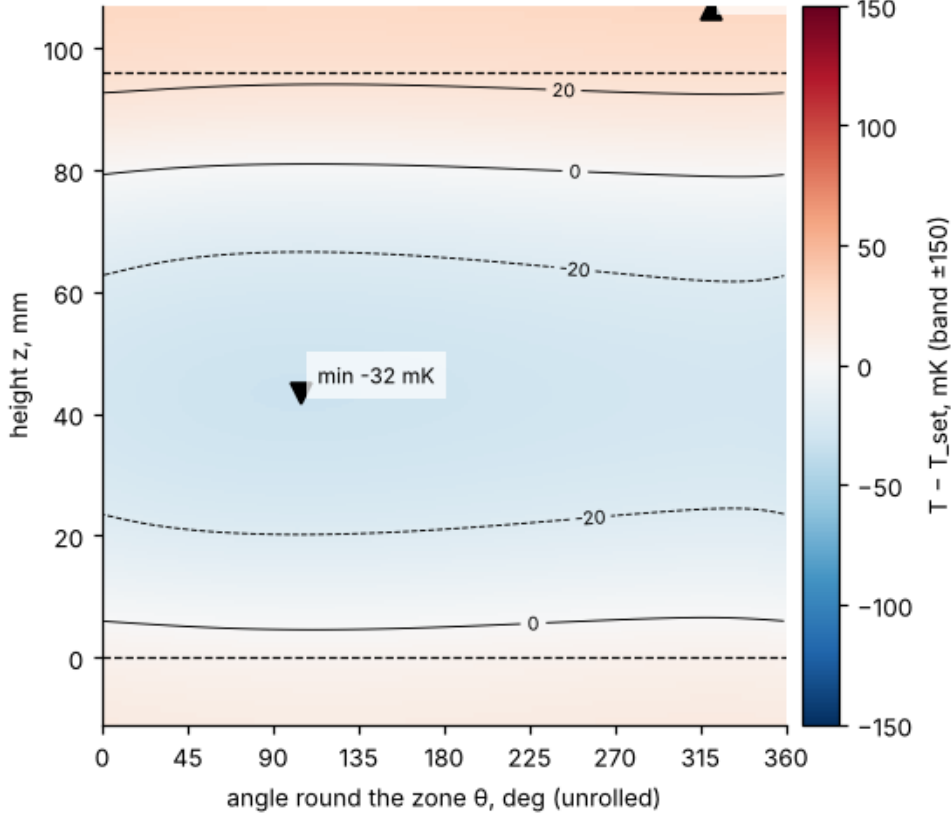
Layout F — stacked ring channels

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.295 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



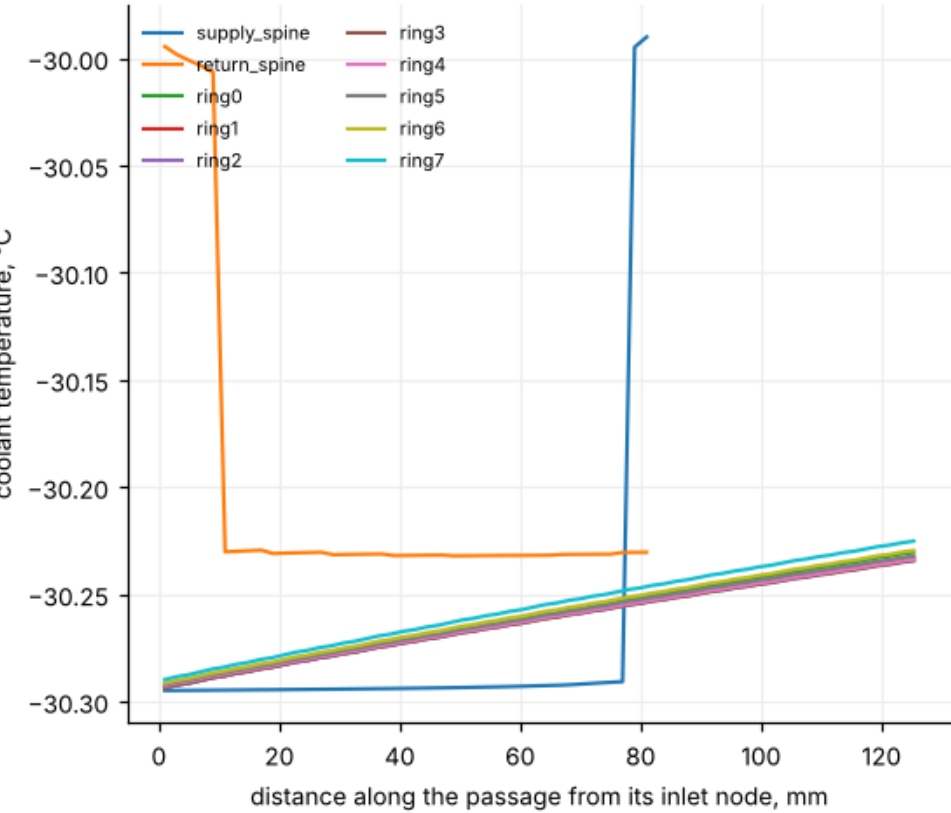
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.288 °C	-30.295 °C
Allowed inlet window, °C	-30.404 to -30.17	-30.413 to -30.177
Inner-wall spread	67.9 mK	63.7 mK
Margin inside ±0.15 K	116.0 mK	118.2 mK
Spread at 5 W (same pattern)	132 mK	123 mK
Coolant rise, mixed outlet	64 mK	65 mK
Pressure drop, worst path	1.91 kPa	0.25 kPa*
Reynolds number, passages	488 – 3126	487 – 3116
Coolant-copper UA	11.1 W/K	10.8 W/K
Energy balance error		3.8e-03 W

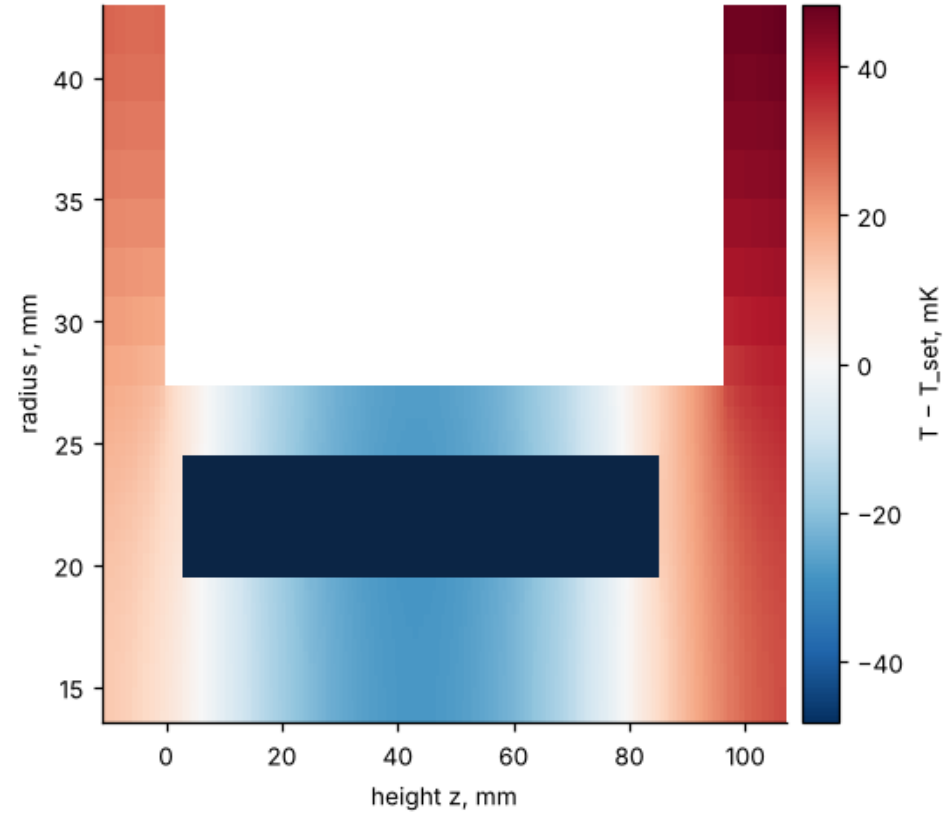
* friction of developed flow only (no bends / fittings)

Δ spread -4.2 mK · Δ inlet -7 mK → SAME

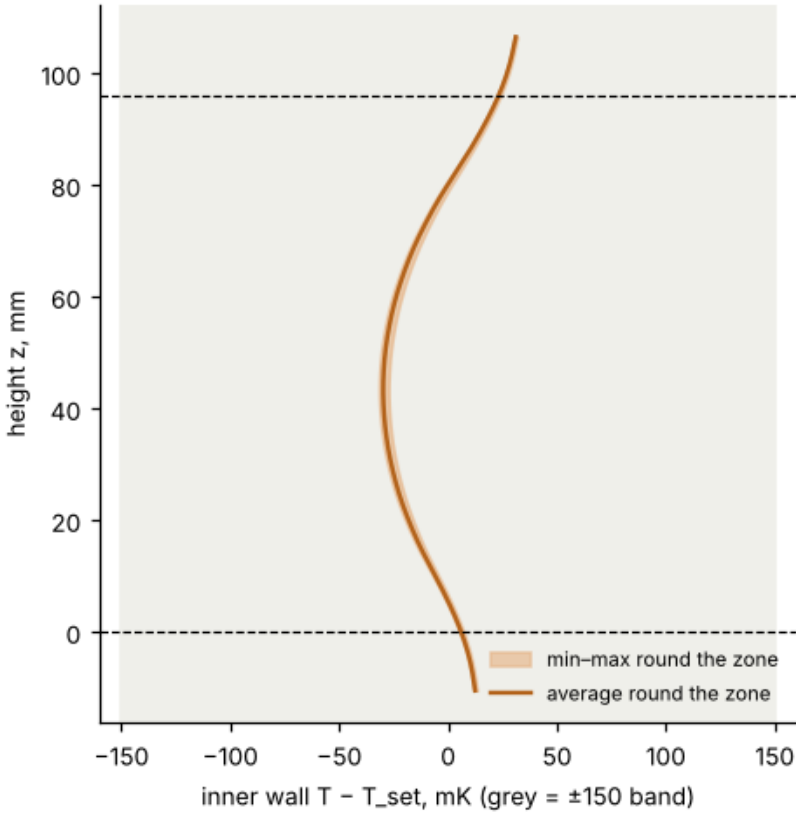
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



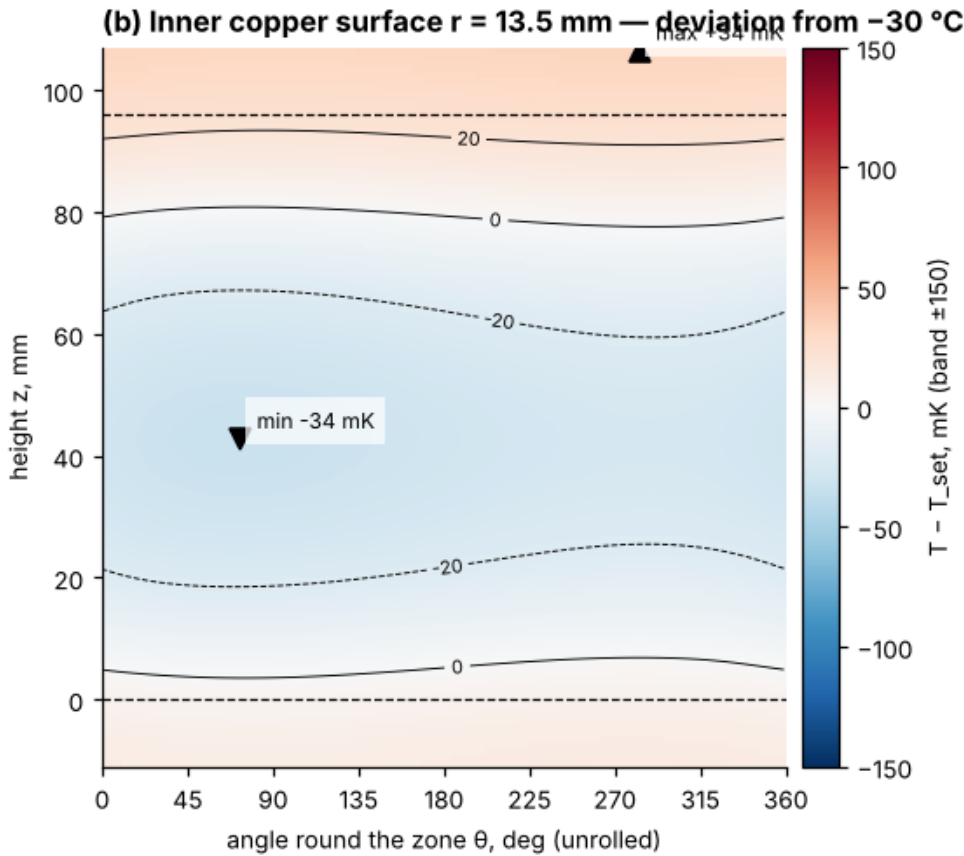
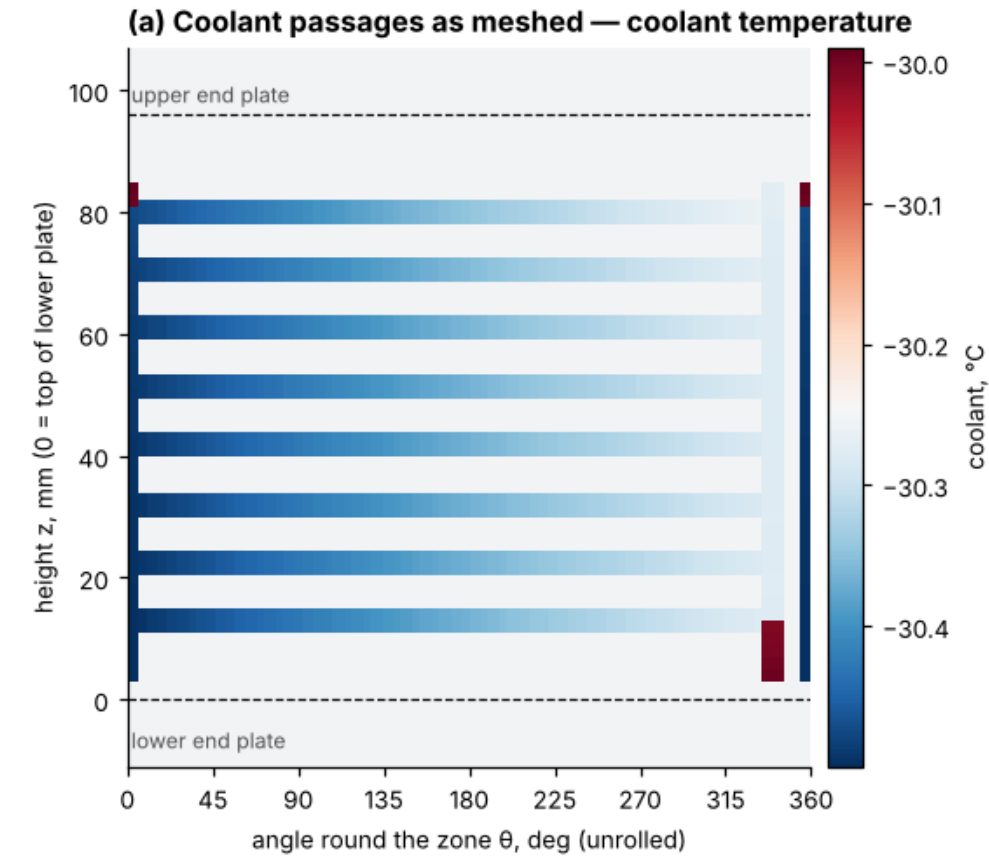
(f) Inner wall along the height



Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

Layout F — stacked ring channels

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.500 °C · 1,348,200 cells · re-validation model

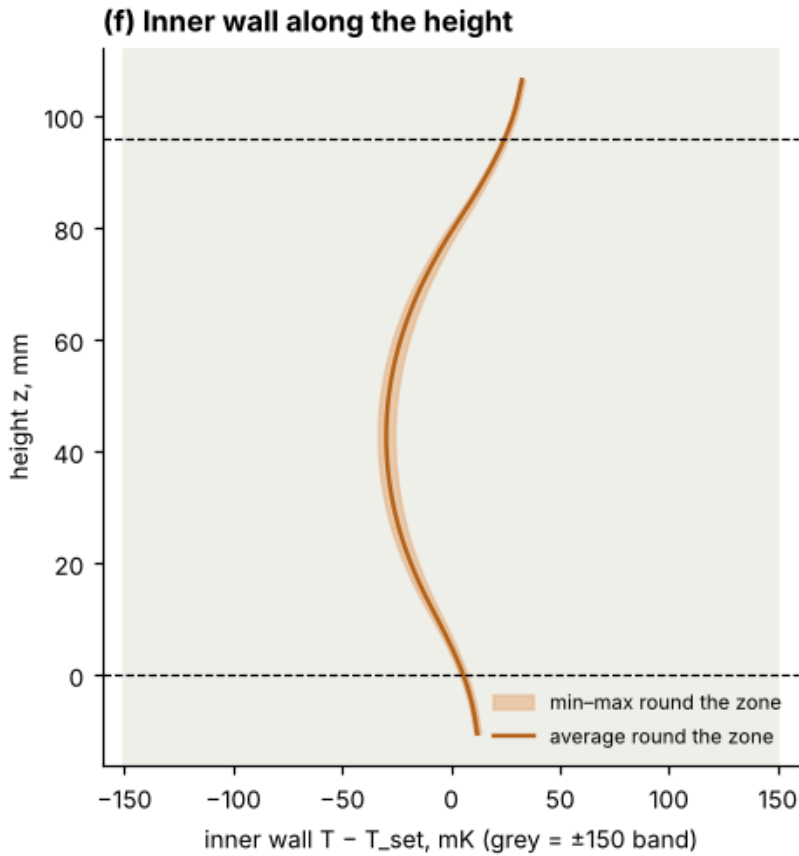
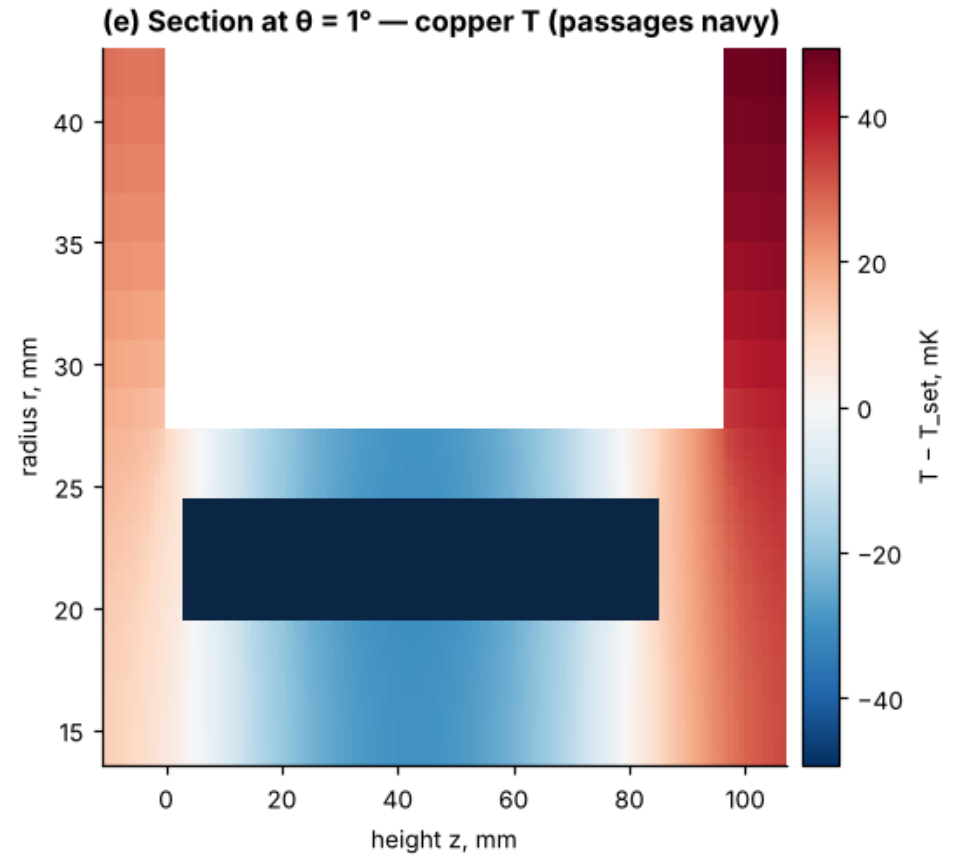
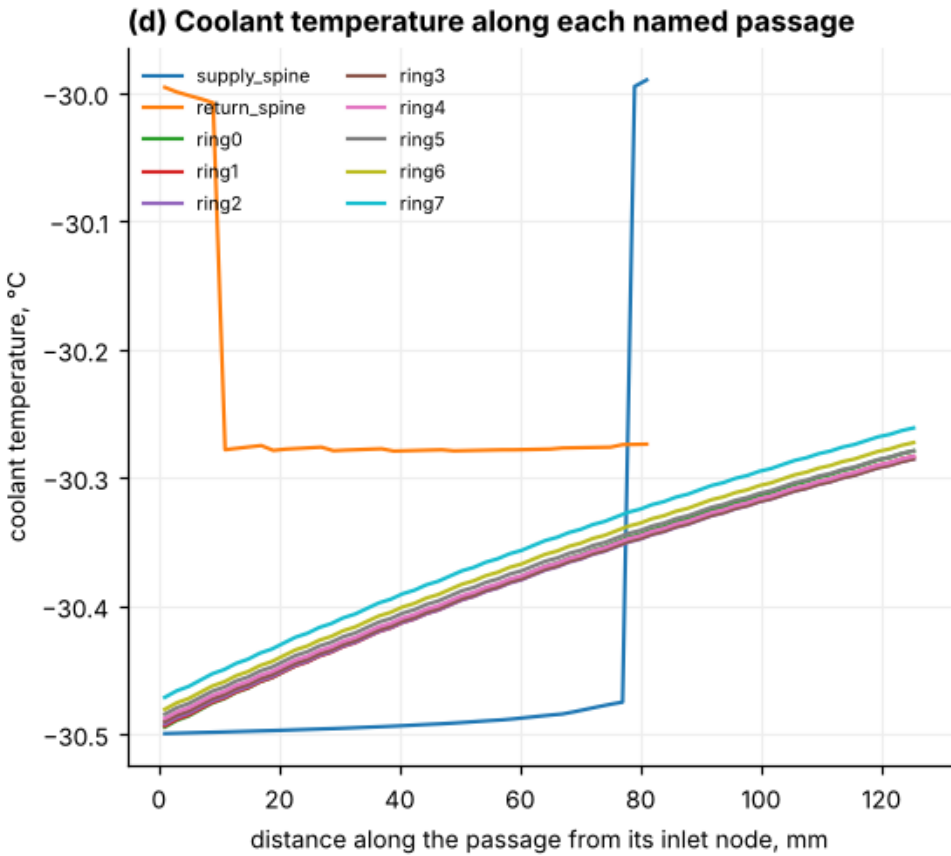


(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.487 °C	-30.500 °C
Allowed inlet window, °C	-30.601 to -30.37	-30.616 to -30.384
Inner-wall spread	71.0 mK	67.2 mK
Margin inside $\pm 0.15\text{ K}$	114.5 mK	116.4 mK
Spread at 5 W (same pattern)	138 mK	130 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	0.24 kPa	0.06 kPa*
Reynolds number, passages	140 – 893	139 – 890
Coolant-copper UA	7.6 W/K	7.3 W/K
Energy balance error		5.4e-03 W

* friction of developed flow only (no bends / fittings)

Δ spread -3.8 mK · Δ inlet -13 mK → SAME

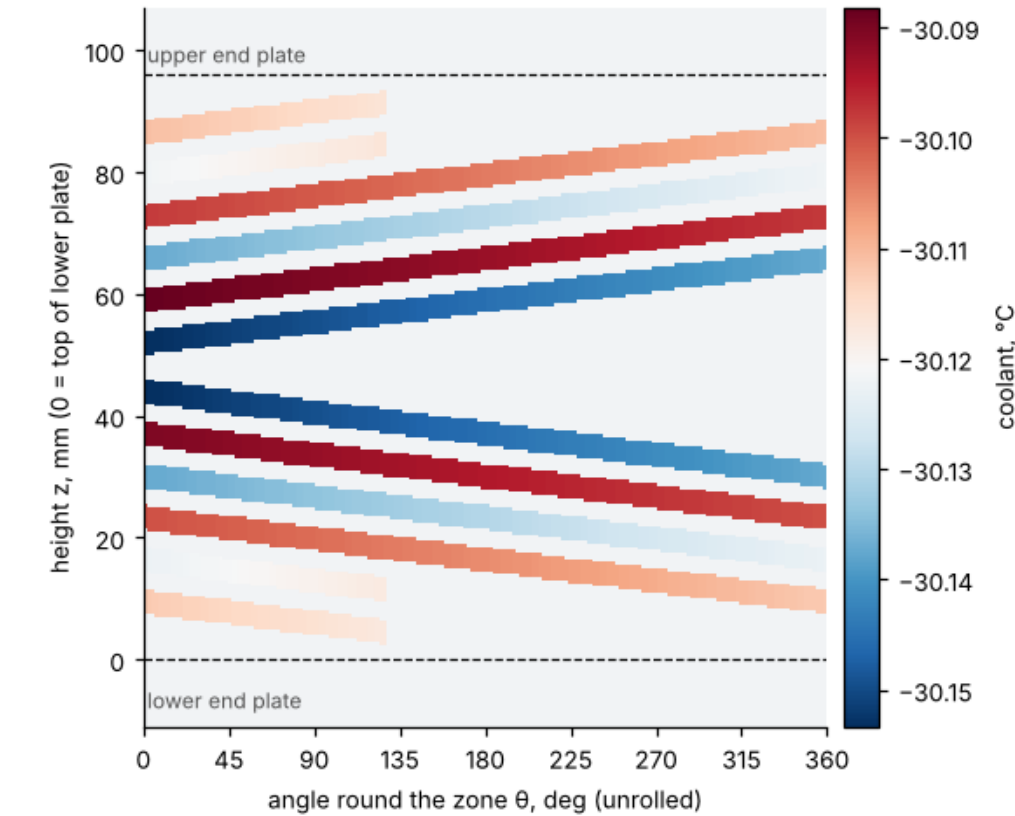


Model: 3-D finite-volume conduction (copper wall $r\ 13.5\text{--}15$, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C , heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C .

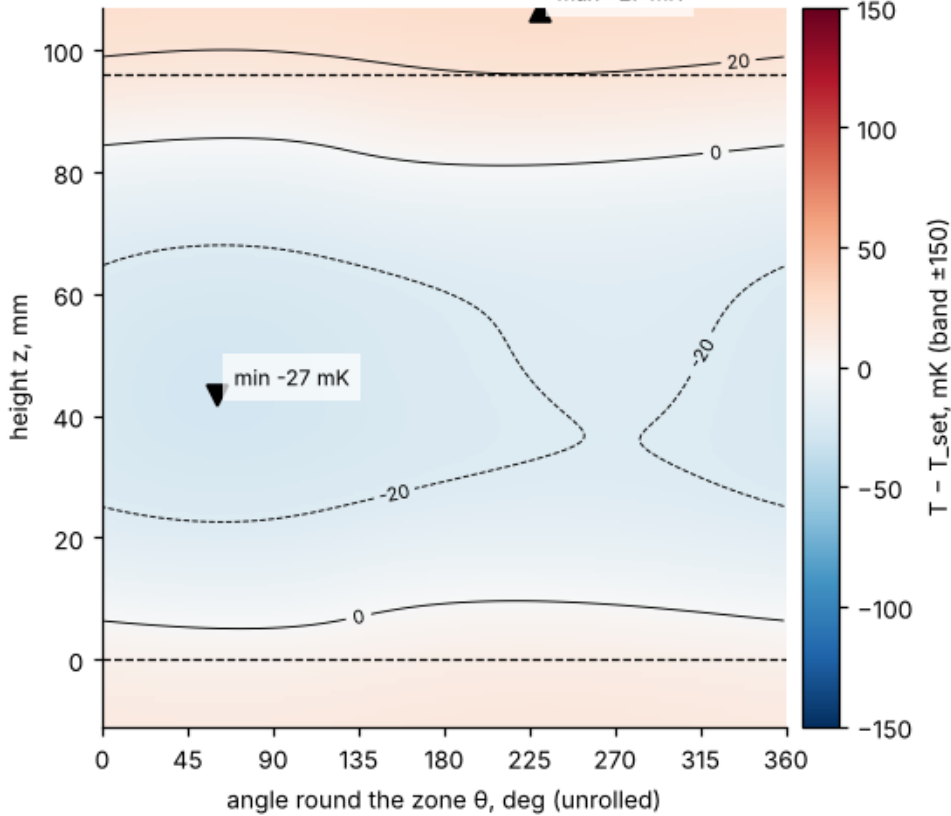
Layout G — two short folded coils

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.154 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



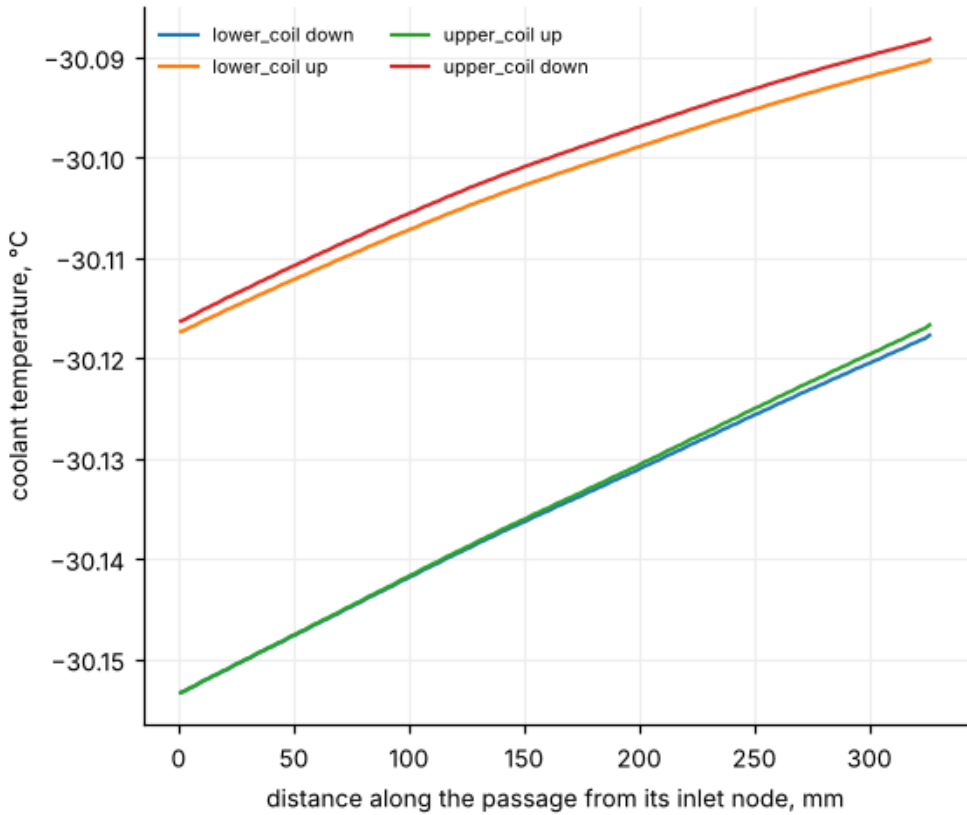
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.157 °C	-30.154 °C
Allowed inlet window, °C	-30.279 to -30.03	-30.276 to -30.031
Inner-wall spread	56.8 mK	55.0 mK
Margin inside ±0.15 K	121.6 mK	122.5 mK
Spread at 5 W (same pattern)	110 mK	106 mK
Coolant rise, mixed outlet	64 mK	64 mK
Pressure drop, worst path	5.21 kPa	4.56 kPa*
Reynolds number, passages	1954 – 1954	1948 – 1948
Coolant-copper UA	24.6 W/K	25.0 W/K
Energy balance error		-2.4e-13 W

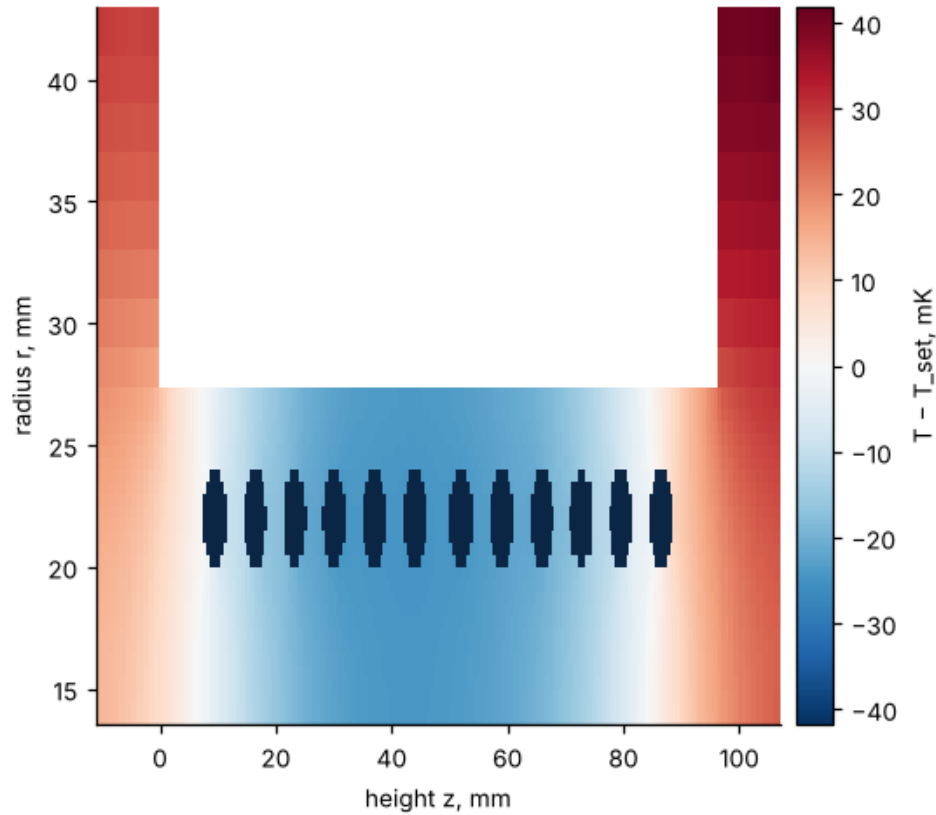
* friction of developed flow only (no bends / fittings)

Δ spread -1.8 mK · Δ inlet +3 mK → SAME

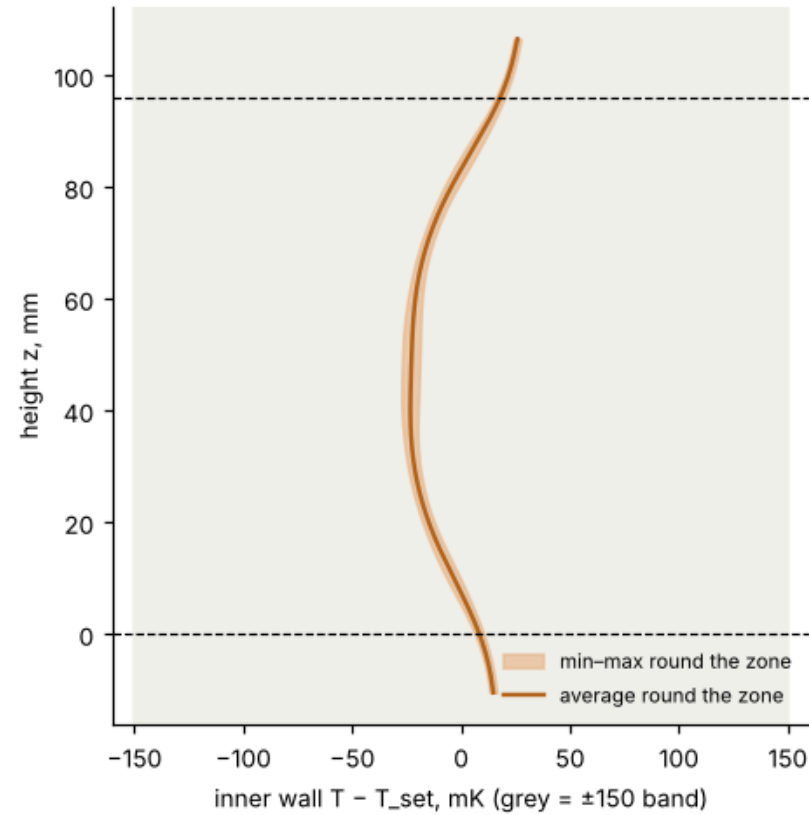
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

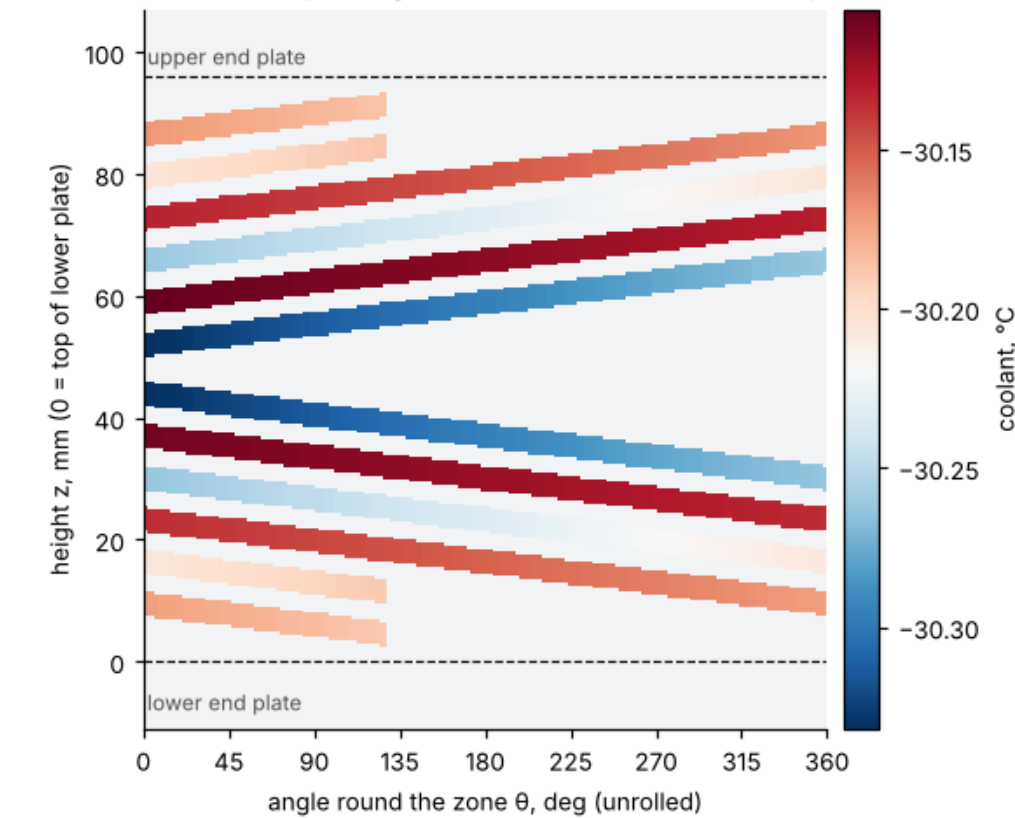


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

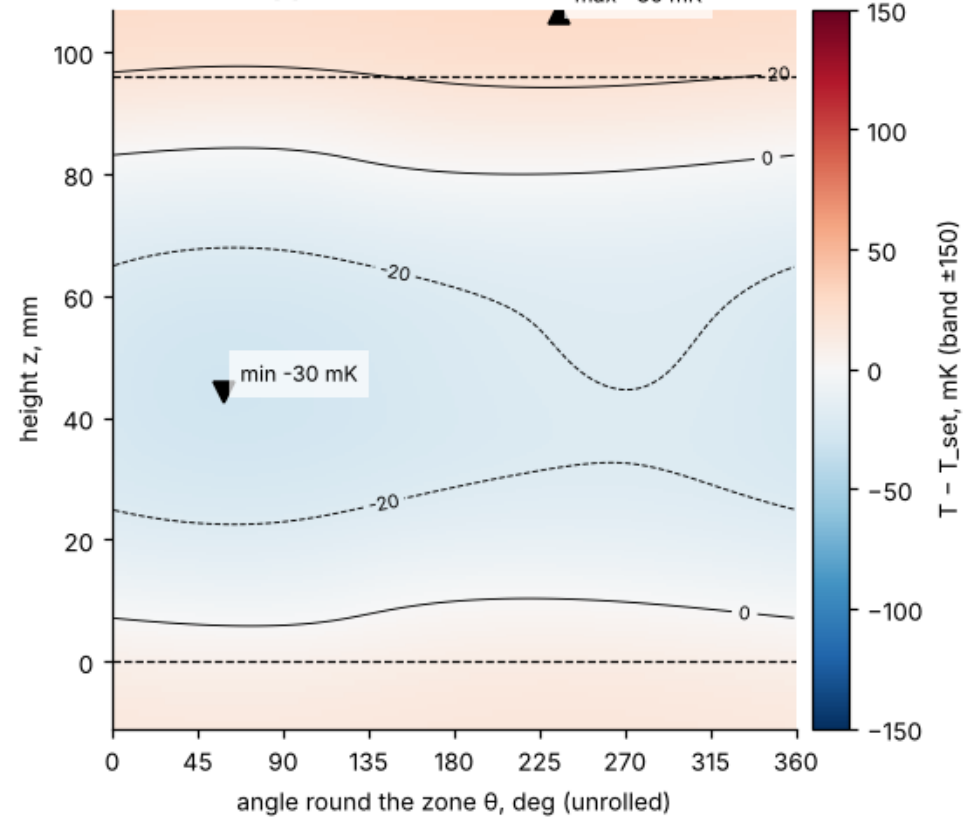
Layout G — two short folded coils

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.334 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



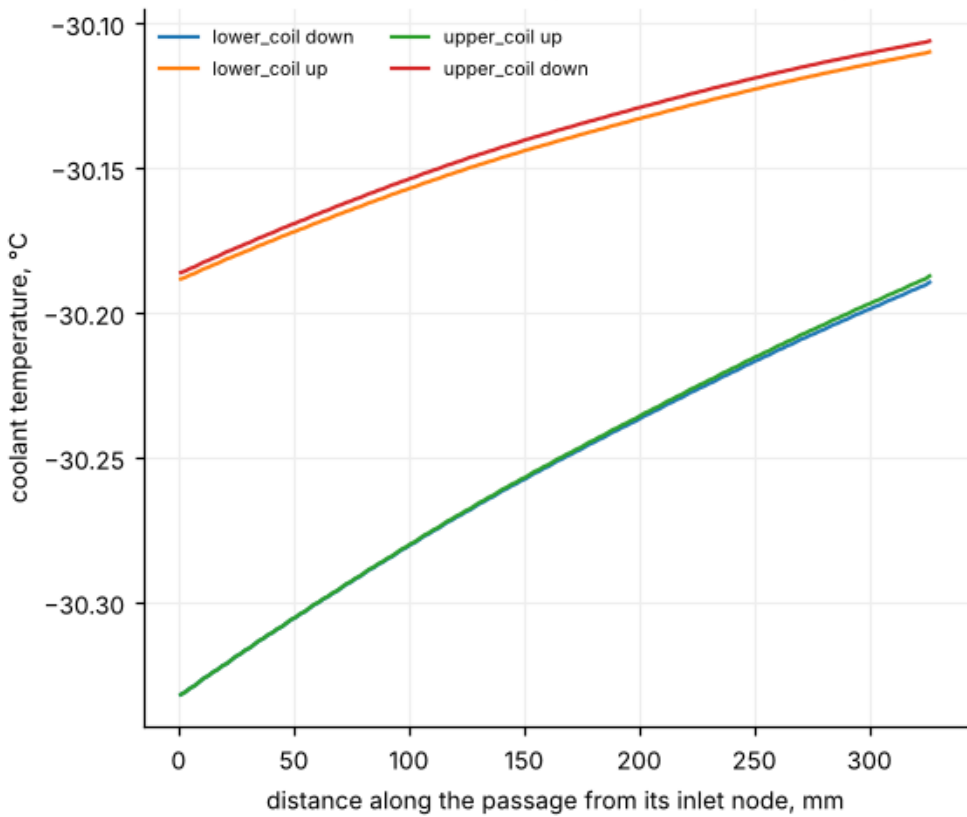
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.338 °C	-30.334 °C
Allowed inlet window, °C	-30.458 to -30.21	-30.454 to -30.213
Inner-wall spread	60.2 mK	59.1 mK
Margin inside ±0.15 K	119.9 mK	120.4 mK
Spread at 5 W (same pattern)	117 mK	115 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	0.83 kPa	0.80 kPa*
Reynolds number, passages	558 – 558	556 – 556
Coolant-copper UA	13.9 W/K	14.1 W/K
Energy balance error		6.3e-13 W

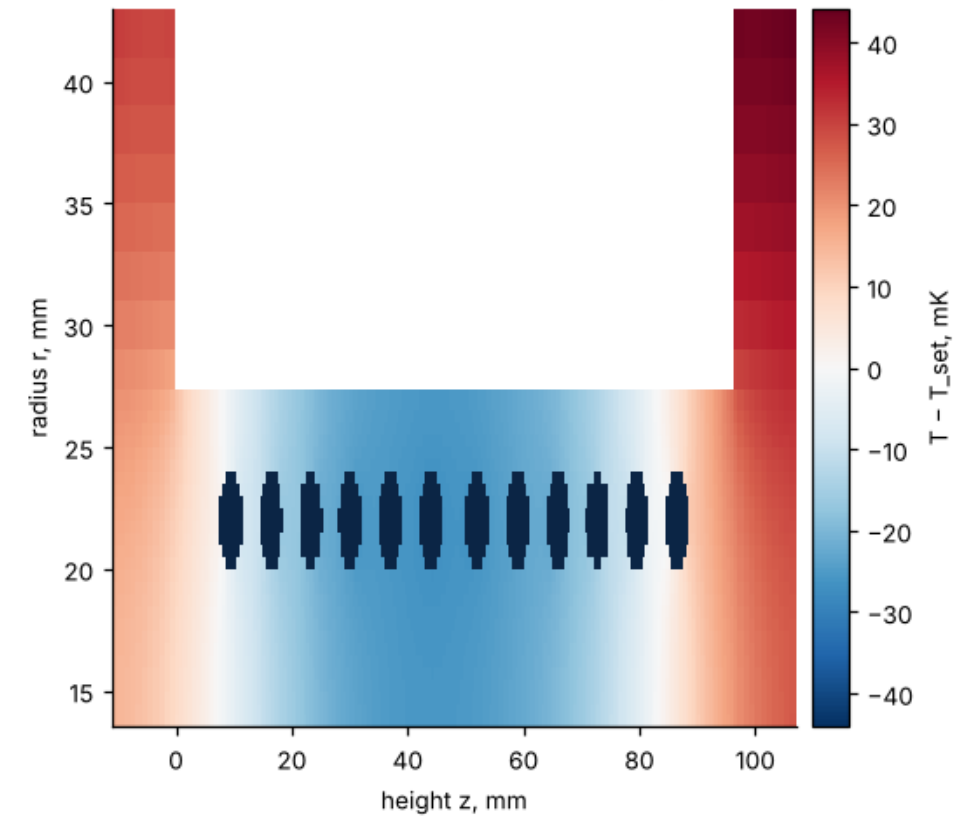
* friction of developed flow only (no bends / fittings)

Δ spread -1.1 mK · Δ inlet +4 mK → SAME

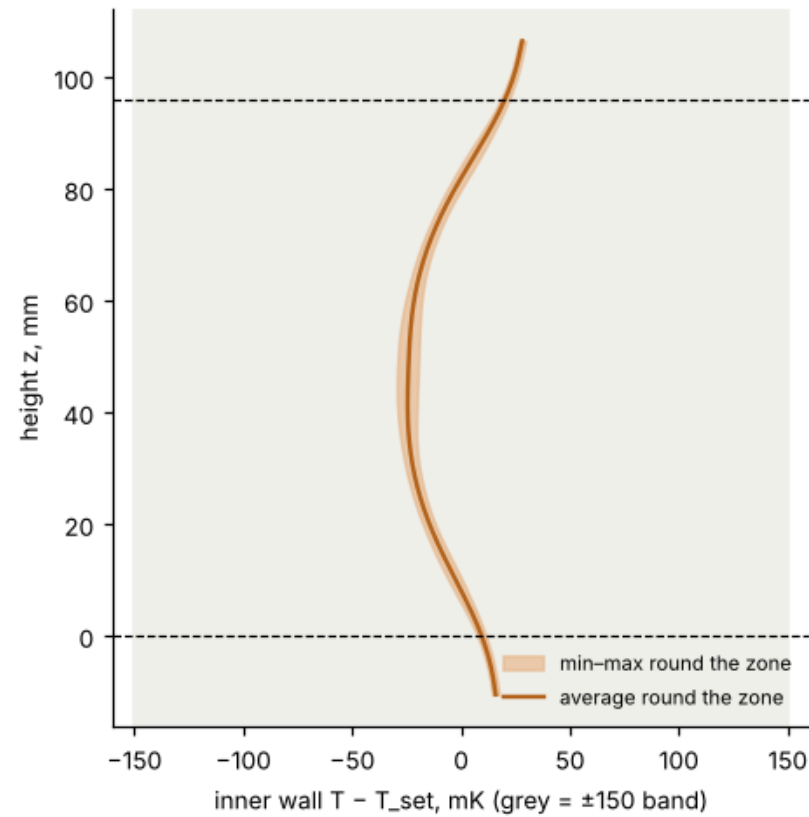
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

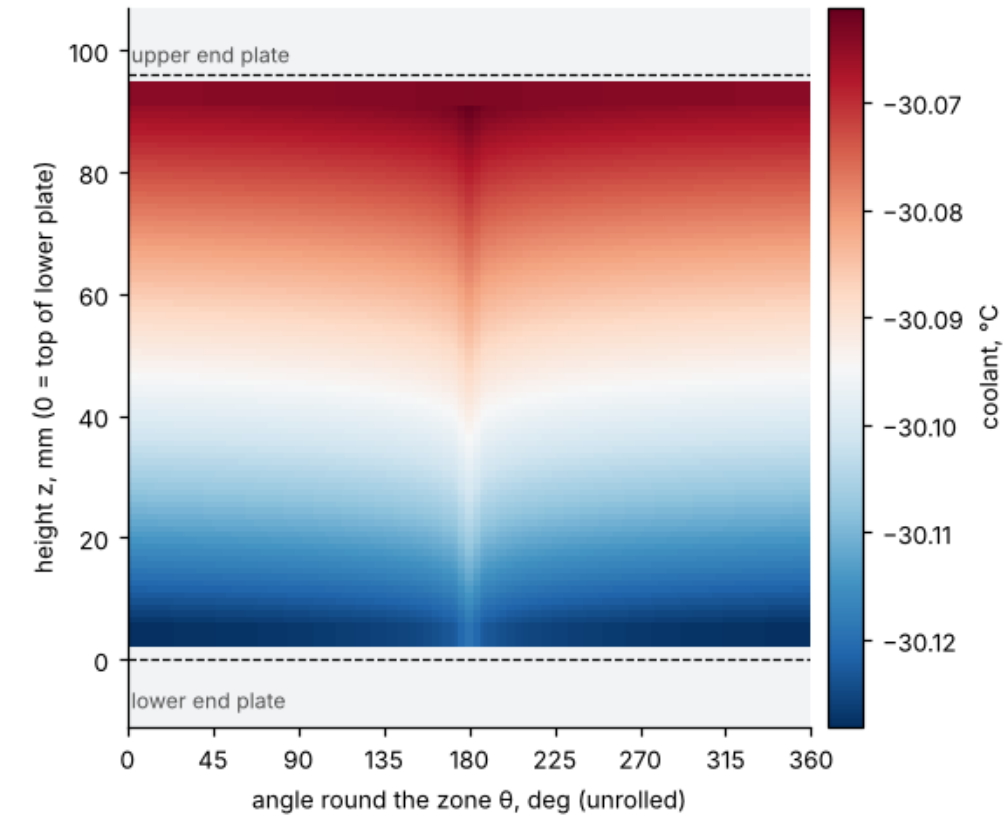


Model: 3-D finite-volume conduction (copper wall r 13.5–15, carrier / buffer, end plates) coupled to a 1-D coolant network (developed-flow h), AMG-preconditioned GMRES. Basis: Zone 1, set point -30 °C, heat from the axisymmetric chamber model scaled to the sheet's total, methanol at -30 °C.

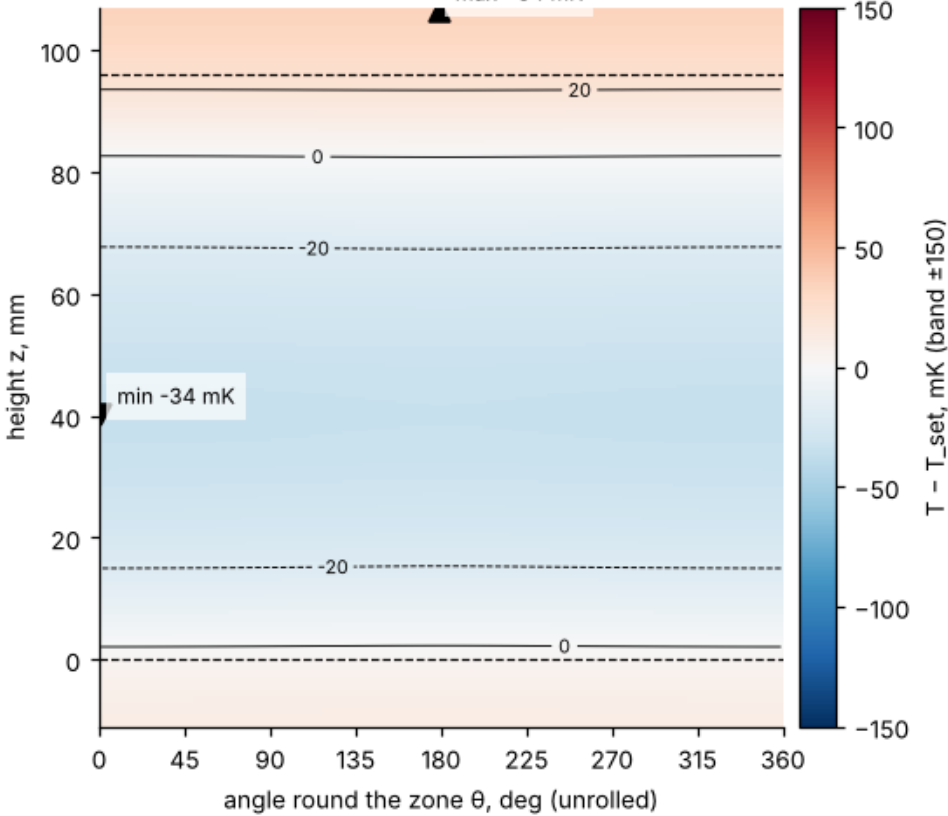
Layout H — full annular jacket

Zone 1 · total coolant flow 17.5 g/s · heat 2.58 W · inlet -30.128 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



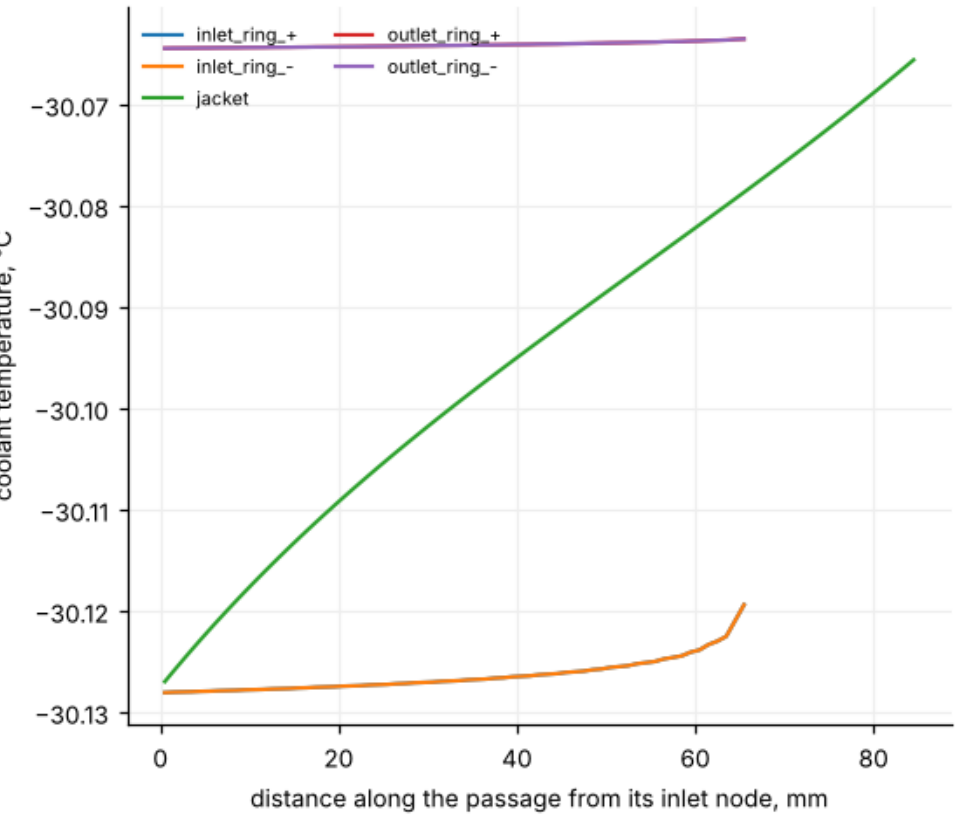
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.239 °C	-30.128 °C
Allowed inlet window, °C	-30.355 to -30.12	-30.244 to -30.013
Inner-wall spread	68.6 mK	69.0 mK
Margin inside ±0.15 K	115.7 mK	115.5 mK
Spread at 5 W (same pattern)	133 mK	134 mK
Coolant rise, mixed outlet	64 mK	65 mK
Pressure drop, worst path	3.35 kPa	1.23 kPa*
Reynolds number, passages	191 – 3069	79 – 2040
Coolant-copper UA	13.2 W/K	30.8 W/K
Energy balance error		2.5e-03 W

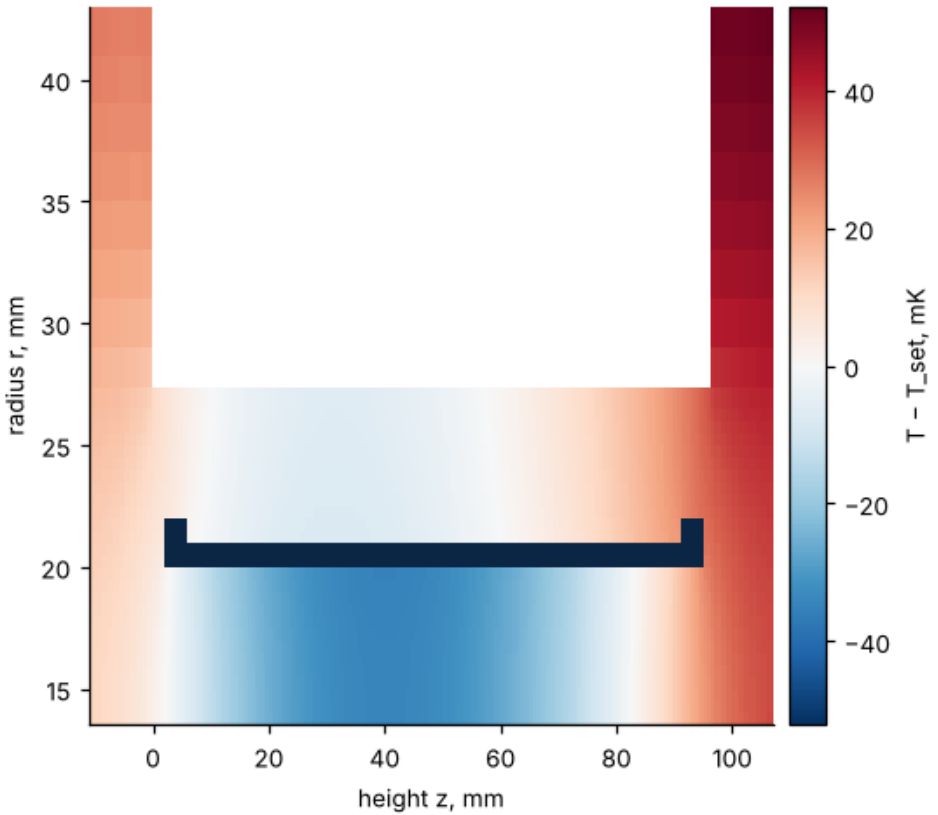
* friction of developed flow only (no bends / fittings)

Δ spread +0.4 mK · Δ inlet +111 mK → CLOSE

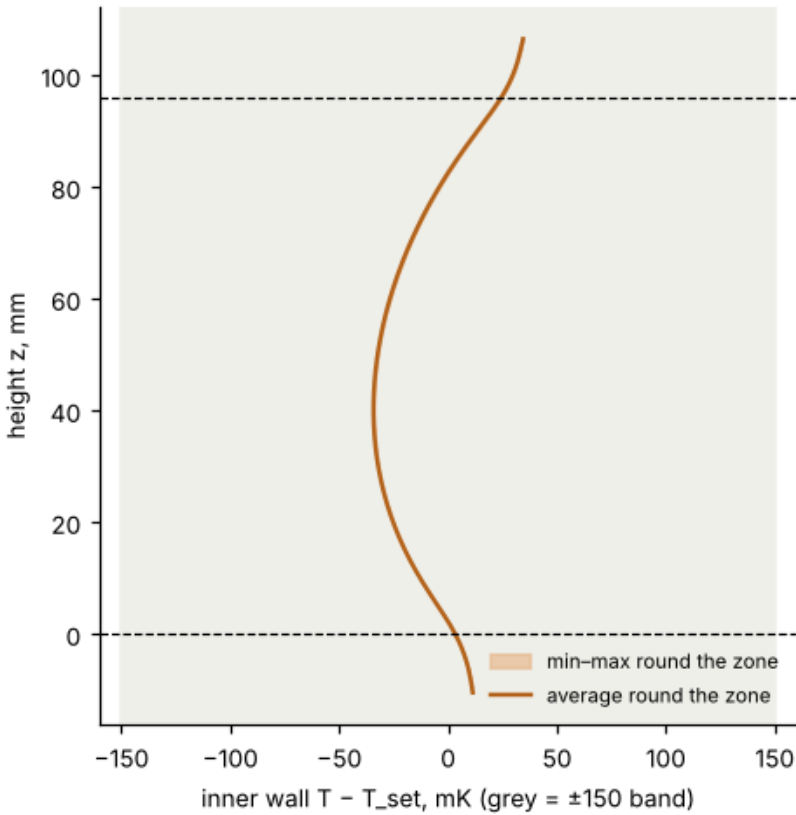
(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



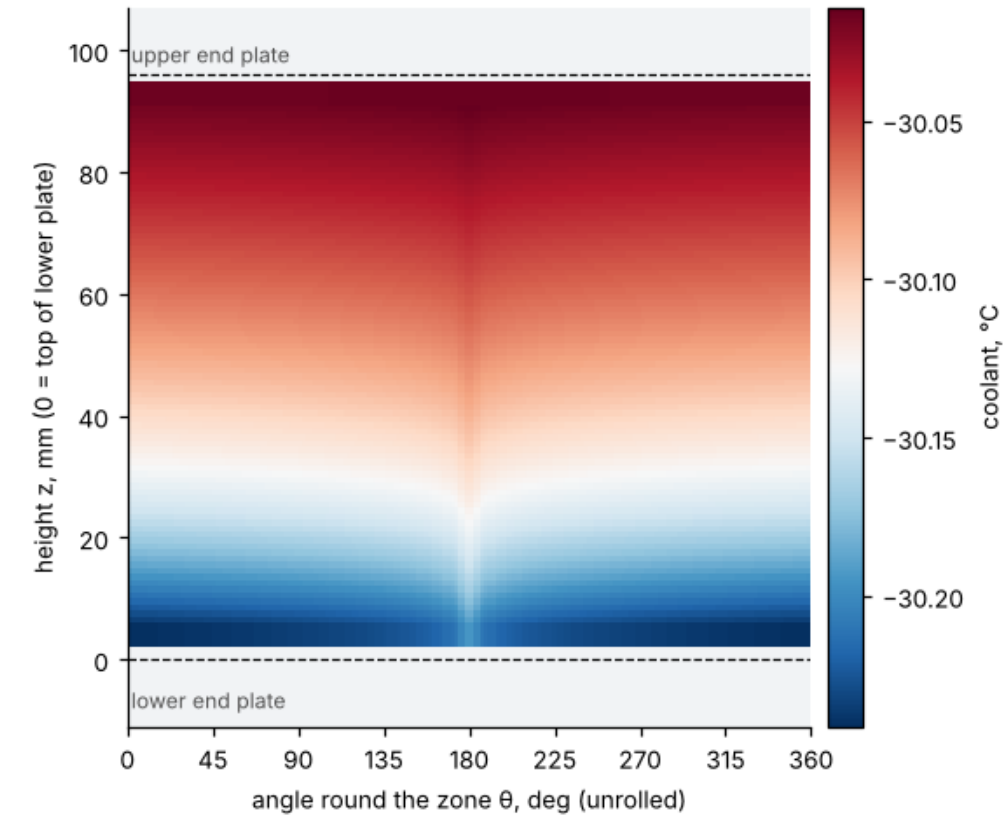
(f) Inner wall along the height



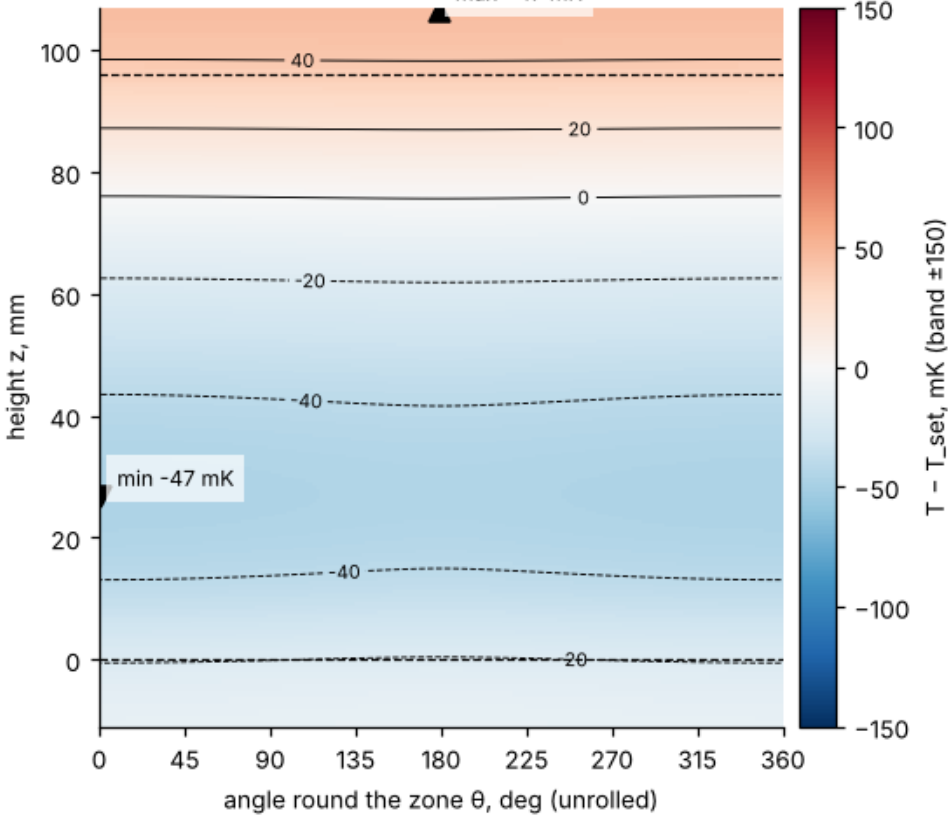
Layout H — full annular jacket

Zone 1 · total coolant flow 5 g/s · heat 2.58 W · inlet -30.241 °C · 1,348,200 cells · re-validation model

(a) Coolant passages as meshed — coolant temperature



(b) Inner copper surface r = 13.5 mm — deviation from -30 °C



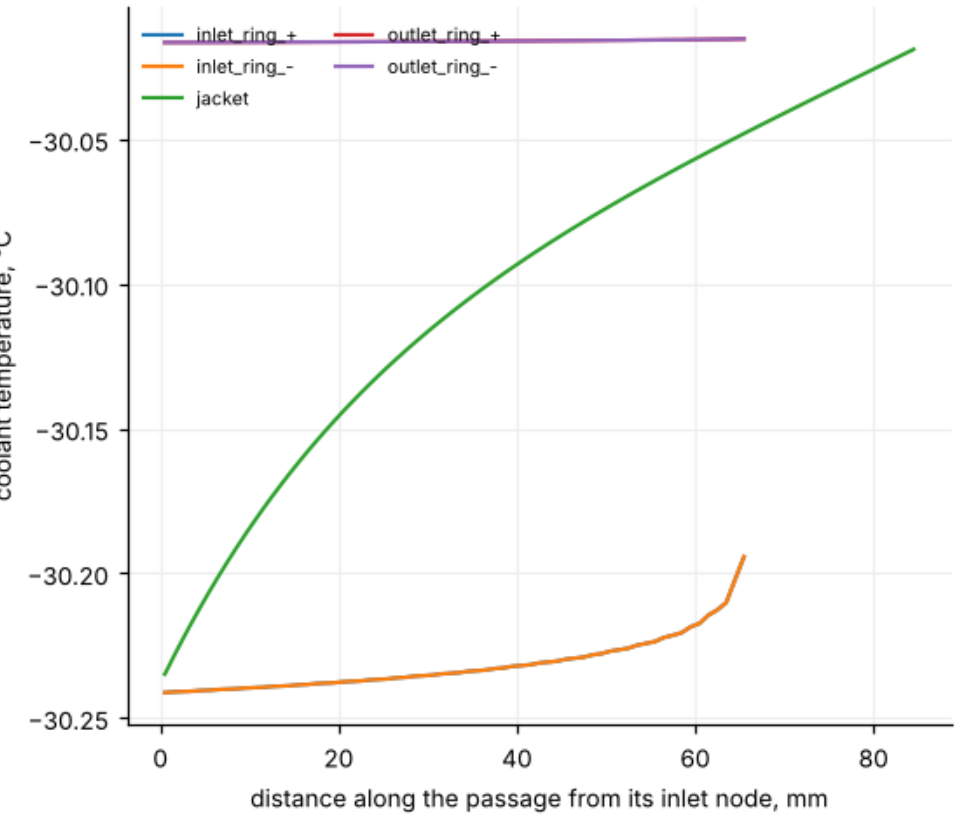
(c) Results — sheet against this model

Quantity	Sheet	This model
Coolant inlet (centres the wall)	-30.339 °C	-30.241 °C
Allowed inlet window, °C	-30.449 to -30.241	-30.345 to -30.138
Inner-wall spread	80.8 mK	93.5 mK
Margin inside ±0.15 K	109.6 mK	103.3 mK
Spread at 5 W (same pattern)	157 mK	181 mK
Coolant rise, mixed outlet	225 mK	226 mK
Pressure drop, worst path	0.52 kPa	0.35 kPa*
Reynolds number, passages	54 – 877	23 – 583
Coolant-copper UA	13.2 W/K	30.8 W/K
Energy balance error		3.9e-03 W

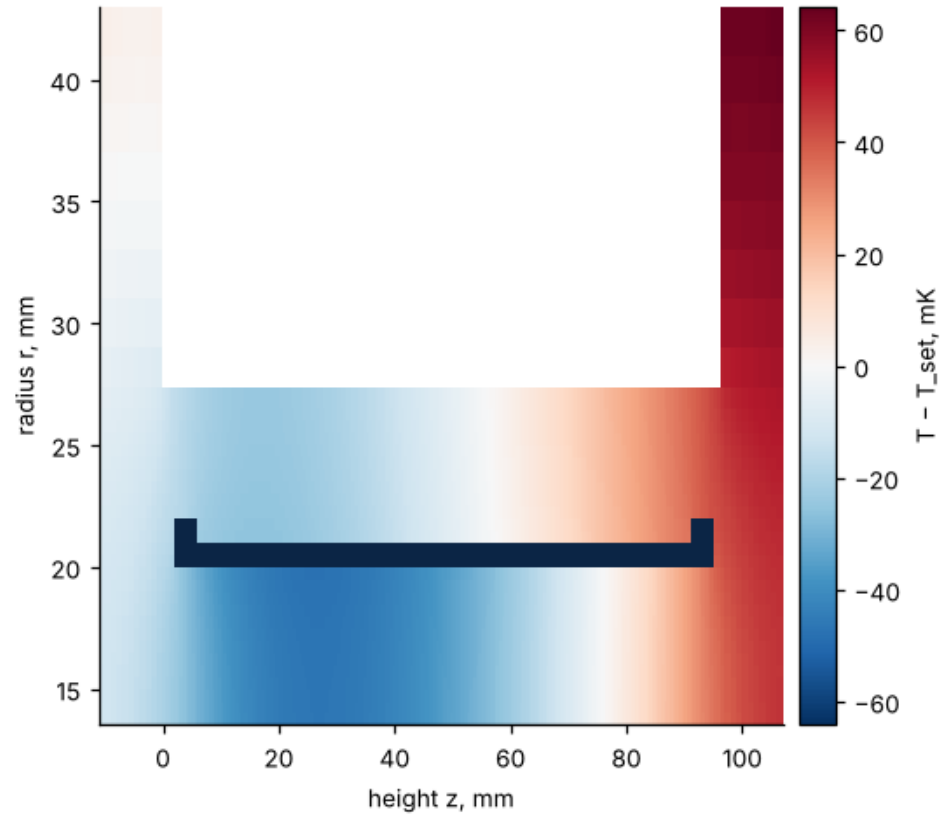
* friction of developed flow only (no bends / fittings)

Δ spread +12.7 mK · Δ inlet +98 mK → CLOSE

(d) Coolant temperature along each named passage



(e) Section at θ = 1° — copper T (passages navy)



(f) Inner wall along the height

